GE Healthcare

Signa Pre–EXCITE 3T94 (Eclipse) to HDx System Upgrade

Technical Publication

5166009
Rev 8

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

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GE Medical Systems, a General Electric Company, going to market as GE Healthcare
3200 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1	May 17, 2006	Initial Release.
2	Jul 17, 2006	Section 2: Added CAUTION note to upgrade flowchart that refers to the IIS Key and SaveINFO operation. Revised s/w load instructions and added LCD monitor adjustments (MRIhc15805, MRIhc16251, MRIhc16325). Added blocks for restoreINFO, ICW, and update of eLicense (MRIhc15873, MRIhc15892, MRIhc15894, MRIhc16209). Section 4: Added sections for installation of cables in equipment and magnet rooms. Added CAUTION note to not install O–ring and grease when connecting RF Body cables to penetration panel. Section 8: Added note informing the installer not to connect additional monitors to host computer prior to operating system software load (MRIhc15870).
3	Oct 22, 2006	Section 6: updated PAC replacement instructions with information on removing the PAC Control Housing (MRIhc17607).
4	Feb 22, 2007	Section 2: Reversed flowchart procedures *Software Option Install w/eLicense* and *Install and Calibration Wizard.* Section 6: Changed Notch Filter part number (CQA 13090674). Removed PAC2B part number. Section 8: Added note that Humidity iButton can be installed in either direction.
5	May 24, 2007	Section 5: Revised de–installation process by adding a Warning and illustration for removing mounting frame from magnet room side because frame may be made of magnetic material. Section 8: Installation instructions for OW table were revised due to introduction of new OW table. Updated iButton positioning and added block diagram in Environmental Sensor installation.
6	Jun 23, 2008	Section 4: Updated cable map to show the cabling of the existing and the new GOC. Updated cable maps to show new Temperature and humidity Sensor installation and cables. Section 6: Added bridge part number. Section 8: Revised entire section to add OW upgrade for systems that are shipped after July 1, 2008 with new GOC.
7	Sep 22, 2008	Section 4: (ECO 2054786) Sub–section 4–4, updated cable maps by adding Run 1382. Sub–section 4–7–1, added Runs 1381 and 1382 to PDU connection table. Section 8: Sub–Sectiion 8–4–7, corrected power and ground cable run numbers connected to OW. Sub–section 8–4–12, Added note about SCSI Tower being an option for HDxt and 15.x.
8	Jul 27, 2010	Section 2: Updated eLicense model type from Signa to MR HDx (per Direction 2397845). Section 4: Updated cable maps for removal of OW components and installation of Z400 GOC (ECO 2096124). Section 8: Updated instructions for removal of obsolete components and GOC. Added instructions for installing Z400 GOC (ECO 2096124).

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* This revision number/letter corresponds to the indicated document's revision control system.

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SECTION 1 – GETTING STARTED

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1–1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® 3T/94 pre–EXCITE system to HDx (Release 14.x). Refer to illustration 1–1 for pictorial overview of upgraded system.

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1–2 SITE READY CHECK

Pre–installation work must be completed before equipment is delivered to avoid delays and confusion.

Tools and test equipment to install and calibrate the Signa 3T94 system are listed in *Signa HDx 3.0T Upgrade Pre–Installation* manual, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46–301450G1 Fiber Optic Cable Connector Repair kit

1–3 PRE–DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa 3T94 HDx Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1–4 UPGRADE SCHEDULE CONSIDERATIONS

FMIs, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man–hour and system down–time total. The IIS key will need to be re–generated. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1–5 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The “Shipping Document” lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1–6 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **“damage in shipment”** written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1–7 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1–7–1 MATERIAL POLICY OVERVIEW

1–7–1–1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1–7–1–2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1–7–2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the EXCITE HD upgrade to GoldSeal which includes:

- All removed cabinets
- Penetration Panel
- SRI module
- PAC module
- All removed Magnet Enclosure parts
- All removed Rear Pedestal parts
- LPCC, Carriage, and/or removed Bridge and related parts
- Octane or Indigo or Operator Console, and Computer cabinet with all associated cables and modules

1–7–3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1–7–4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1–7–5 REGION SPECIFIC RETURN INFORMATION

1–7–5–1 Sites in Americas

A GoldSeal return pre–printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1–877–251–4468 or 262–513–4147

1–7–5–1 Sites in Americas (Continued)

For non–GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)–747–6997 or Dial Comm 8*579–6997
Fax: (414)–747–6855
Field ASPEN: 1–800–525–1516, Box #26041

1–7–5–2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS–Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1–7–5–3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS–E – GOLD SEAL OPERATION & RECYCLING CENTER
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL
Quai 45 à 49 – 91090 LISSES – FRANC
Tel: (+33) 1–60–91–57–35 or (+33) 1 64 97 55 41
Fax: (+33) 1–60–86–12–31

For non–Gold Seal returns in Europe:

European Renewable Resources Center
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2
Quai 45 à 49 – 91090 LISSES – FRANCE
TEL = (+33)–1–64–97–55–41 FAX = (+33)–1–60–86–12–31

1–7–6 PRODUCT LOCATOR INORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1–8.

1–8 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans–shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
   http://gein2.med.ge.com/gib
   The FE Site Verification consists of three components that are available on the web from the main menu. They are:
   - Install/deinstall product locator model and serial numbers
   - Add/modify ship to address information
   - Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page – http://supportcentral.ge.com/15563

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans–shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1–9 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on–site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the equipment room is un–cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1–10 UPGRADE CATALOGS

Pre−EXCITE Signa With 3T/4 to Signa HDx upgrade catalogs are listing in Illustration 1–19. The equipment modified and added by the catalogs listed in the flowchart are shown in the illustration 1–20.

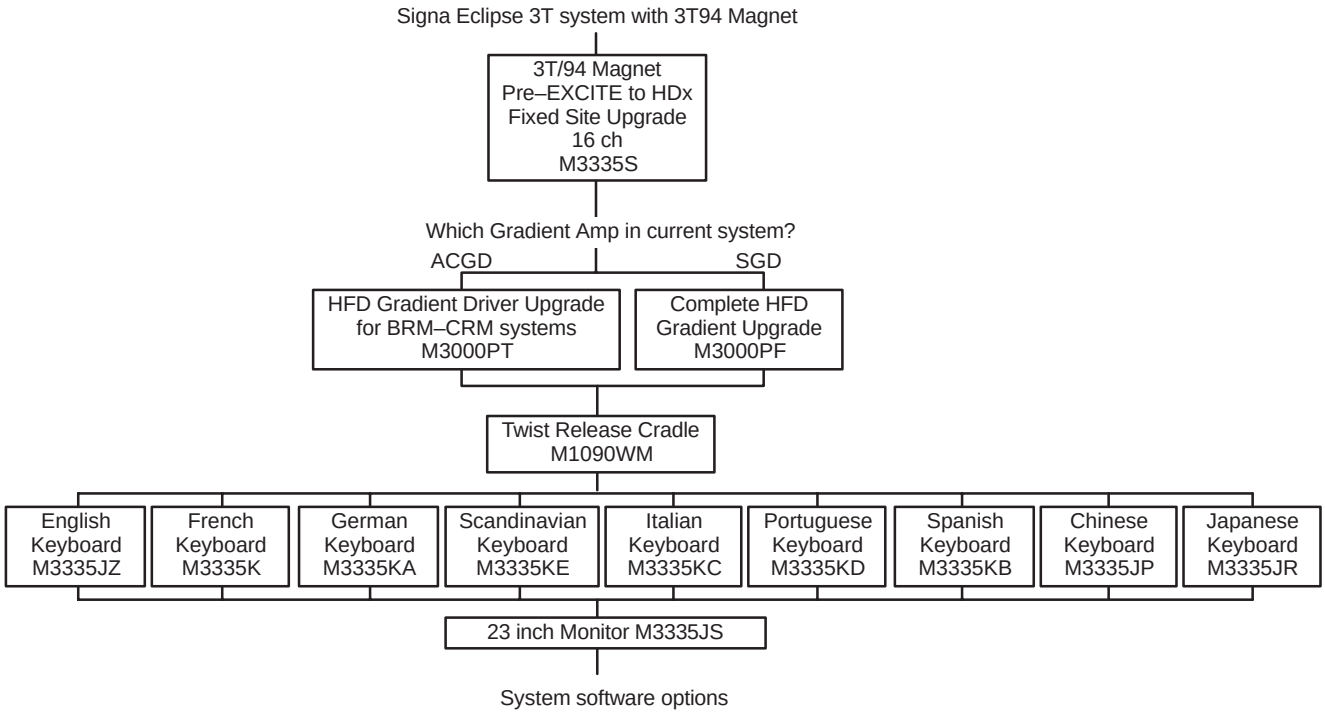


ILLUSTRATION 1–19
Pre−EXCITE Signa With 3.0T/94 To Signa HDx Upgrade Catalogs

1–10 UPGRADE CATALOGS (Continued)

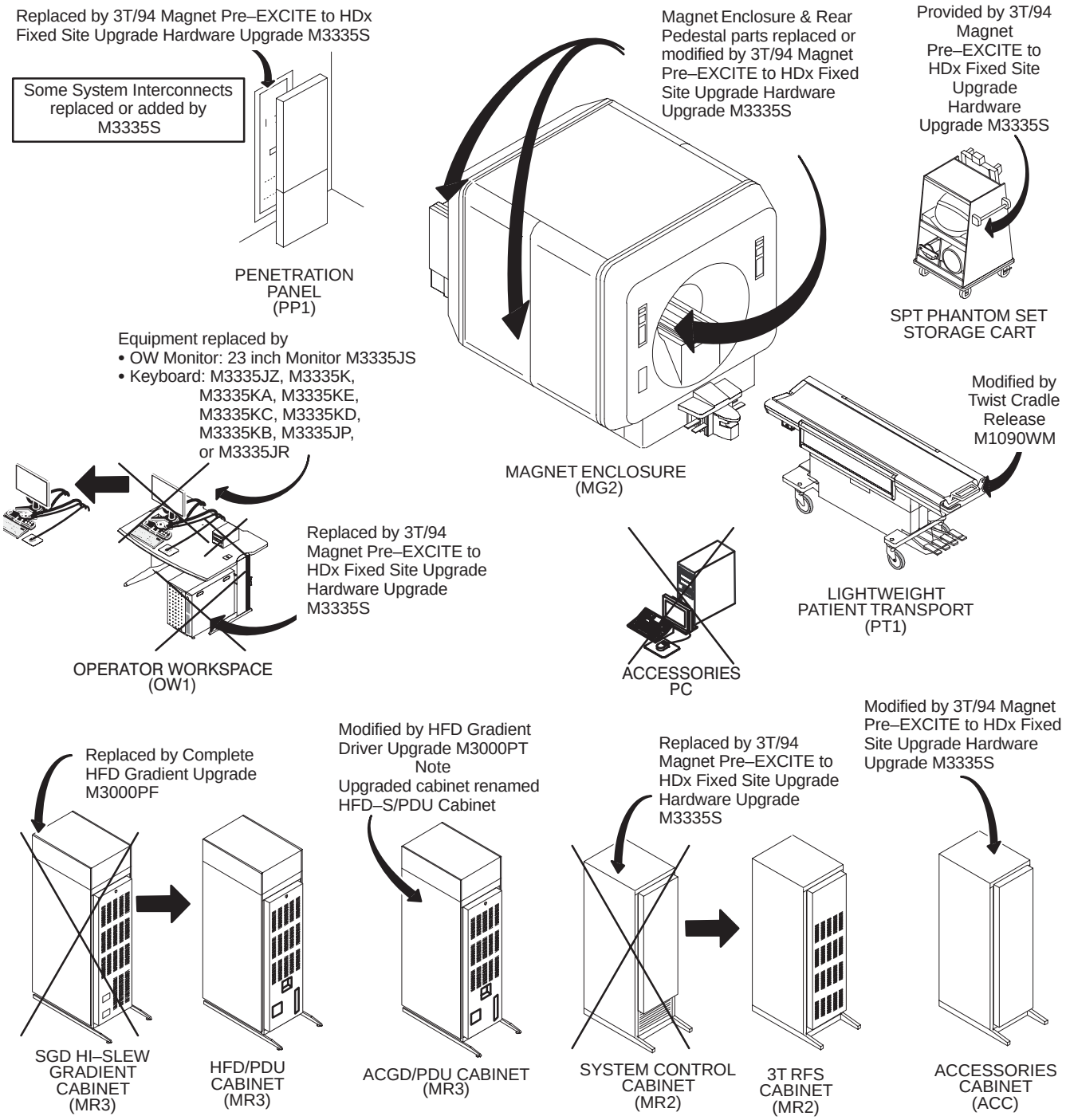
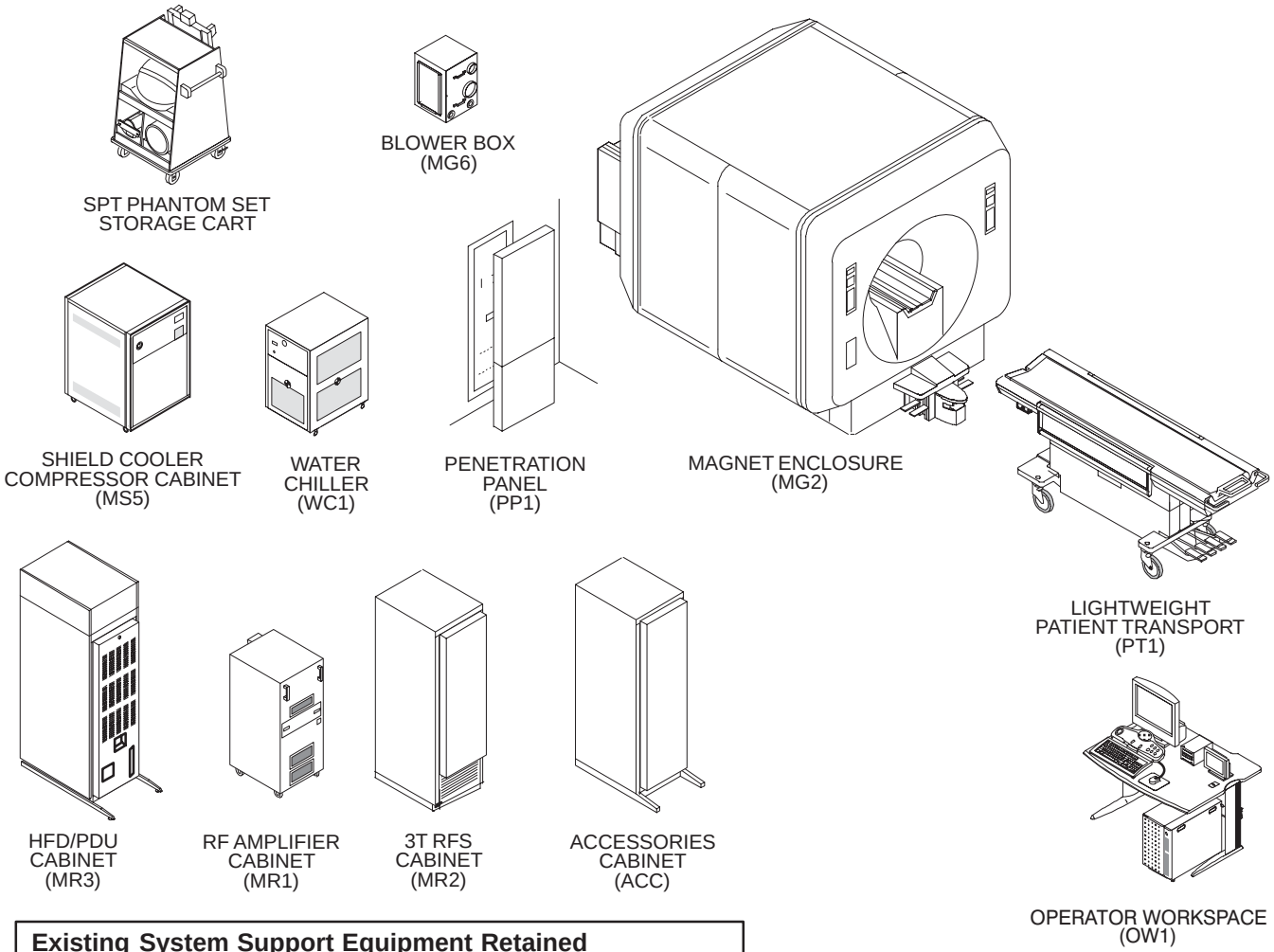


ILLUSTRATION 1–20

Pre–EXCITE Signa 3T/94 VH3 Existing Hardware Replaced Or Modified By HDx Upgrade

1–11 UPGRADE SYSTEM CONFIGURATION

Illustration 1–1 shows the major equipment installed for a Signa 3T/94 Pre–EXCITE to Signa HDx Upgrade.



Existing System Support Equipment Retained (not shown)

System retains their exisiting equipment for:

- Main Disconnnect Panel
- Shield Cooler Compressor water cooling equipment

ILLUSTRATION 1–1
PRE–EXCITE SIGNA 3T/94 VH3 UPGRADED TO Release 12.x SYSTEM MAJOR EQUIPMENT

SECTION 2 – UPGRADE INSTALLATION PROCESS

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2–1 SITE READINESS

All on–site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

2–2 UPGRADE PROCEDURE

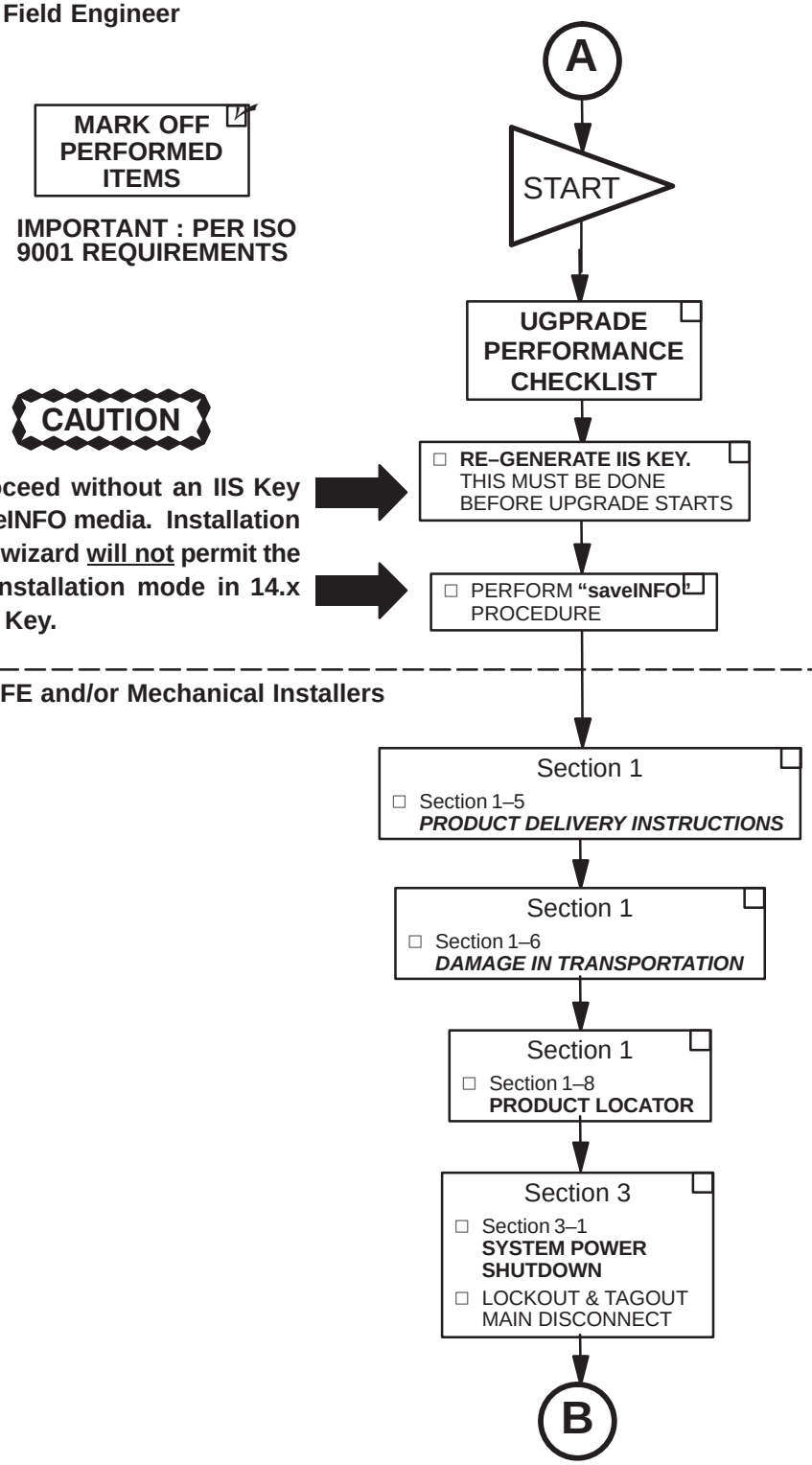
The following upgrade flow charts contain information for upgrading to Signa 3T/94 to EXCITE HD 3.0T Release 12.x.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa EXCITE HD equipment has been delivered together. **Make sure that every part required is available before starting.**

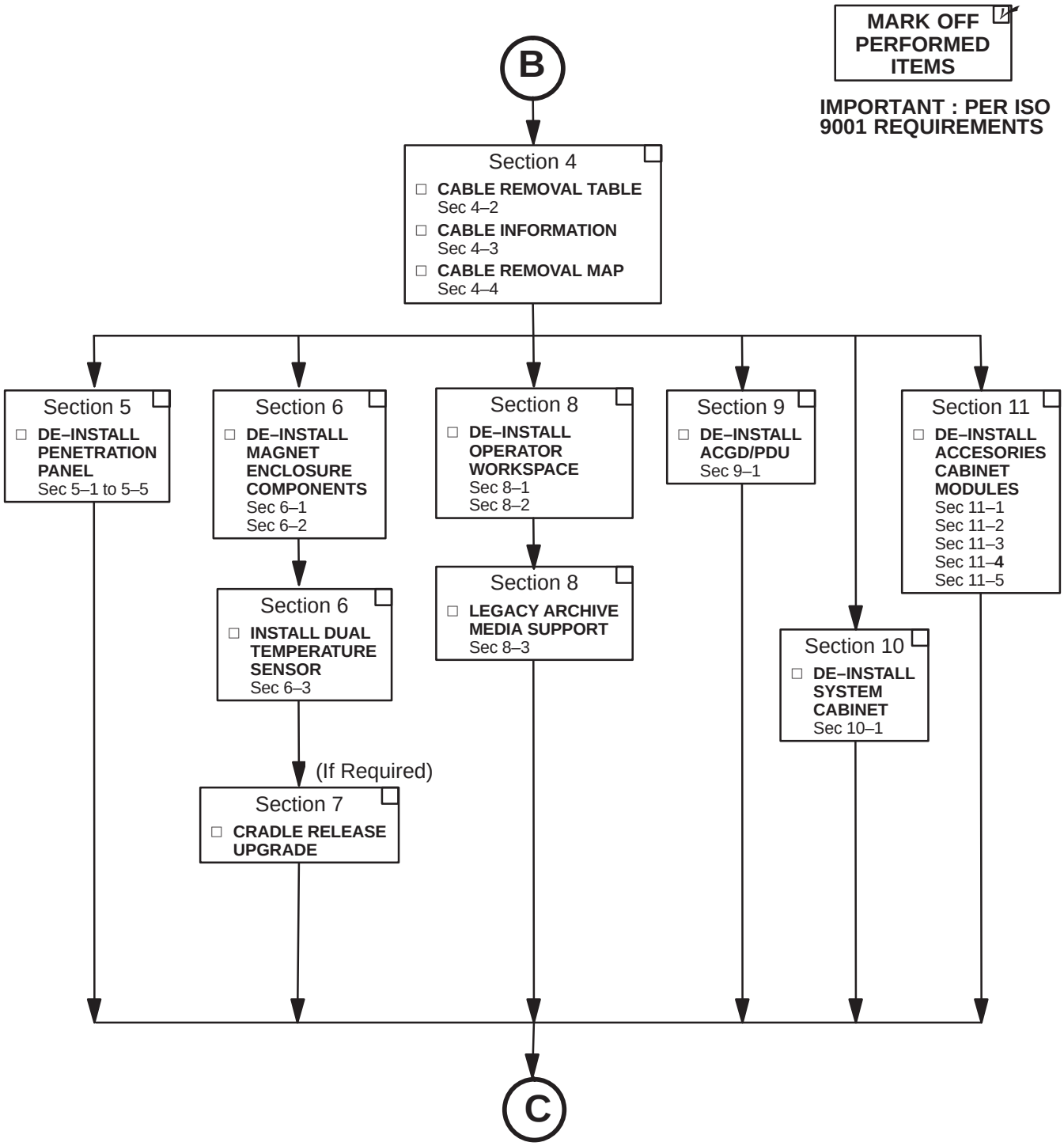
2–3 SYSTEM MECHANICAL UPGRADE FLOWCHART

Procedures to be Completed by Field Engineer



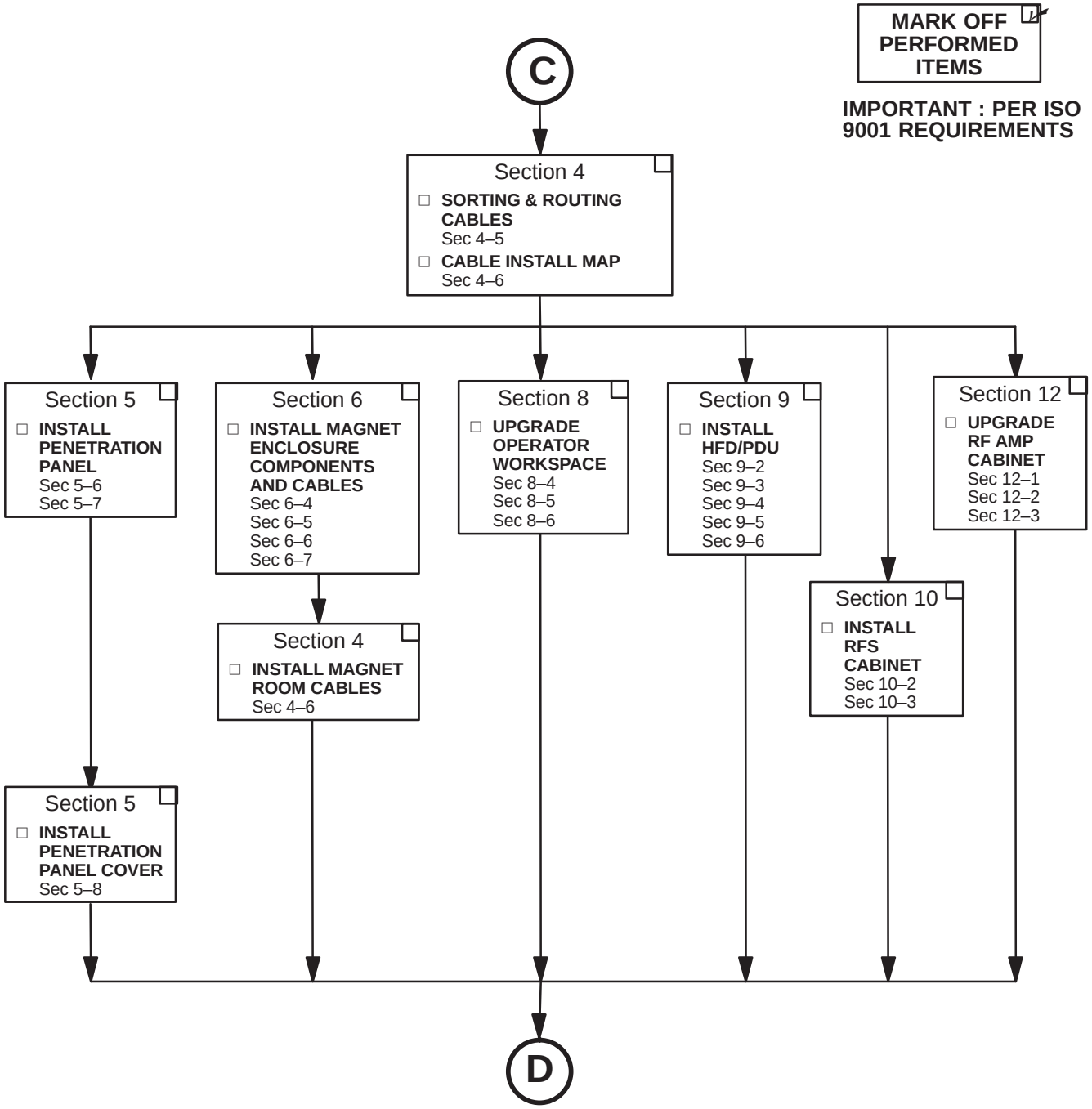
2–3 SYSTEM MECHANICAL UPGRADE START FLOWCHART (Continued)

De–Installation Procedures to be Completed by FE and/or Mechanical Installers

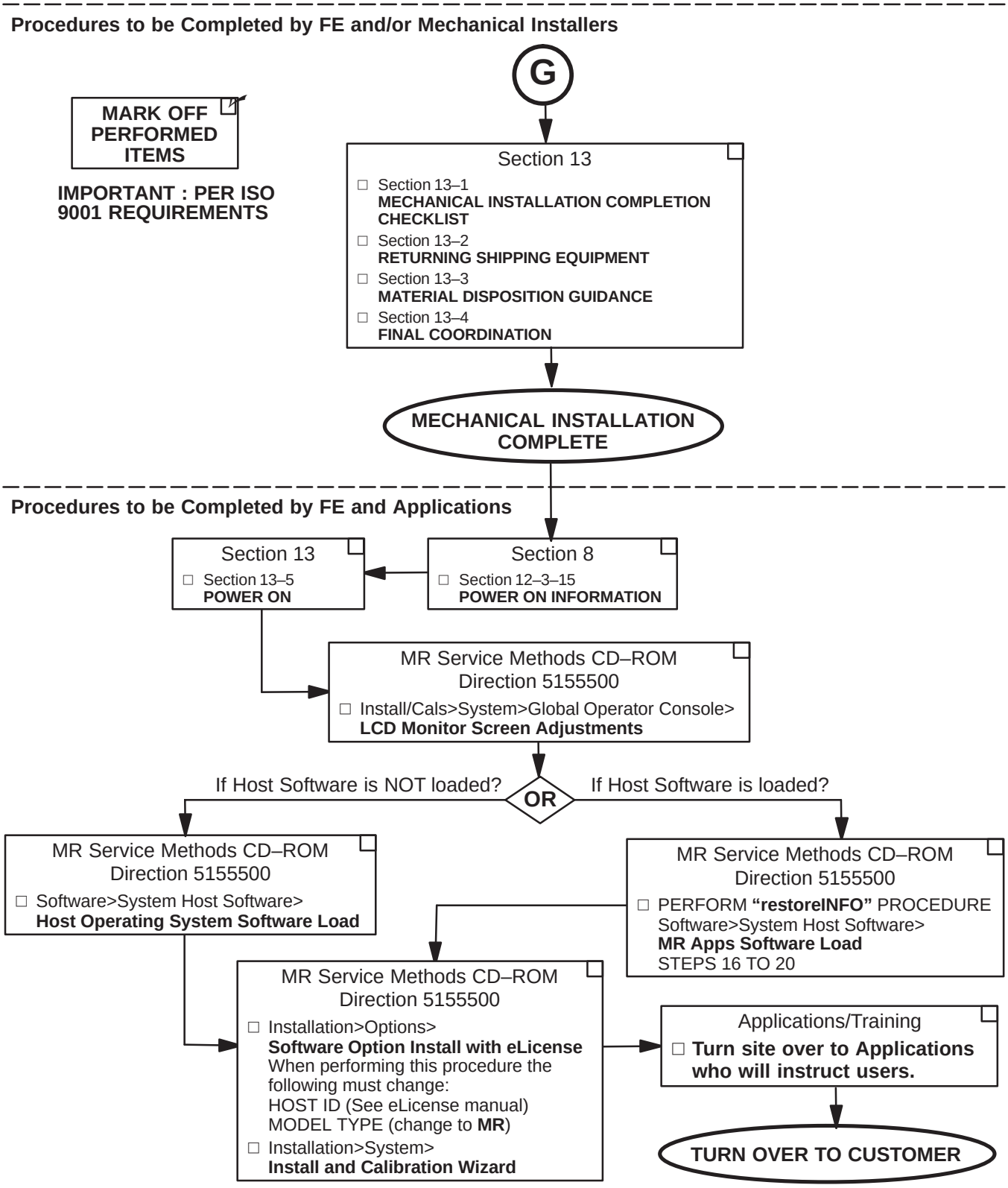


2–3 SYSTEM MECHANICAL UPGRADE START FLOWCHART (Continued)

Installation Procedures to be Completed by FE and/or Mechanical Installers



2–3 SYSTEM MECHANICAL UPGRADE START FLOWCHART (Continued)



SECTION 4 – SYSTEM CABLES

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4–1 INTRODUCTION

This section provides information on removing existing system cables and installing new cables for upgrading a Signa 3T/94 system to HDx.

4–2 CABLES DE–INSTALLED FOR UPGRADING TO SIGNA EXCITE HD

Table 4–1 lists all cables to be removed from 3T/94 system configurations upgrading to a Signa HDx system. Additional cable information is located in Section 4–3. Use table along with “Affected Cables” interconnect map for removal information.

TABLE 4–1
3T ECLIPSE CABLES REMOVED FOR UPGRADE TO HDx

RUN#	FROM	EQUIPMENT	TO	EQUIPMENT
037	PD1–GND BUS	PDU	MR2–A11 GND	System Cabinet
044	PD1–GND BUS	PDU	MR1–A7 GND	Acc Cabinet
045	PD1–GND BUS	PDU	MG6–GND	Fan Cabinet
048	PD1–GND BUS	PDU	OW1–A2–A5–GND	OW Cabinet
049	OW1–A1–A1	OW Table	OW1–A2–A5–J2	OW Cabinet
231	MR2–A11–J2	System Cabinet	PP1–J8	Pen Panel
235	MR1–A7–J13	RF Amp Cabinet	PP1–J44	Pen Panel
236	MR1–A7–J12	RF Amp Cabinet	PP1–J4	Pen Panel
257	MG3–A3–J1	Hybrid Splitter	PP1–J44	Pen Panel
258	MG3–A17–J1	Rear Pedestal	PP1–J4	Pen Panel
262	MG3–A3–A5–OUTPUT	BodyCoil Preamp	PP1–J8	Pen Panel
282	MG3–A2–J8	Rear Pedestal	PP1–J12	Pen Panel
318	MG3–A7–LONG DRIVE	Rear Pedestal	PP1–J11	Pen Panel
387	MG2–A29–P1	Dock	PP1–J47	Pen Panel
404	MG3–A17–J1	Rear Pedestal	MG2–A1– A5–J6	LPCC
485	MG3–A17–J2	Rear Pedestal	MG2–A16–A3 OUT	LPCC
486	PP1–A14–J85	Pen Panel	MR1–A7–J22	Acc Cabinet
487	PP1–A15–J78	Pen Panel	MR1–A7–J20	Acc Cabinet
488	PP1–J80	Pen Panel	MR2–A11–J8	System Cabinet
489	PP1–J81	Pen Panel	MR2–A11–J9	System Cabinet
490	PP1–J82	Pen Panel	MR2–A11–J10	System Cabinet
491	PP1–J80	Pen Panel	MG3–A36–J7	MC Select Sw
492	PP1–J81	Pen Panel	MG3–A36–J8	MC Select Sw
493	PP1–J82	Pen Panel	MG3–A36–J9	MC Select Sw
494	PP1–A14–J87	Pen Panel	MG3–A36–J6	MC Select Sw
495	PP1–A15–J78	Pen Panel	MG3–A36–J10	MC Select Sw
499	PP1–A14–J84	Pen Panel	MR2–A11–J5	System Cabinet
566	APC–CPU–PWR	Acc PC	MR1–A7–J231	Acc Cabinet
568	HOS Option Cable	GOC	J1 COMM 1	Acc Cabinet
571	APC–CPU–J2	Acc PC	OW1–A2–A4–PORT3	OW Cabinet

4–2 CABLES DE–INSTALLED FOR UPGRADING TO SIGNA EXCITE HD (Continued)

TABLE 4–1 (Continued)
3T ECLIPSE CABLES REMOVED FOR UPGRADE TO 3T with EXCITE TECHNOLOGY

RUN#	FROM	EQUIPMENT	TO	EQUIPMENT
700	PP1 A14 J86	Pen Panel	MG2/MG3 A17 J2	Rear Pedestal
702	MR2–J11–J18	System Cabinet	MR1–A7–J5	Acc Cabinet
708	MR2–A11–J16	System Cabinet	MR1–A7–J27	Acc Cabinet
709	MR1–A7–J26	Acc Cabinet	MR3–A9–J1	ACGD Cabinet
710	MR2–A11–J17	System Cabinet	MR3–A9–J2	ACGD Cabinet
711/712	PP1–A17–J91	Pen Panel	MR2–A11–J13	System Cabinet
726	PP1–A16–J77	Pen Panel	MR1–A7–J18	Acc Cabinet
727	PP1–A16–J77	Pen Panel	MG3–A36–J5	MC Select Sw
728	MG3–A36–J11	MC Select Sw	MG2–A16–OUT–J1	LPCC
729	MG3–A36–J12	MC Select Sw	MG2–A16–OUT–J2	LPCC
730	MG3–A36–J13	MC Select Sw	MG2–A16–OUT–J3	LPCC
731	MG3–A36–J14	MC Select Sw	MG2–A16–OUT–J4	LPCC
732	MG3–A36–J15	MC Select Sw	MG2–A16–OUT–J5	LPCC
733	MG3–A36–J16	MC Select Sw	MG2–A16–OUT–J6	LPCC
734	MG3–A36–J17	MC Select Sw	MG2–A16–OUT–J7	LPCC
735	MG3–A36–J18	MC Select Sw	MG2–A16–OUT–J8	LPCC
736	MG3–A36–J19	MC Select Sw	MG2–A16–A8–J9	LPCC
749	MG2–A33–J10	SRI Module	MG2–A36	Temp Sensor
768	MR1–A7–J31	Acc Cabinet	PP1–J51	Pen Panel
769	MR1–A7–J34	Acc Cabinet	PP1–J20	Pen Panel
770	MR1–A7–J41	Acc Cabinet	PP1–J15	Pen Panel
771	MR1–A7–J36	Acc Cabinet	PP1–J11	Pen Panel
772	MR1–A7–J40	Acc Cabinet	PP1–J47	Pen Panel
773	MR1–A7–J35	Acc Cabinet	PP1–J14	Pen Panel
774	MR1–A7–J32	Acc Cabinet	MR2–A11–J25	Pen Panel
775	MR1–A7–J42	Acc Cabinet	PP1–A11–J92	Pen Panel
776	MR1–A7–J44	Acc Cabinet	PP1–A11–J93	Pen Panel
777	MR1–A7–J43	Acc Cabinet	PP1–A11–J94	Pen Panel
778	MG2–A12–A2–J3	RF Coil	PP1–A11–J72	Pen Panel
779	MG2–A12–A2–J4	RF Coil	PP1–A11–J73	Pen Panel
780	MG3–A3–J6	Hybrid Splitter	PP1–A11–J74	Pen Panel
781	MG2–A12–A2–J5	RF Coil	PP1–A11–J75	Pen Panel
782	MG2–A12–A2–J6	RF Coil	PP1–A11–J76	Pen Panel
783	MR1–A7–J37	Acc Cabinet	PP1–F5,F6	Pen Panel
784	MG6–J1	Fan Cabinet	PP1–F5,F6	Pen Panel
785	MR1–A7–J47	Acc Cabinet	PP1–F7,F8	Pen Panel

4–2 CABLES DE–INSTALLED FOR UPGRADING TO SIGNA EXCITE HD (Continued)

TABLE 4–1 (Continued)
3T ECLIPSE CABLES REMOVED FOR UPGRADE TO 3T with EXCITE TECHNOLOGY

RUN#	FROM	EQUIPMENT	TO	EQUIPMENT
786	MG6–J12	Fan Cabinet	PP1–F7,F8	Pen Panel
788	MR2–A11–J30	OW Table	OW1–A1–A4–J7	OW Table
789	MR2–A11–J26	System Cabinet	OW1–A1–A2–J18	OW Table
790*	MR2–A11–J21	System Cabinet	OW1–A2–A6–X0,R0,X1,R1	OW Cabinet
791	MR2–A11–J30	System Cabinet	OW1–A16–PORT1	OW SCSI
792	OW1–A2–A1–MIC/AUDIO	OW Cabinet	OW1–A1–A4–J11,J13	OW WIM
797	OW1 A2 A1 PC	OW Cabinet	OW1 A3 LCD	LCD Display
805	OW1–A1–A4–J6	OW Table	OW1–A7	Keyboard
807	OW1–A16	OW Table	OW1–A5–DASM	OW Table
810	OW1–A15–MD,KYBD	Octane	OW1–A1–A4–J8	OW WIM
811	OW1–A2–A4–PORT2	OW WIM	OW1–A15–ETHERNET	Octane
812	OW1–A15–COM1	Octane	OW1–A13–MODEM	OW Table
813	OW1–A2–A1–COM2	OW Cabinet	OW1–A15–COM2	Octane
814	OW1–A1–A4–J9	OW WIM	OW1–A2–A1–KBD,MOUSE	OW Cabinet
815	OW1–A16–SCSI	OW SCSI	OW1–A15–SCSI (PCI)	Octane
816*	OW1–A2–A5–J1	OW Cabinet	OW1–A15–BIT3	Octane
817	OW1–A2–A5–J3	OW Cabinet	OW1–A1–A4–J17	OW Table
818	MR2–A11–J31	System Cabinet	OW1–A16–PORT6,7,8	OW SCSI
819	OW1–A2–A7–SCSI	OW Cabinet	OW1–A15	Octane
821	OW1–A6	OW Table	OW1–A15–MONITOR	Octane
836*	MR1–A11–J21	System Cabinet	OW1–A15–BIT3	Octane
861	MG2–A4–J4	Front Cover	MG2–A33–J5,6,7,8,9	SRI
968	MR2–A11–J1	Systen Cabinet	MR3–PD1–4,5,6,GND	PDU
969	OW1–A2–A5–GND	OW Cabinet	MR3–PD1–18,19,GND,N	PDU
970	WC1	Water Chiller	MR3–PD1–16,17,GND	PDU
971	MR2–A11–J23	System Cabinet	MR3–PD1–J2	PDU
972	MR2–A11–J14	System Cabinet	MR3–PD1–J3	PDU
996	NB–AMP–J18	RF Amp Cab	DRSM–J7	Acc Cabinet
998	MR2–A11–J15 (RUN 701)	System Cabinet	3T CONTROL–J56	Acc Cabinet
1000	NB–AMP–J20	RF Amp Cab	DRSM–J16	Acc Cabinet
1002	NB–AMP–J14	RF Amp Cab	DRSM–J18	Acc Cabinet
1003	MR1–A11–J1	System Cabinet	DRSM–J19	Acc Cabinet
1006	NB–AMP–J10	RF Amp Cab	SSM–J408	Acc Cabinet
1007	NB–AMP–J9	RF Amp Cab	SSM–J409	Acc Cabinet

* Prior to Run 836, sites were installed with Runs 790 and 816 routed to Bit3 at bottom of OW Cabinet. If so for this site, remove Runs 790 and 816.

4–3 ADDITIONAL CABLE INFORMATION

The following items refer to handling of existing cables for this upgrade.

1. A new RF Door Switch cable, Run 701, is provided for upgrade. If existing cable is routed within wall, it may be reused. If existing cable in labeled as Run 207, replace with new Run 701 or relabel per instructions in Section 4.
2. If optional Oxygen Monitor is installed on site, retain Runs 457 and 458 for connection to new Penetration Panel.
3. Magnet Room side of Gradient cables, Runs 762, 763, and 764 may be reused. Butt Splices and Shrink Wrap Tubing are not provided with upgrade for reterminating Gradient cables at rear of magnet.
4. Existing runs 762, 763, and 764 in Equipment Room can also be reused and reconnected to new HFD/PDU Cabinet and Penetration Panel.
5. Existing microphone cable will be replaced with upgrade.
6. Existing Emergency Off Switch cables, Runs 296 and 297, may be retained and reused.
7. Early system configurations installed Run 612, later configurations installed Run 1152 for Shim option.

4–4 CABLE REMOVAL INTERCONNECT MAP

A double set of “Cable Removal” interconnect maps is provided. One set for the equipment room and the other for the magnet room. Tape maps in a convenient location in the Magnet Room and the Equipment/Operator Room to check off affected “Removed” cables.

The “Removed” Cable maps are located on the following pages.

- Page 4–7 Affected (removed) Cables for Upgrading to Signa 3.0T HDx (Equip. Room)
- Page 4–9 Affected (removed) Cables for Upgrading to Signa 3.0T HDx (Magnet Room) *(Duplicate)*

For the convenience of the mechanical installers, a duplicate System Interconnect is provided. This will allow the install team to post one interconnect map in the Magnet Room and the other map in the Equipment/Operator Room. This will improve the efficiency of the installation.

Upon completion of connecting cables, return fold–out maps to the binder for future reference.

Affected (removed) Cables for Upgrading to Signa 3T94 HDx from 3.0T VH3 ECLIPSE

- with ACGD/PDU
- with Octane Computer
- with Multicoil

NOTES:

1. Customer furnished.
2. Helium meter provided by magnet vendor.
3. Adaptors selected according to coil used (HEAD, SURFACE) and MAGNET POLARITY.
4. Typical connection for normally closed emergency off switch.
5. OPTIONAL: May not be present on all systems.
6. Customer supplied telephone line for insite modem.
7. Ramp and shim cabling provided by Magnex.
8. Wall mounted MRDU.
9. Flexible extension cables
10. Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. If so for this site, remove Runs 790 & 816.
11. For this upgrade, new cable is provided. Installer has option of relabeling existing cables or installing new.

LEGEND:

RELABELED/RELOCATED CABLE (OLD) XXX (NEW) YYY

REMOVED CABLE XXX

DISCONNECTED AT BOTH ENDS AND REUSED XXX

DISCONNECTED AT ONE END AND REUSED XXX

MODIFIED CABINET OR MODULE

REMOVED CABINET OR MODULE

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Affected (removed) Cables for Upgrading to Signa 3T94 HDx from 3.0T VH3 ECLIPSE (*Duplicate of Page 4–7*)

- with ACGD/PDU
- with Octane Computer
- with Multicoil

NOTES:

1. Customer furnished.
2. Helium meter provided by magnet vendor.
3. Adaptors selected according to coil used (HEAD, SURFACE) and MAGNET POLARITY.
4. Typical connection for normally closed emergency off switch.
5. OPTIONAL: May not be present on all systems.
6. Customer supplied telephone line for insite modem.
7. Ramp and shim cabling provided by Magnex.
8. Wall mounted MRDU.
9. Flexible extension cables
10. Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. If so for this site, remove Runs 790 & 816.
11. For this upgrade, new cable is provided. Installer has option of relabeling existing cables or installing new.

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4–5 SORTING AND ROUTING CABLES

CAUTION

Ty–wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

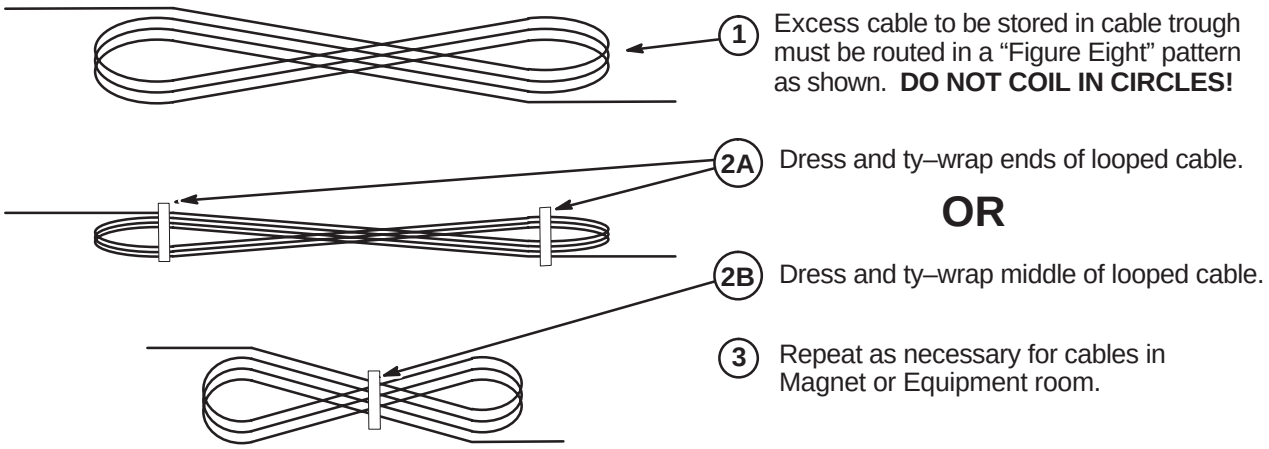
System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.

3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 2–1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber–optic cables.



PROPER STORAGE OF EXCESS CABLES
ILLUSTRATION 2–1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

4–6 HDx CABLE INTERCONNECT MAPS

The following cable map illustrates the Signa 3T/94 upgrade to HDx system interconnect for all supplied cables and interface with customer supplied wiring.

- Interconnects for additional cables supplied with other options are illustrated in the installation manual shipped with the option.
  - – Instructions for connections for optional laser cameras are supplied with option.
  - – Connection procedures for other options are covered in the installation manual shipped with the option.
  - – Spectroscopy option cabling can be found on page 4–17.
- Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation section within this manual.

A double set of “Added” cables maps is provided. Tape maps in a convenient location to check off affected “Added” cables. This will improve the efficiency of the upgrade process.

- One set if site has a 25kW RF Amp and the other set is if site has a 25/35kW NB RF Amp. Select one or the other.
- One set for the equipment room and the other for the magnet room.

The Affected “Added” Cable maps are located on the following pages.

- Page 4–13 For Upgrading to Signa HDx with 25kW RF Amp (Equipment Room)
- Page 4–14 For Upgrading to Signa HDx with 25/35kW NB RF Amp (Equipment Room)
- Page 4–15 For Upgrading to Signa HDx with 25kW RF Amp (Magnet Room) *(Duplicate of page 4–13)*
- Page 4–16 For Upgrading to Signa HDx with 25/35kW NB RF Amp (Magnet Room) *(Duplicate of page 4–14)*

Upon completion of connecting cables, return fold–out map to the binder for future reference.

Signa 3T/94 HDx System Interconnect Upgraded from 3.0T VH3 ECLIPSE with 25kW NB RF Amp

NOTES:

1. CUSTOMER FURNISHED
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Refer to next page for cooling system that's designed for your site. Source of water supply for Cold Head Compressor depends on system used.
12. Refer to Tab 6, Section 6–2–3, for instructions on ty–wrapping ground cables to each gradient cable between TAC and PP1.
13. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethermet connection at OW1 A7 PORT 8.

There are two types of GOC.
– GOC shipped after July 1, 2008
– GOC shipped before July 1, 2008

For detail Internal wiring, refer to page 4–18.

Basically, cables from GOC are pre–connected before shipment for forward production.

Some cable Run number from GOC is different between two types of GOC.
Please read the cable Run Number as follows.

A /B
A: GOC shipped after July 1, 2008
B: GOC shipped before July 1, 2008

Signa 3T/94 HDx System Interconnect Upgraded from 3.0T VH3 ECLIPSE with 25/35kW NB RF Amp

NOTES:

1. CUSTOMER FURNISHED
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Refer to next page for cooling system that's designed for your site. Source of water supply for Cold Head Compressor depends on system used.
12. Refer to Tab 6, Section 6–2–3, for instructions on ty–wrapping ground cables to each gradient cable between TAC and PP1.
13. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethermet connection at OW1 A7 PORT 8.

There are two types of GOC.
– GOC shipped after July 1, 2008
– GOC shipped before July 1, 2008

For detail Internal wiring, refer to page 4–18.

Basically, cables from GOC are pre–connected before shipment for forward production.

Some cable Run number from GOC is different between two types of GOC.
Please read the cable Run Number as follows.

A /B
A: GOC shipped after July 1, 2008
B: GOC shipped before July 1, 2008

Signa 3T/94 HDx System Interconnect Upgraded from 3.0T VH3 ECLIPSE with 25kW NB RF Amp

(Duplicate of Page 4–13)

NOTES:

1. CUSTOMER FURNISHED
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Refer to next page for cooling system that's designed for your site. Source of water supply for Cold Head Compressor depends on system used.
12. Refer to Tab 6, Section 6–2–3, for instructions on ty–wrapping ground cables to each gradient cable between TAC and PP1.
13. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethermet connection at OW1 A7 PORT 8.

There are two types of GOC.
– GOC shipped after July 1, 2008
– GOC shipped before July 1, 2008

For detail Internal wiring, refer to page 4–18.

Basically, cables from GOC are pre–connected before shipment for forward production.

Some cable Run number from GOC is different between two types of GOC.
Please read the cable Run Number as follows.

A /B
A: GOC shipped after July 1, 2008
B: GOC shipped before July 1, 2008

Signa 3T/94 HDx System Interconnect Upgraded from 3.0T VH3 ECLIPSE with 25/35kW NB RF Amp

(Duplicate of Page 4–14)

There are two types of GOC.
– GOC shipped after July 1, 2008
– GOC shipped before July 1, 2008

For detail Internal wiring, refer to page 4–18.

Basically, cables from GOC are pre–connected before shipment for forward production.

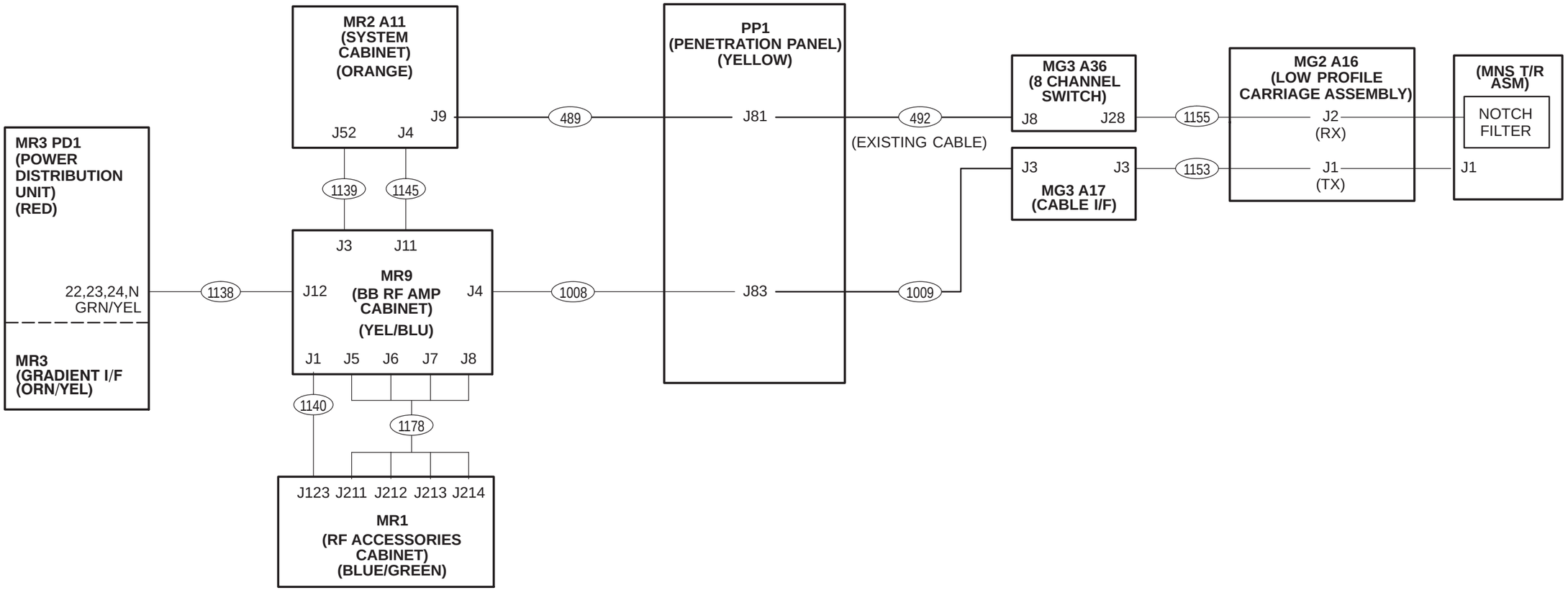
Some cable Run number from GOC is different between two types of GOC.
Please read the cable Run Number as follows.

A /B
A: GOC shipped after July 1, 2008
B: GOC shipped before July 1, 2008

NOTES:

1. CUSTOMER FURNISHED
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on PP1 plate.
7. Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. Customer supplied telephone line for Insite Modem.
11. Refer to next page for cooling system that's designed for your site. Source of water supply for Cold Head Compressor depends on system used.
12. Refer to Tab 6, Section 6–2–3, for instructions on ty–wrapping ground cables to each gradient cable between TAC and PP1.
13. Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethermet connection at OW1 A7 PORT 8.

2.5KW MNS Option Cable Map



4–7 EQUIPMENT ROOM CABLE INSTALLATION

The equipment room cables and cabinets can be installed simultaneously. Cable locations depend on where cabinets are installed.

4–7–1 HFD CABINET CABLE CONNECTIONS

Note: If site has an ACGD/PDU cabinet and is being upgraded to HFD–S/PDU, refer to Section 4–7–1–C for connection of cables to PDU module.

DANGER!!

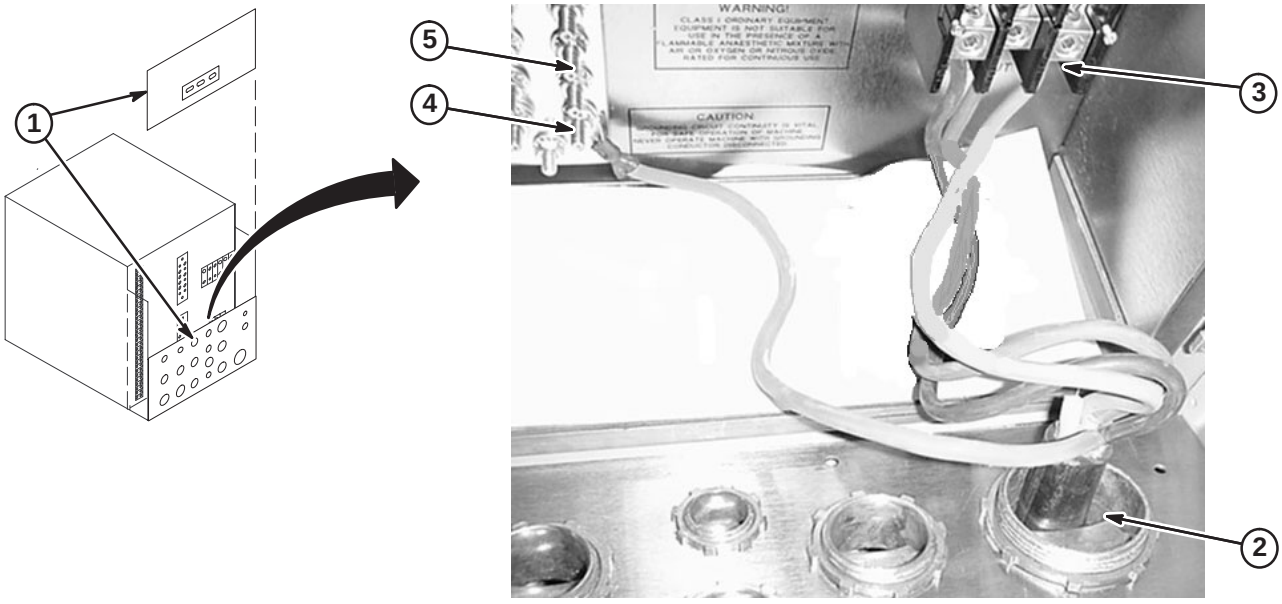
MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE CONNECTING WIRES.

WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE HFD CABINET.

4–7–1–A HFD Cabinet Power and Ground Cable Installation

1. Remove rear PDU module covers.
2. Route facility input power cable and ground wire thru interface panel.
3. Connect L1–L3 conductors to terminal block.
4. Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.
5. Connect RF screen room ground conductor to Ground Bus Bar.

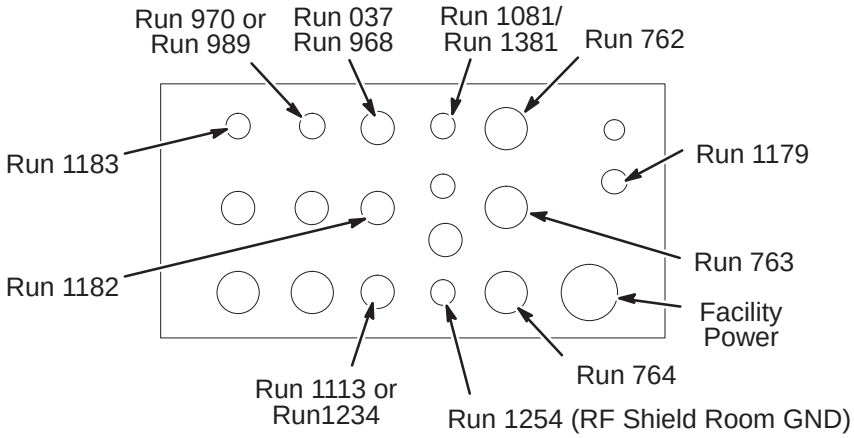


4–7–1–B Route Power Cables Thru Interface Panel

1. Route Runs thru interface panel as marked.

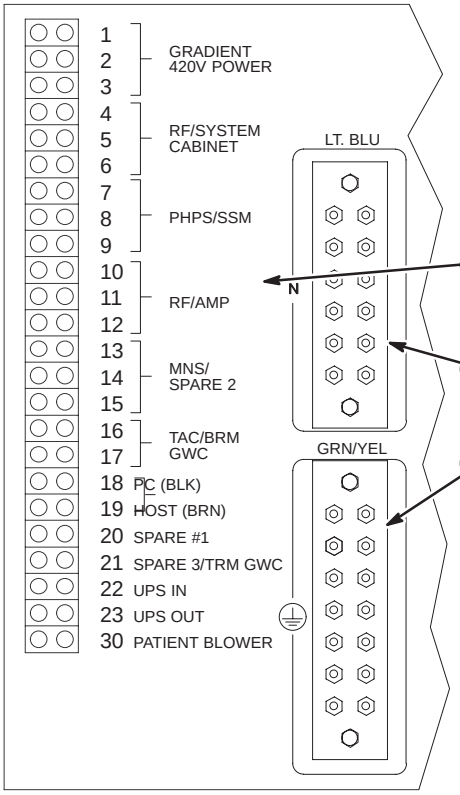
Note: The GRN/YEL ground wire, Run 037, **MUST** be routed with Run 968 power cable thru the same port.

2. Route any option Runs as marked.



4–7–1–C Connect Power Cables to PDU Module

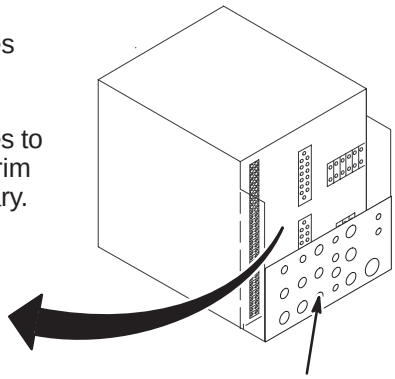
Note: Connection to Main Disconnect Panel to be completed by customer electrician.



WARNING!

ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN HFD CABINET.

1. Route/connect power cables to pressure connectors and studs on back of PDU module as shown in applicable Table below
2. Connect neutral wire cables to Neutral bus connectors.
3. Connect ground wire cables to Ground bus connectors. Trim and reterminate if necessary.



4. Tighten clamps of interface panel to secure cables in place.

CONNECT TO	RUN	PHASE	DESTINATION	
4,5,6	968	ϕ A, B, C	RFS Cabinet	3ϕ, Neutral, GND
7,8,9*	966	ϕ A, B, C	Accessory Cabinet	3ϕ, GND
10,11,12	1234	ϕ A, B, C	RF Amp in RFS Cabinet	3ϕ, Neutral, GND
16,17	970	ϕ A,B	Water Chiller	Single ϕ GND
18,19	1081	ϕ C,A	Operator Workspace (Old GOC)	Single ϕ, Neutral, GND
19	1381	ϕ A	Operator Workspace (New GOC)	Single ϕ, Neutral, GND
20*	1182	ϕ A	PHPS Module	Single ϕ Neutral, GND
30	1183	ϕ A	Blower Box	Single ϕ, Neutral, GND
GRN/YEL	1382		GOC Ground Wired (New GOC)	
GRN/YEL	037		System Cabinet Ground Wire	
GRN/YEL	037A		Accessory Cabinet Ground Wire	
GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)	

COLOR CODE: for remainder of cables:

3ϕ = Black, Red, Orange Neutral = Light blue Single ϕ = Brown, Black Gnd = Grn/Yel

* If necessary, relabel circuit breakers at front of cabinet to reflect actual connections of Acccesory Cabinet and PHPS Module.

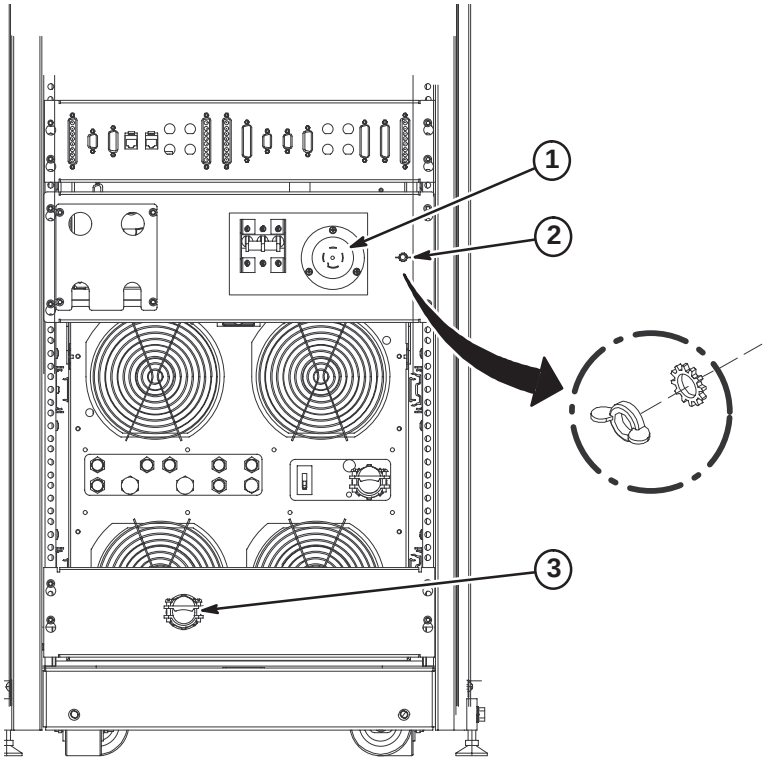
4–7–2 EQUIPMENT ROOM CABINET POWER CABLE CONNECTIONS

WARNING!

CABINET POWER AND GROUND CABLES MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN HFD CABINET.

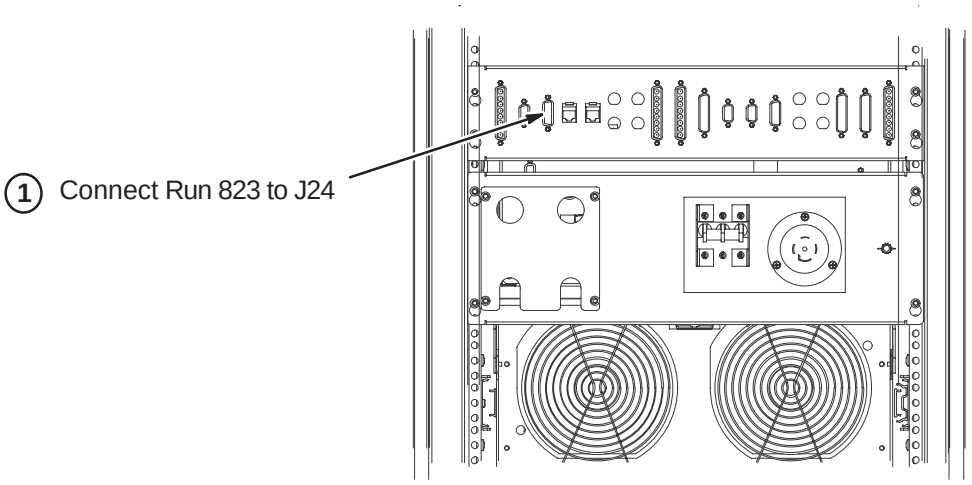
4–7–2–A RF/System Cabinet Power and Ground Cable Connections

1. Connect Run 968 cable plug to “J1”. Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.
3. Route Run 1234 thru lower I/F panel and clamp in place. Connect as marked to RF Amp module.



4–7–3 INSTALLATION OF MAGNET MONITORING CABLES

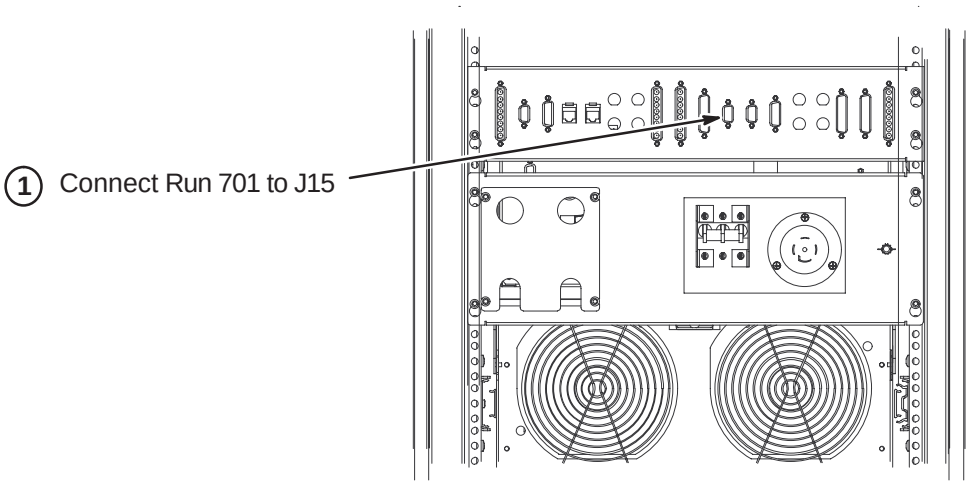
Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual,* for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.



4–7–4 INSTALL RUN 701 – RF DOOR SWITCH (If Required)

Run 701 is a 100 ft cable with a 9–pin subminature “D” plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminature D plug.

1. Connect Run #701 to J15 on System Cabinet and route to RF Door Switch. Refer to site engineering drawings for cable routing path.
2. Connect black lead on Run #701 to RF Door Switch COM (common) contact.
3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
4. Connect red lead on Run #701 to RF Door Switch N.O. (normally open) contact.
5. If RF door switch is not normally open, connect red lead on Run #701 to RF Door Switch N.C. (normally closed) contact.

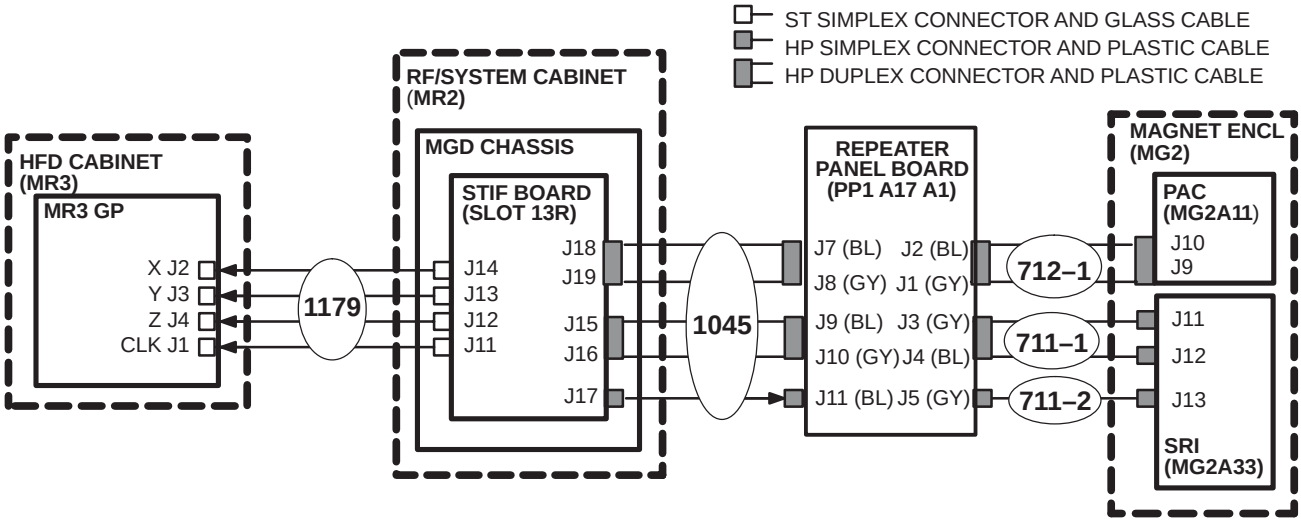


4–7–5 INSTALLATION OF FIBER OPTIC CABLES

Plan for storage of excess Run 1179 conduit, as it cannot be reterminated. For most Horizon amd LX configurations, Runs 711/712 in the magnet room will be retained for the upgrade to EXCITE HD.

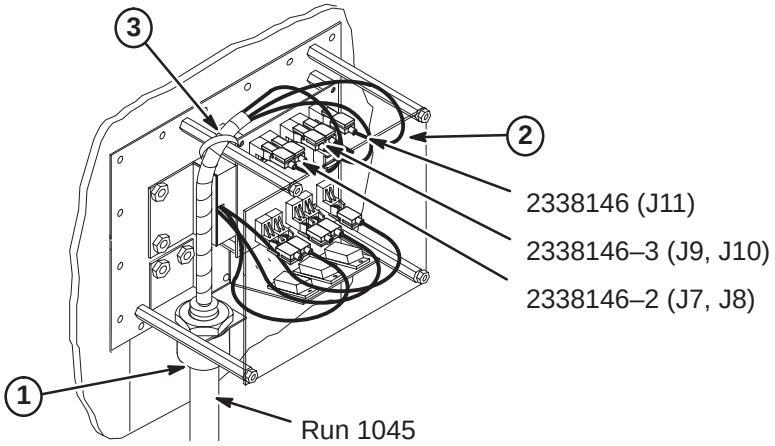
4–7–5–A Routing of Fiber Optic Cables

1. Unroll coiled fiber optic conduits for Runs 711/712, 1045, and 1179 along intended routes.
2. Connection of Runs is shown in diagram below.



4–7–5–B Connect Run 1045 to Penetration Panel

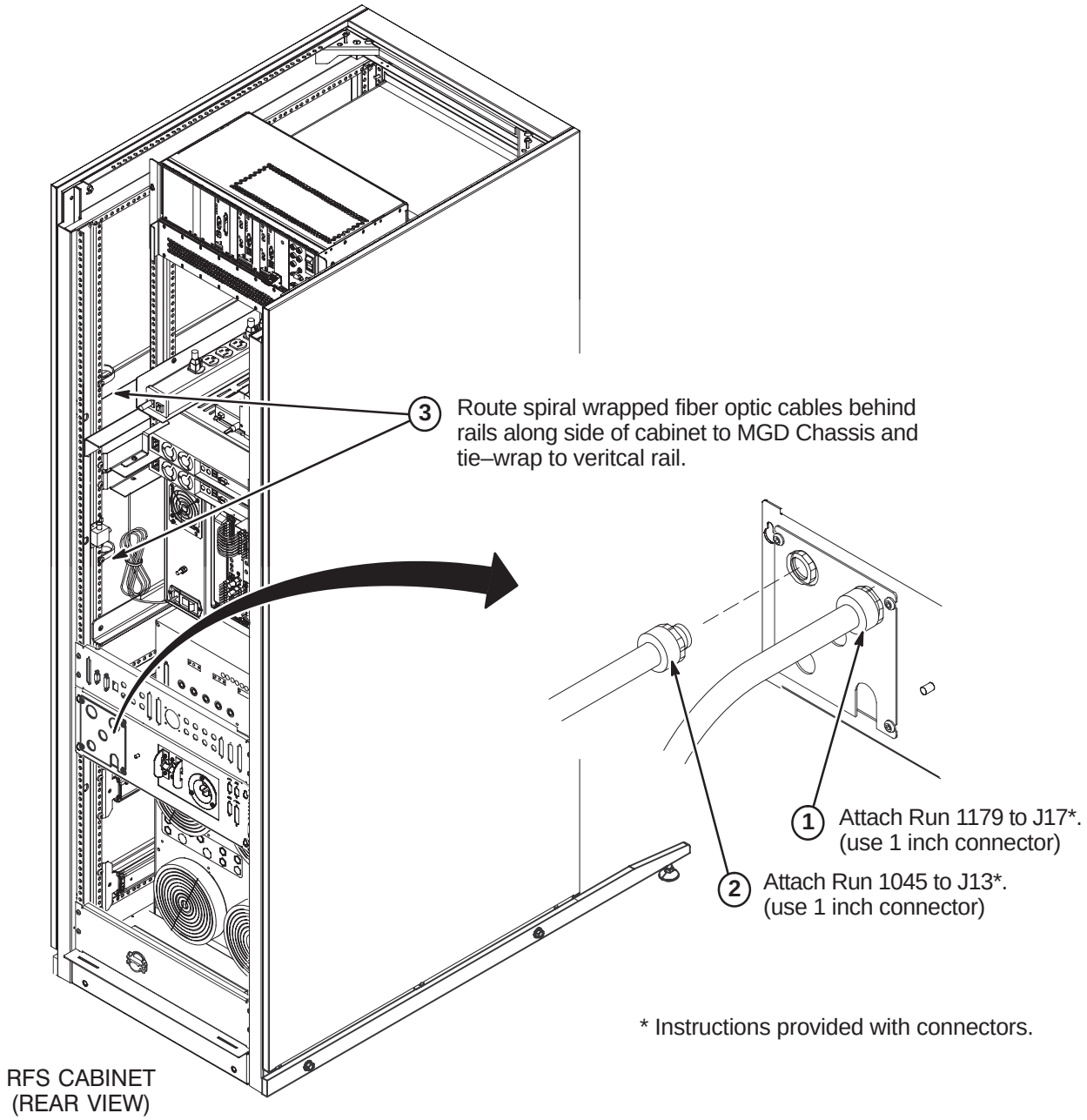
1. Move Run 1045 to the Equipment Room side and attach “PP1–A17” end to access bracket on Penetration Panel. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
2. Adjust route of individual fiber optic runs to maintain large radius and connect Run 1045 as shown.
3. Adjust position of protective spiral wrap and secure with tie–wrap.
4. Route other end of Run 1045 to System Cabinet.



4–7–5–C Route Fiber Optic Runs 1045, 1083, and 1179 in RF/System Cabinet

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

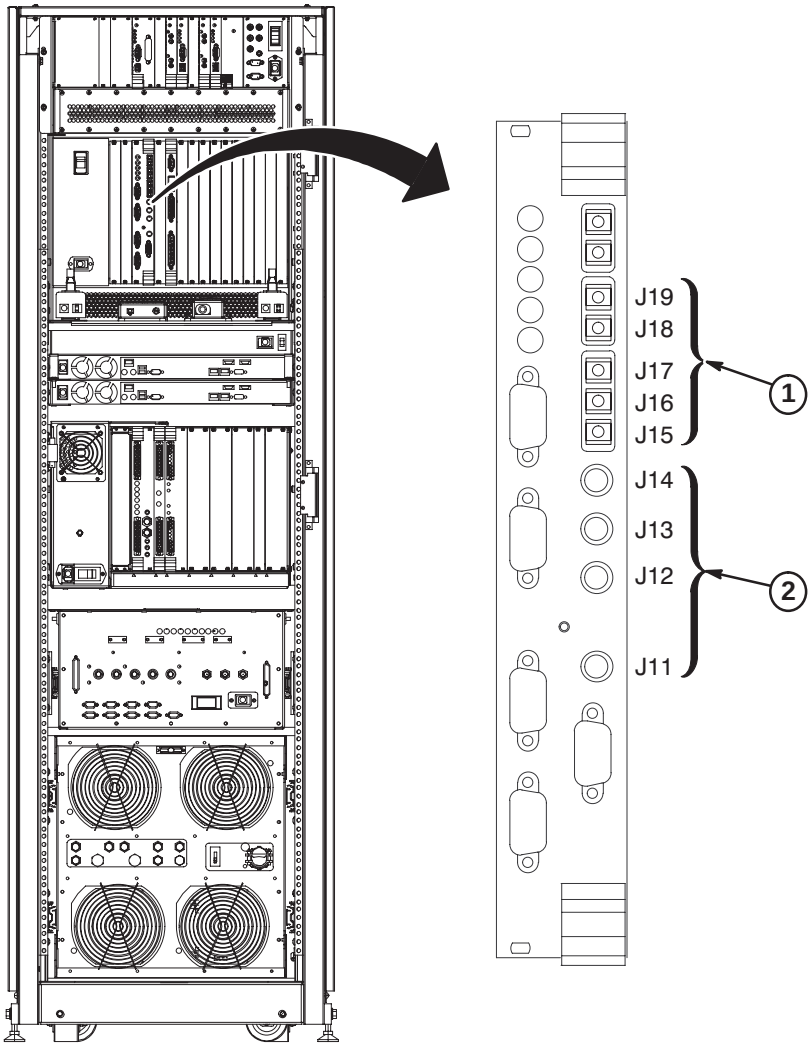


4–7–5–D Route/Connect Fiber Optic Runs1179 and 1045 to STIF Board

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
2. Route/Connect Run 1179 fiber optic cables to J11, J12, J13, and J14 on the STIF Board.



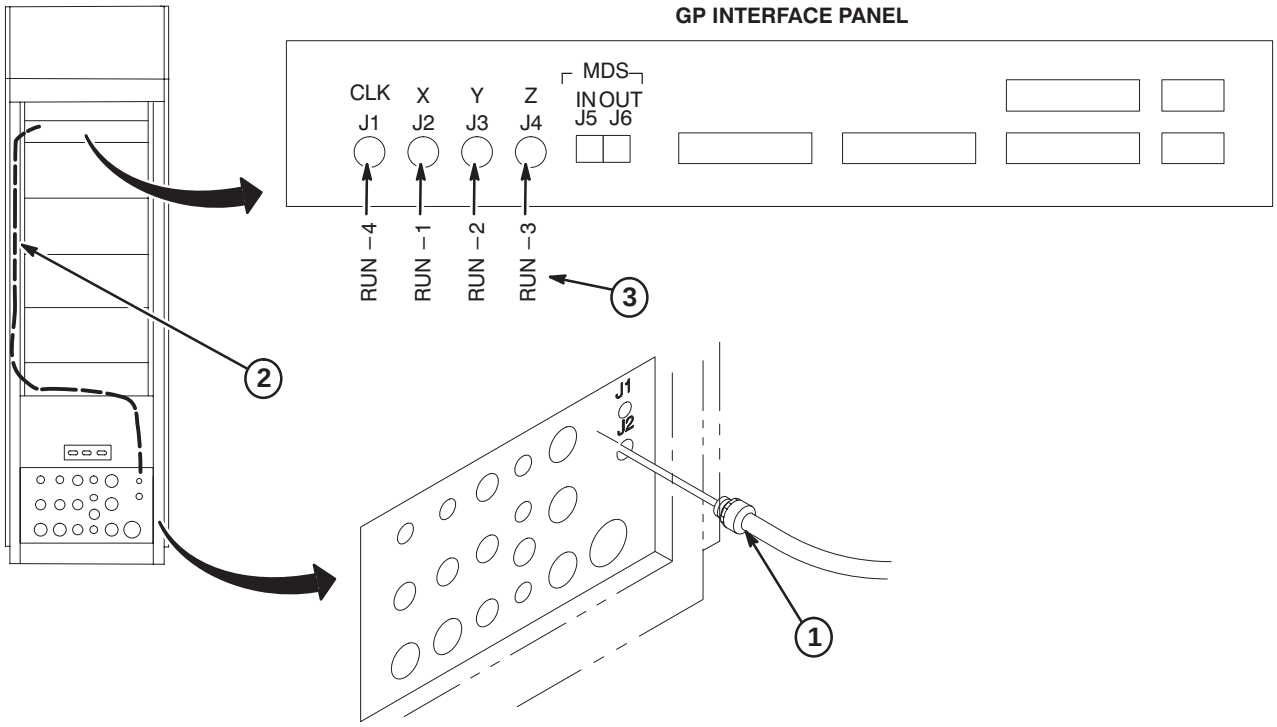
Note:
Use tie-wraps to secure fiber optic cables to MGD Chassis cables.

4–7–5–E Route/Connect Fiber Optic Run 1179 to GP3 Module in HFD

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.

1. Attach Run 1179 to I/F J2.
2. Route fiber optic cables to GP3 Module. Leave enough slack for extending GP module forward.
3. Connect fiber optics to GP3 module according to table.

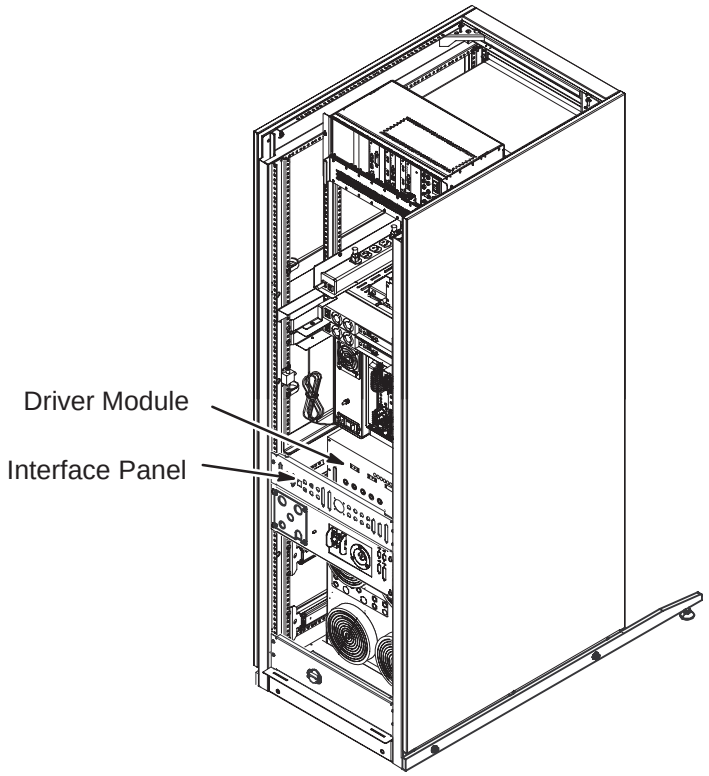


4–7–6 INSTALLATION OF REMAINING EQUIPMENT ROOM CABLES

Connect remaining Runs within equipment room. Dress and secure all cables after connections are made. Reference the applicable interconnect map in Section 4 and Cable Table in Appendix for to/from designator information and Run number reference.

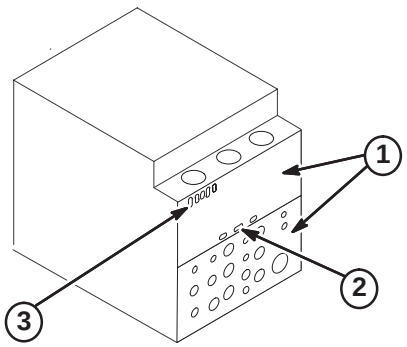
4–7–6–A RF/System Cabinet Connections

1. Connect routed cables to Interface Panels at bottom and middle of cabinet and to the Driver Module as marked. Refer to cable map.
2. Dress and secure cables. Adjust as necessary.



4–7–6–B HFD Cabinet PDU Module Connections

1. Attach rear PDU module covers removed earlier**.**
2. Connect Run 1179 to J2.
3. Connect Runs 974, 1180, and 1181 to J5, J7, and J8.



4–8 MAGNET ROOM CABLE INSTALLATION

4–8–1 OVERVIEW

The magnet enclosure and magnet room cables must be installed simultaneously. Also refer to Section 6 for additional cable routing and connection.

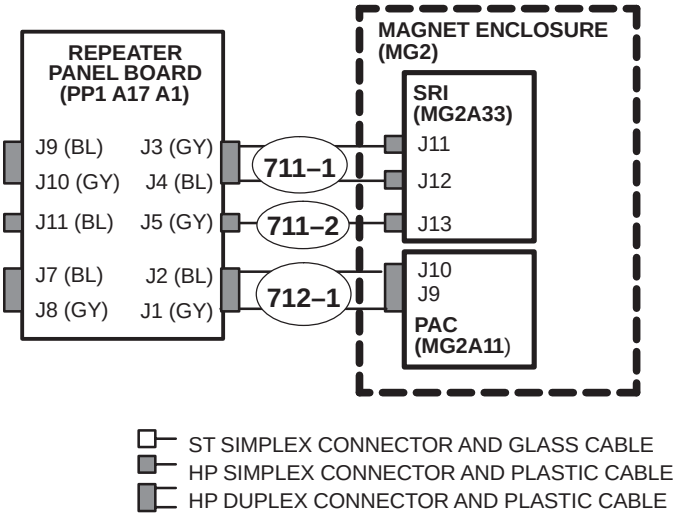
Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.

FERROUS MATERIAL HAZARD! PARTS REQUIRED FOR THIS UPGRADE INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AND MATERIALS AT LEAST 10 FEET AWAY FROM THE MAGNET.

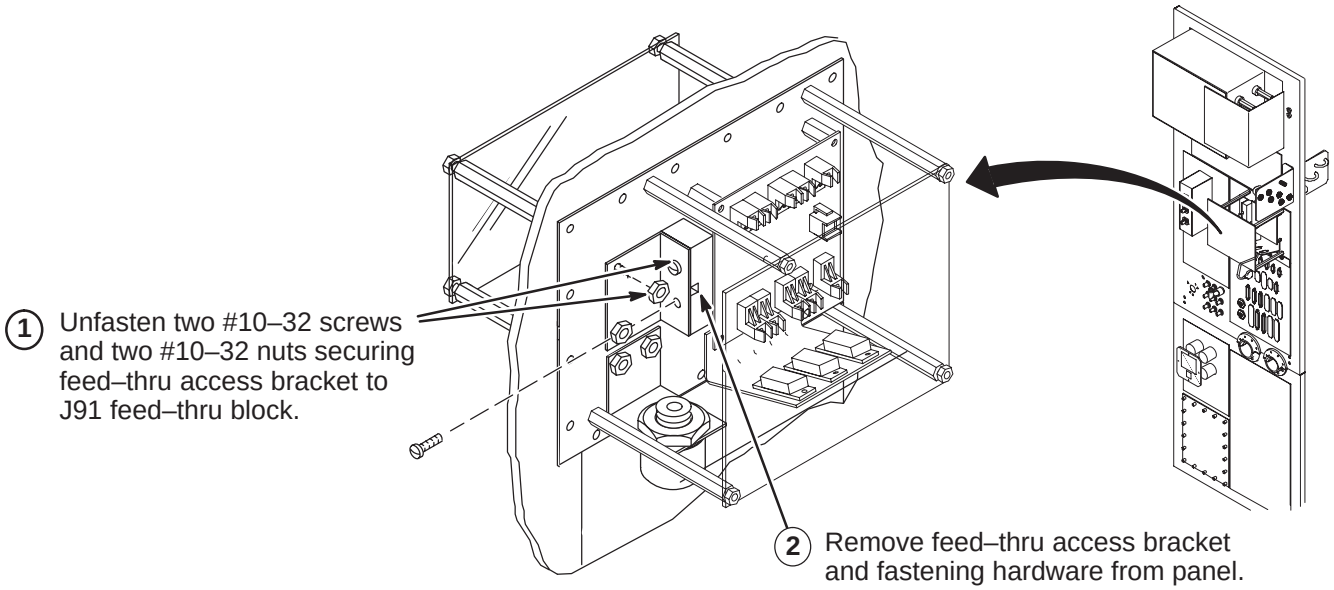
4–8–2 MAGNET ROOM FIBER OPTIC PAC/SRI INSTALLATION

For Horizon and LX configuratoins, existing Run 711/712 will be retained and connected to the new HDx Penetration Panel.

Shown below are the connections for the fiber optics cables routed in the magnet room from the Penetration Panel, thru the cable trough to the Rear Pedestal, and under the magnet to the PAC/SRI platform.



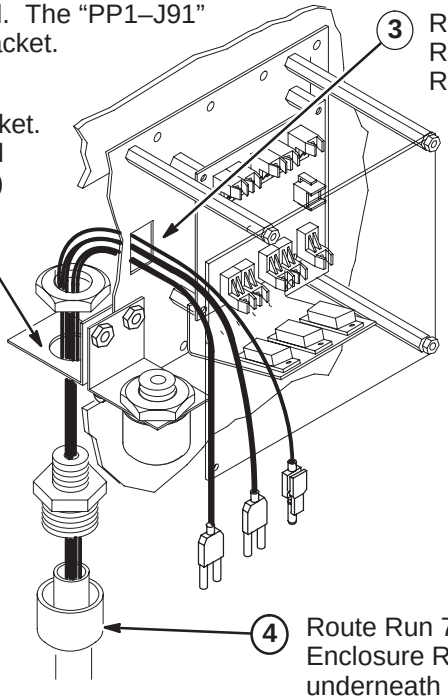
4–8–2–A Repeater Panel Feed–thru Bracket Removal



4–8–2–B Routing Run 711/712 in Magnet Room

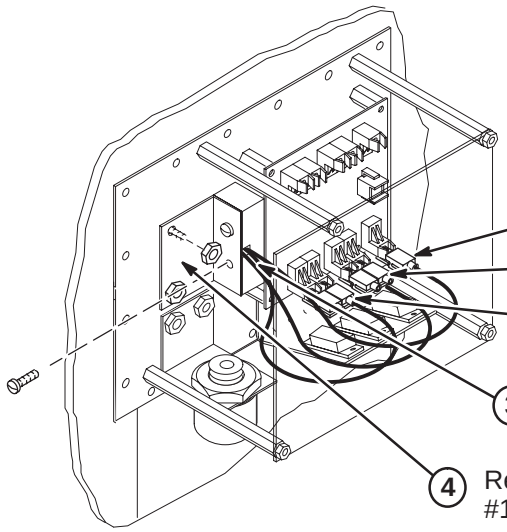
Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cut and ends terminated with the Fiber Optic Connector Repair Kit. Refer to Appendix for instructions. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

1. Move Run 711/712 (46–328079G9) to the Magnet Room side of the Penetration Panel. The “PP1–J91” end will be attached to interface bracket.
2. Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
3. Route ends of individual fiber–optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
4. Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.



4–8–2–C Connecting Fiber Optic Run 711/712 to Repeater Panel

Note: Take care to maintain the minimum 2 inch bend radius.



1. Adjust route of individual fiber optic runs to maintain large radius.
2. Connect Runs 712–1, 711–1, and 711–2 as shown.

Run 711–2 (J5)

Run 711–1 (J3, J4)

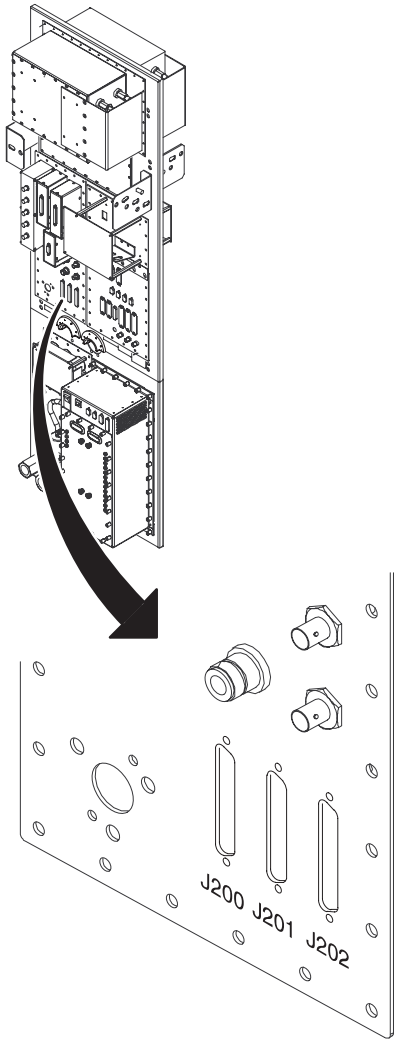
Run 712–1 (J1, J2)

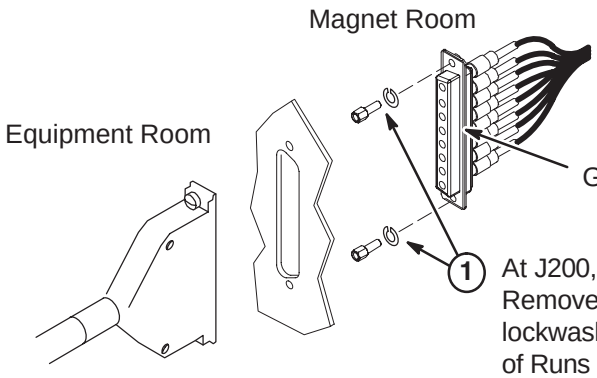
3. Gather fiber optic runs and route through wave guide notch in feed–thru block.
4. Replace feed–thru access bracket and fasten with two #10–32 screws and two #10–32 nuts removed in step A1.

4–9 INSTALLATION OF CABLES THAT RUN BETWEEN EQUIPMENT ROOM AND MAGNET ROOM

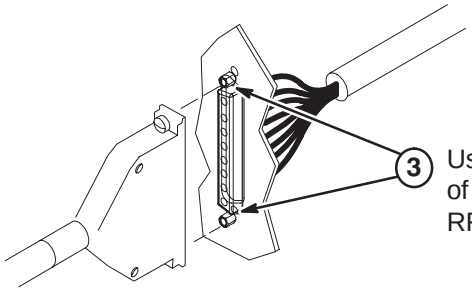
4–9–1 CONNECTION OF CABLES AT PENETRATION PANEL J200, J201, and J202

Note: Two people are required, one on each side of the Penetration panel, to install the cables in Step 1.

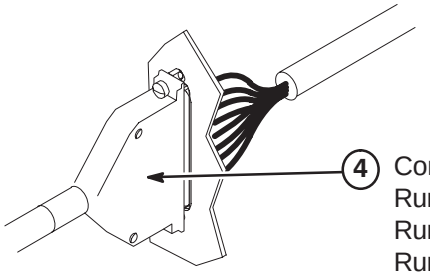




1. At J200, J201 and J202: Remove jackscrews and lockwashers from housing plugs of Runs 1200, 1201, and 1202.
2. Discard the lockwashers.



3. Using jackscrews, mount plugs of Runs 1200, 1201, and 1202 to RF Upgrade panel as marked.



4. Connect:
   Run 1185 to J200
   Run 1186 to J201
   Run 1187 to J202
5. Refer to cable map for connection of remaining cables to RF Panel.

4–9–2 GRADIENT CABLE INSTALLATION

NOTE: Sites that already have a BRM–D Body Coil installed with leads crimped to gradient cables routed from Penetration Panel WILL NOT replace the existing cables with new gradient cables in Magnet Room. Sections **4–9–2–E** thru **6–4–2–H** will not be completed for these sites.

Three cables have been delivered to site. All have three lead wires with terminals at each end. The leads on this cable end are connected to the Equipment room and Magnet room sides of the Gradient Filter on the Penetration Panel. This procedure cuts the cables to length and connects them to the HFD cabinet and the cable leads from the Gradent Coil.

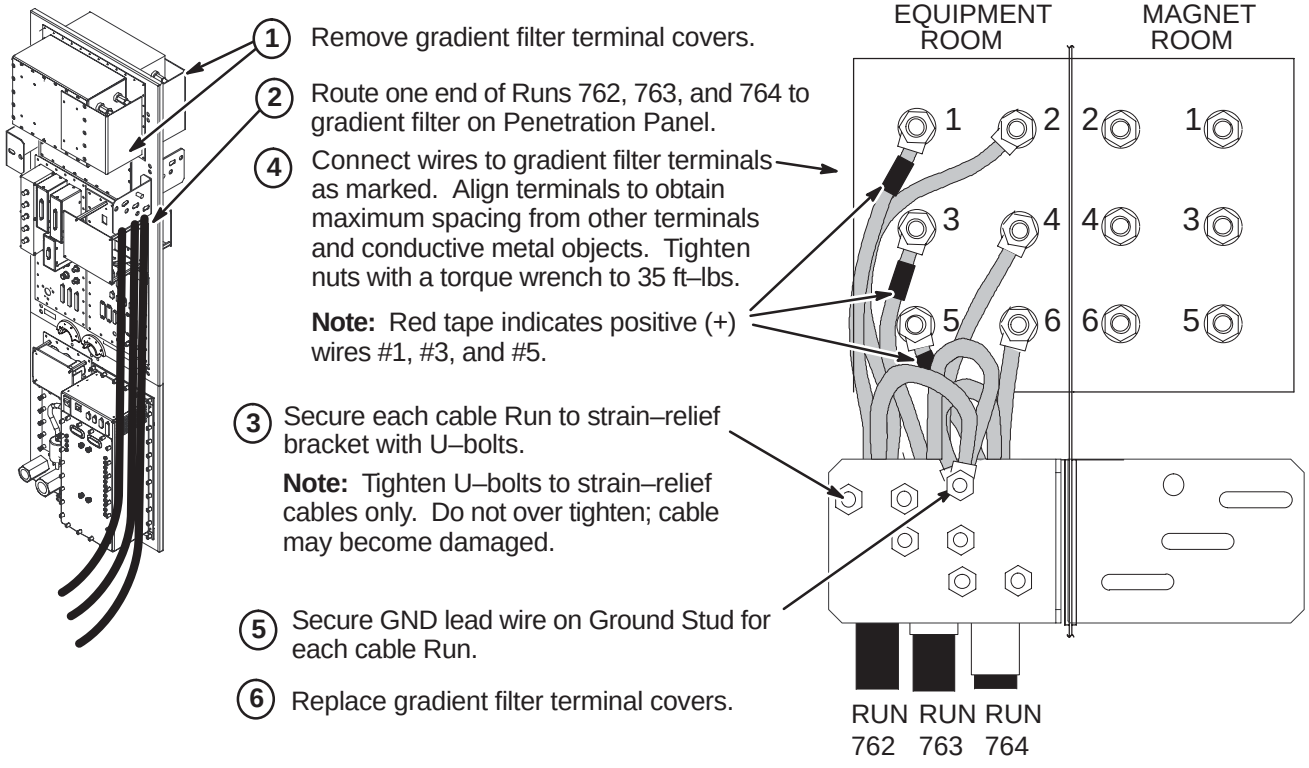
Verify that the length of supplied Run 762, 763, and 764 cable is sufficient to route from HFD Cabinet to Penetration Panel and, if required, Penetration Panel to Body Coil Gradient Cables at rear of magnet. **Each cable has identical terminations at each end for connection to both sides of the Penetration Panel Gradient Filter**.

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

4–9–2–A Connect Gradient Cables to Equipment Room Side of Penetration Panel

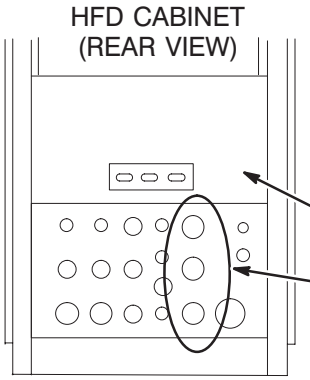


1. Remove gradient filter terminal covers.
2. Route one end of Runs 762, 763, and 764 to gradient filter on Penetration Panel.
4. Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft–lbs.

   Note: Red tape indicates positive (+) wires #1, #3, and #5.
3. Secure each cable Run to strain–relief bracket with U–bolts.

   Note: Tighten U–bolts to strain–relief cables only. Do not over tighten; cable may become damaged.
5. Secure GND lead wire on Ground Stud for each cable Run.
6. Replace gradient filter terminal covers.

4–9–2–B Route Runs 762, 763, and 764 to HFD Cabinet

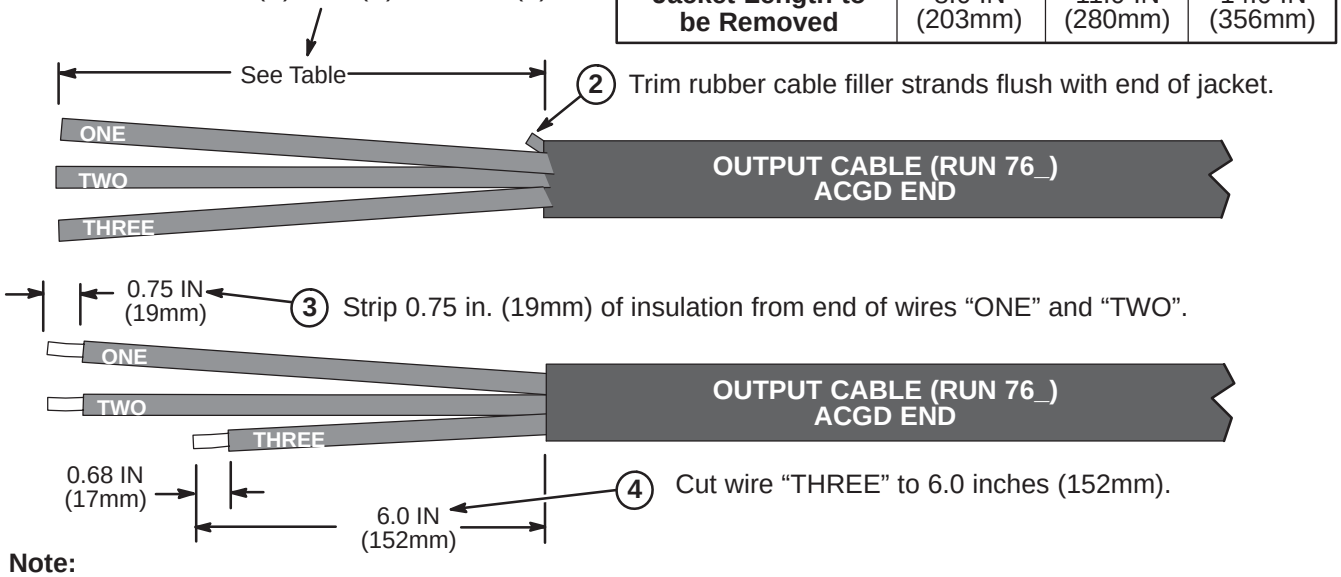


1. Unroll and route each cable from equipment room side of Penetration Panel to HFD Cabinet. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at HFD Cabinet.
2. Mark location for cutting.
3. Cut cables to length.
4. Remove Rear Cover Plate by removing seven screws that hold it in place.
5. Loosen screws on clamps for gradient output cables. After termination of cables in Procedure **4–9–2–C**, insert ends of cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

4–9–2–C Terminate Runs 762, 763, and 764 for HFD Cabinet

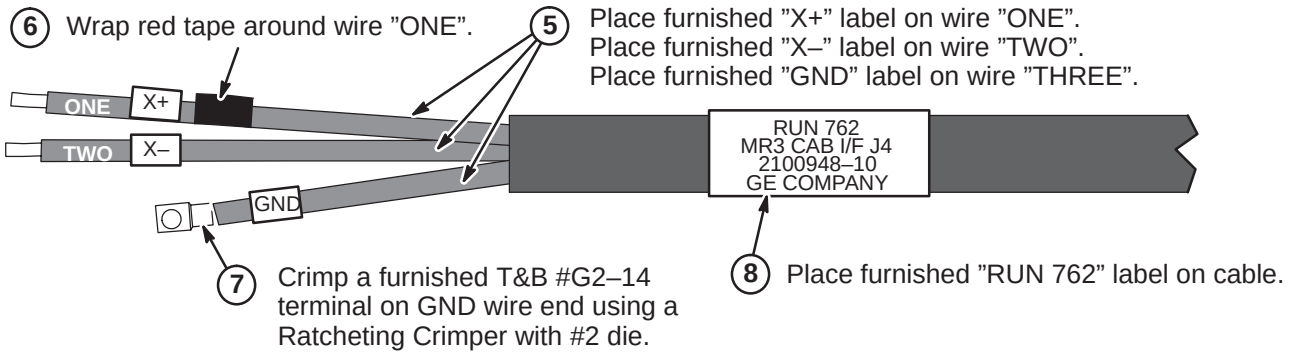
1. Refer to above table to determine length of jacket and foil to be removed from end of cable Runs 762 (X), 763 (Y), and 764 (Z).

	Run 762	Run 763	Run 764
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)

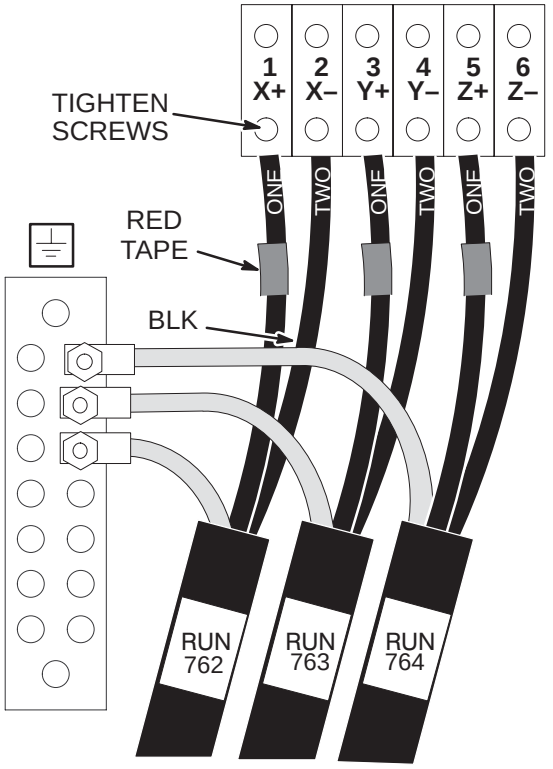


Note:
Steps below show completion of Run 762. Repeat same process for Runs 763 (Y+, Y–) and 764 (Z+, Z–).

Red tape, Y+ (RUN 763) **and Z+** (RUN 764) **labels are attached to wire ”ONE”.**



4–9–2–D Connect Runs 762, 763, and 764 at HFD Cabinet



GRADIENT CABLE CONNECTIONS

1. Connect gradient output cables with red–taped (+) #1 wires to + output terminals and black (–) #2 wires to – output terminals.

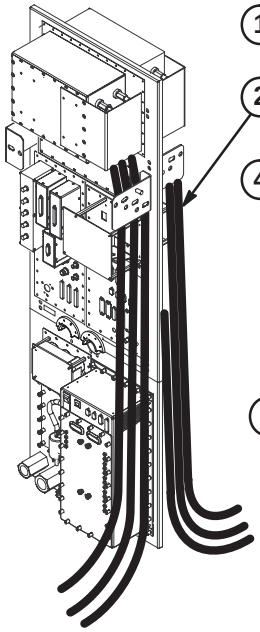
   NOTE: OVER TIGHTENING OF SCREWS IN STEP 2 CAN DAMAGE TERMINAL BLOCK.

2. Tighten screws to 52–65 inch–lbs (5,88–7,34 N–M). This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.

3. Connect the GND leads of each run to the Ground Bus Bar studs.

4. Tighten clamps on Interface Panel to secure gradient output cables.

4–9–2–E Connect Gradient Cables to Magnet Room Side of Penetration Panel



1. Move remainder of gradient cables to Magnet Room.

2. Route Run 762, 763, and 764 to gradient filter on Penetration Panel.

4. Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft–lbs.

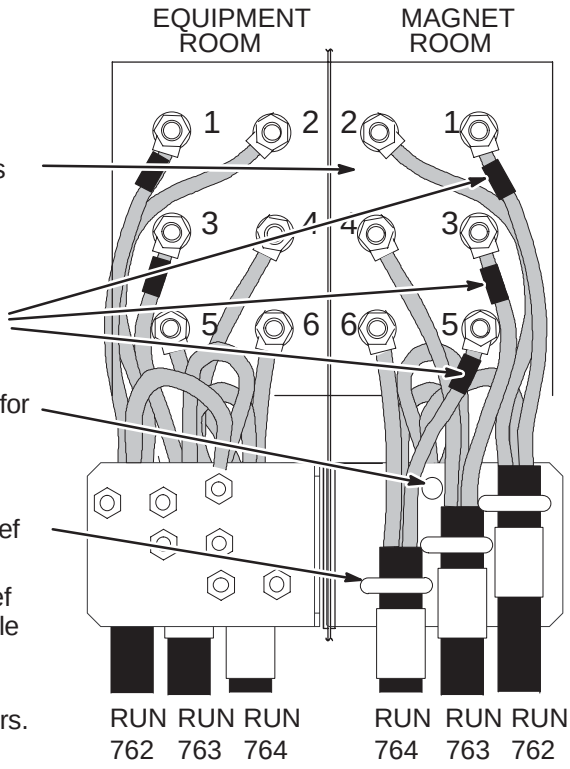
   Note: Red tape indicates positive (+) wires #1, #3, and #5.

5. Secure GND lead wire on Ground Stud for each cable Run.

3. Secure each cable Run to strain–relief bracket with U–bolts.

   Note: Tighten U–bolts to strain–relief cables only. Do not over tighten; cable may become damaged.

6. Replace gradient filter terminal covers.



4–9–3 TIMES MICROWAVE RF CABLES INSTALLATION

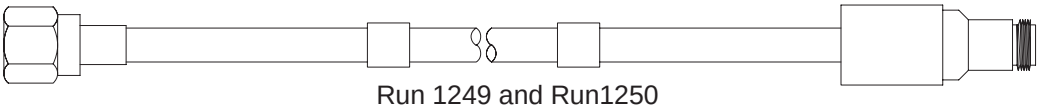
4–9–3–A Cable Information

Five cables, three extension cables, and other associated parts to be installed are provided for this procedure. Also provided are two Times Microwave stripping tools for LMR 1200 (2384858) and the LMR 600 (2352193), and the Times Microwave cutting tool (5111565).

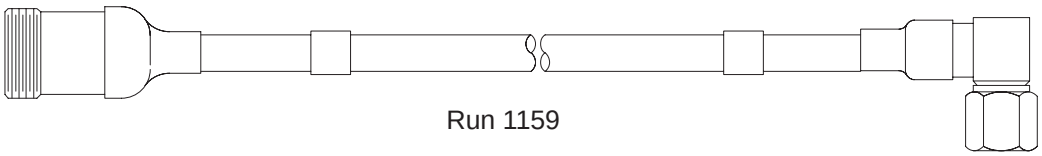
CAUTION

Do not subject LMR–600FR cable (Head) to a bend radius less than 3 inches (75mm). Do not subject LMR–1200FR cable (Body) to a bend radius less than 6.5 inches (165mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

Two extension cables, 5116213–2, Run 1249 (Head) and 5121269, Run 1159 (Body) are connected to the RF Amplifier module in the RF/System Cabinet. The other extension cable, Run 1250, 5121245, is connected to the Body Hybrid Splitter in the Rear Pedestal.

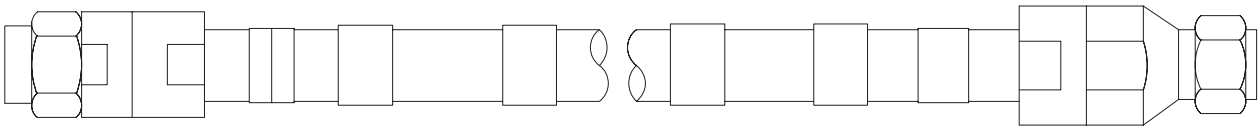


Run 1249 and Run1250



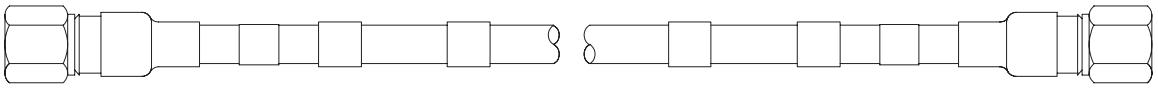
Run 1159

Cable 2316584–2 (Body, Runs 235 & 257), will be cut to length to make two cables and terminated for routing in the equipment and scan rooms.



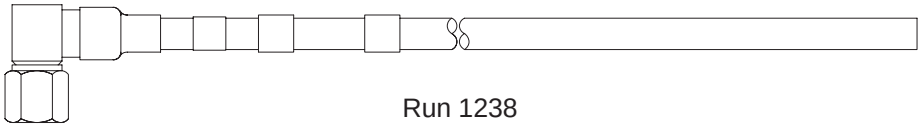
Run 235 and Run 257

Cable 5115915 (Head, Runs 1198 & 1199), will be cut to length to make two cables and terminated for routing in the scan room.



Run 1198 and Run 1199

Cable 5115916 (Head, Run 1238), will be cut to length and terminated for routing in the equipment room.



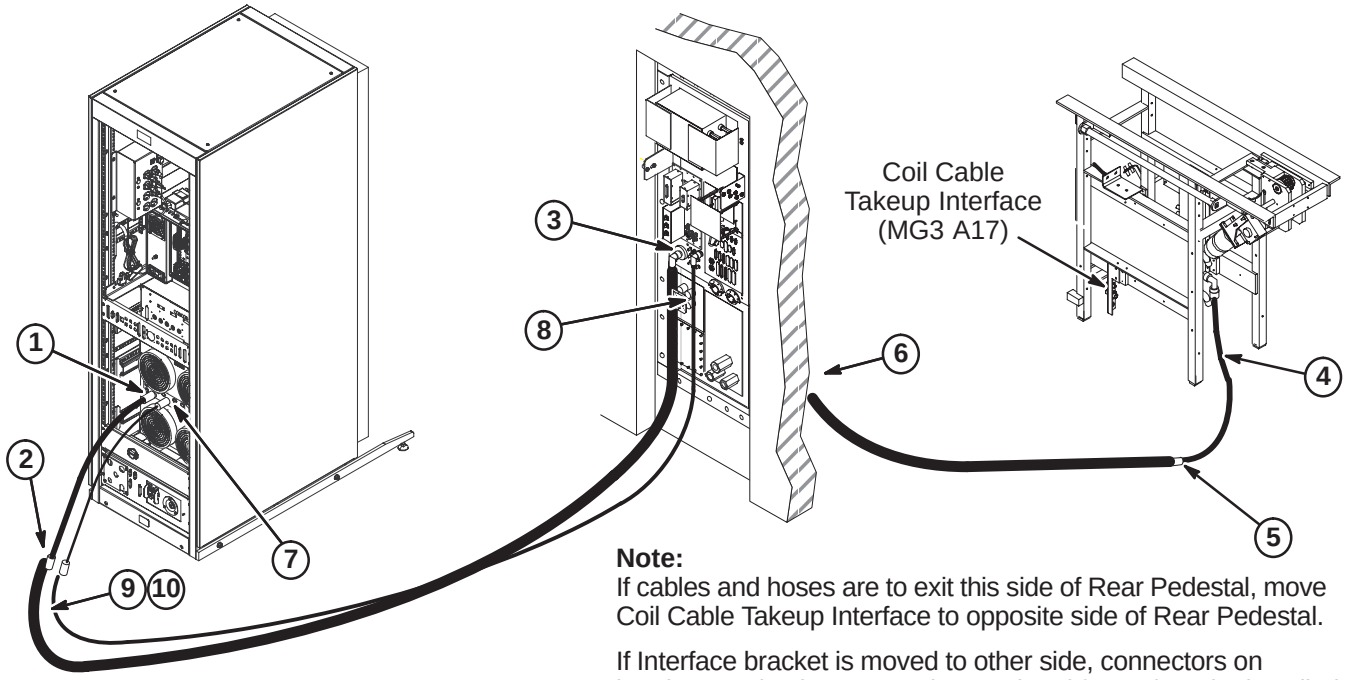
Run 1238

The remaining connectors, fasteners, and labels will be installed as directed in the following steps.

4–9–3–B Install Runs 235, 257, 1238, 1249, 1250, and 1259

RF cables will be routed and connected as indicated in the following steps.

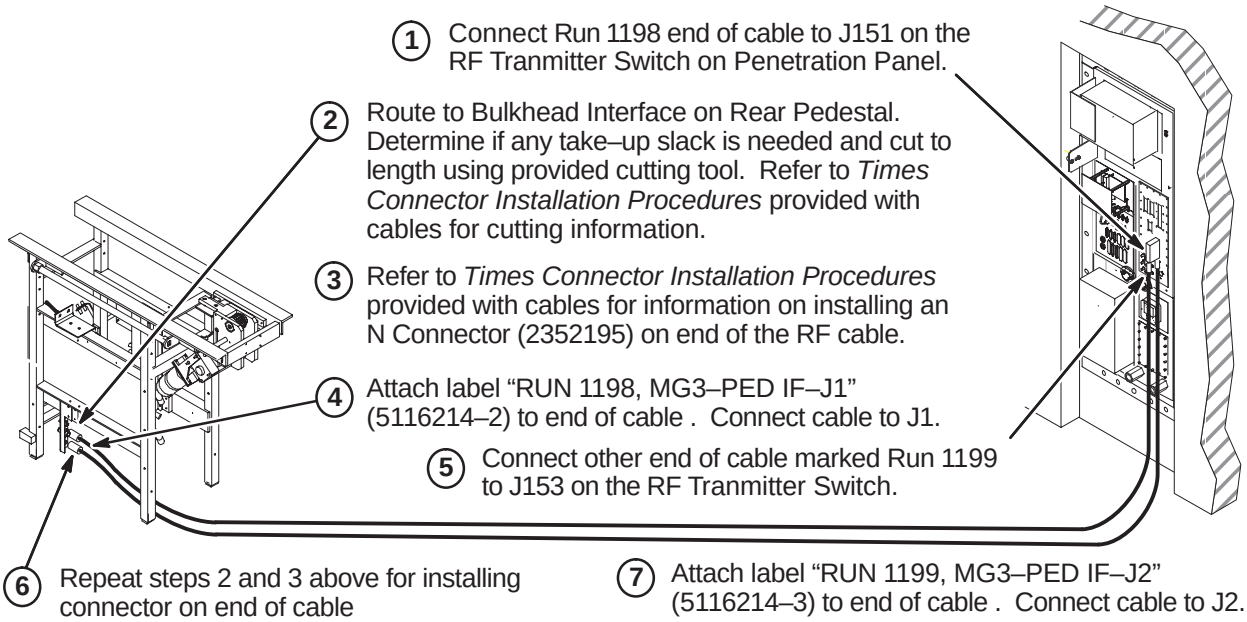
1. Connect extension cable Run 1259 to J4 Body Output at rear of RF Amp in RFS cabinet.
2. Connect end of Times RF Body cable marked “(RUN 235) FJ5” to Run 1259, Body extension cable. Attach label “RUN 235, Connect to Run 1259” (5121244) over existing label.
3. Route cable to J44 at Penetration Panel. Determine if any take–up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **4–9–3–D** for termination procedure of cut end.
4. Connect Run 1250 to J1 on Hybrid Splitter.
5. Move remainder of RF Body cable to Scan Room and connect other end of cable marked “(RUN 257) MG3A3–J1” to Run 1250 at Rear Pedestal as shown below. Attach label “RUN 257, Connect to Run 1250” (5121244–2) over existing label.
6. Route remainder of cable to J44 at Penetration Panel. Determine if any take–up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **4–9–3–D** for termination instructions of cut end.
7. Connect extension cable Run 1249 to J3 Head Output at rear of RF Amp IN RFS cabinet.
8. Connect end of Times RF Head cable marked “(RUN 1238) PP1–J4” to J4 at Penetration Panel. Route to RF Head extension cable at rear of SRF cabinet. Determine if any take–up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information.
9. Attach label “RUN 1238, CONNECT TO RUN 1249” (5116214) to end of cable after termination.
10. Connect cable to Run 1249.



Note:
If cables and hoses are to exit this side of Rear Pedestal, move Coil Cable Takeup Interface to opposite side of Rear Pedestal.

If Interface bracket is moved to other side, connectors on bracket need to be removed, turned end for end, and reinstalled.

4–9–3–C Install Runs 1198 and 1199



4–9–3–D Terminate and Connect RF Body Cables at Penetration Panel

Perform the following steps for terminating and connecting the RF Body cables at the Penetration Panel J44.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an EIA Right Angle Connector (2384859) on each end of the RF cables. Use provided cutting tool (5111565) and the LMR900 Stripping Tool (2384858).

Note

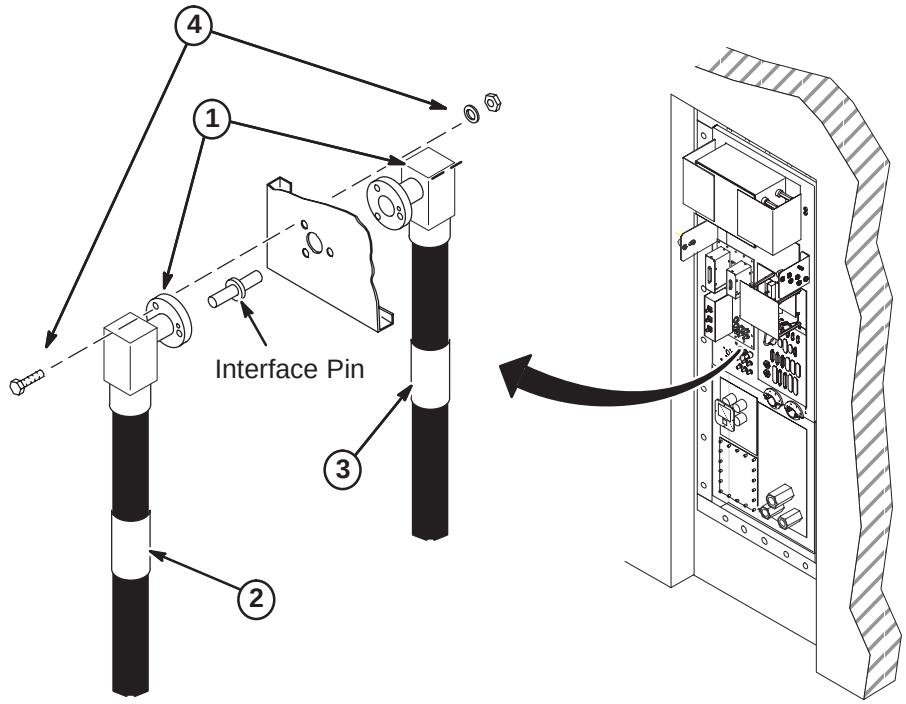
Make sure the fastener holes in the connector flanges align with the holes in the Penetration Panel so the cables will route toward floor when installed.

2. Attach label “RUN 235, PP1–J44” (2357230) to cable as shown on Equipment room side of Penetration Panel.
3. Attach label “RUN 257, PP1–J44” (2357230–2) to cable as shown on Scan room side of Penetration Panel.

CAUTION

Do NOT install O–Ring or Grease that may be provided with Cable Termination Kit. Installation of these parts will prevent the ground from making contact resulting in RF leaks.

4. Insert Interface Pin into one of the connectors. At J44, align connectors on each side of Penetration Panel and push together. Fasten together using three of each of the following:
   - 1/4–20 x 1.25 Hex Head Bolt (brass)
   - 1/4 Screw Size Washer (phos bronze)
   - 1/4–20, 7/16” Hex x 3/16” thk Nut (brass)



SECTION 5 – PENETRATION PANEL

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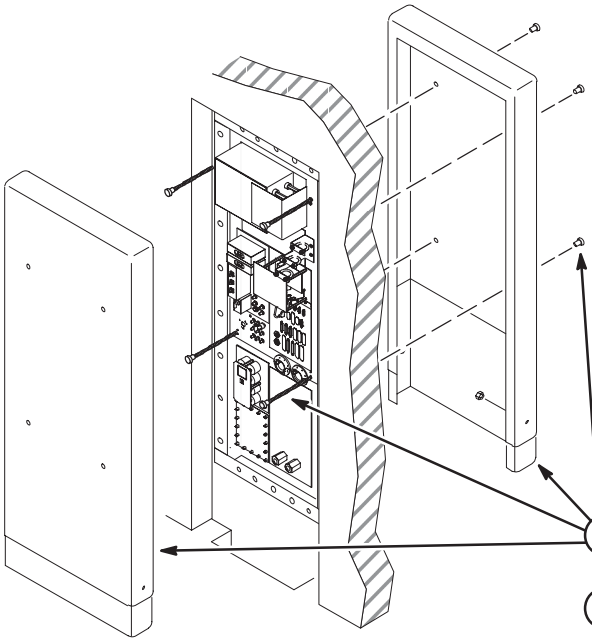
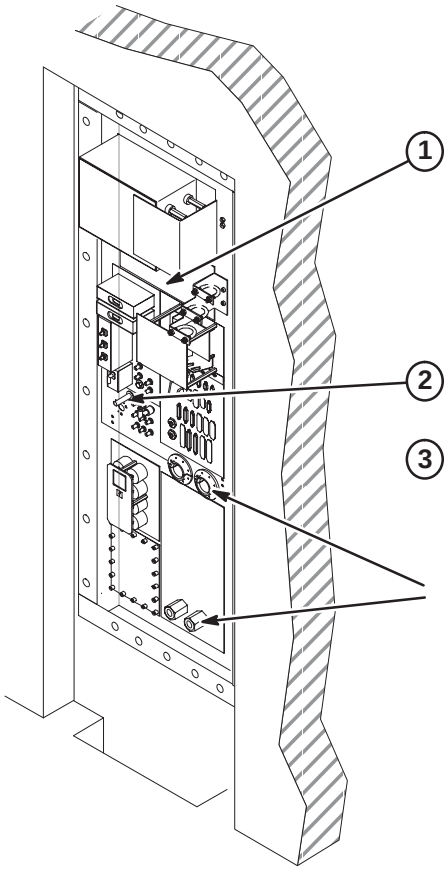
5–1 REMOVE PENETRATION PANEL COVERS



1. If installed, unfasten and save Penetration Panel covers, stand–offs, and mounting hardware.
2. Move above items to a safe storage area. They will be reinstalled at end of upgrade.

5–2 REMOVE PARTS FOR TRANSFER TO NEW PENETRATION PANEL



1. If installed on Penetration Panel, transfer the Shim Filter, only if it's a 46–271282G1, (if not this model, order one) to new Penetration Panel.
2. Disconnect and remove Patient Alert tubing. Transfer Patient Alert Wave Guide to new Penetration Panel re–using mounting hardware.
3. If Patient Alert Extender Box is mounted on Penetration Panel, remove and re–install on new Penetration Panel.

CAUTION! – OLDER SYSTEMS USED BERYLIUM PASTE ON WAVE GUIDES AND GAS LINES. WASH HANDS IF YOU COME IN CONTACT WITH THIS MATERIAL.

5–3 DISCONNECT COOLANT HOSES

CAUTION

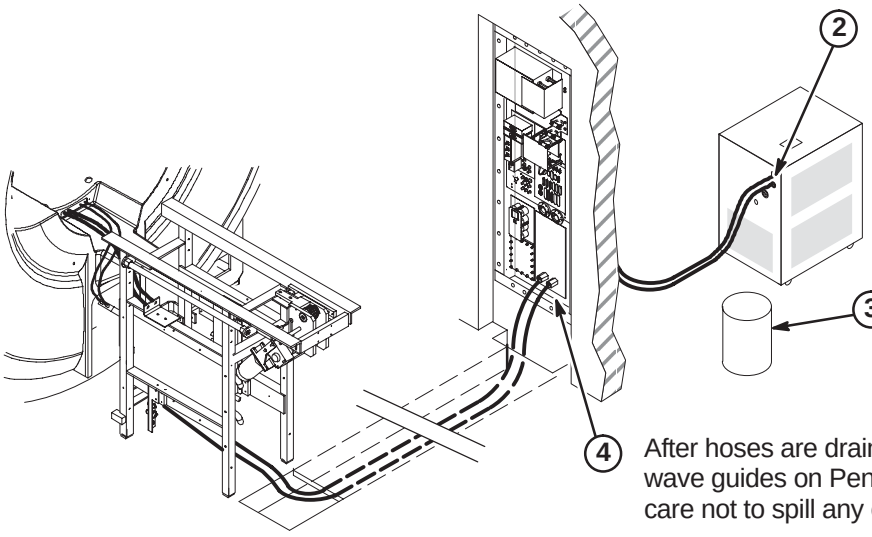
Coolant spillage possibility. The gradient coil cooling system is pressurized. Turn off the heat exchanger or water chiller before disconnecting the hoses. This takes pressure off the system and helps avoid coolant from being sprayed all over the immediate area.

WARNING!

POSSIBLE EYE INJURY! DRAINING OF COOLANT (ANTI–FREEZE) MAKES SPLASHING POSSIBLE. THIS COULD CAUSE EYE INJURY. BE SURE TO WEAR LATEX GLOVES AND SAFETY GLASSES.

Note: Four hands are better than two – While it is possible for one person to disconnect the coolant hoses, it is easier if two people do it. There would be less chance of spilling the coolant.

(1) Prior to disconnection of coolant hoses, calculate the amount of coolant that will be drained from system to determine if the 5 gallon bucket provided with upgrade will be sufficient for collection of coolant to be drained. Approximately two gallons of coolant will remain in gradient coil after disconnection. If more containers are needed for collection of drained coolant, they must be obtained locally.



(2) Remove Hose Clamps (2128542–2) and disconnect two coolant hoses.

Note: If existing system has a Heat Exchanger, it will be deinstalled and a new Water Chiller will replace it. Refer to Section **12**.

(3) Place ends of coolant hoses in provided bucket (2324086–2) for collection of coolant.

(4) After hoses are drained, pull hoses out of wave guides on Penetration Panel. Take care not to spill any excess coolant.

5–4 DISCONNECT PENETRATION PANEL CABLES AND UNFASTEN UPPER & LOWER PANELS

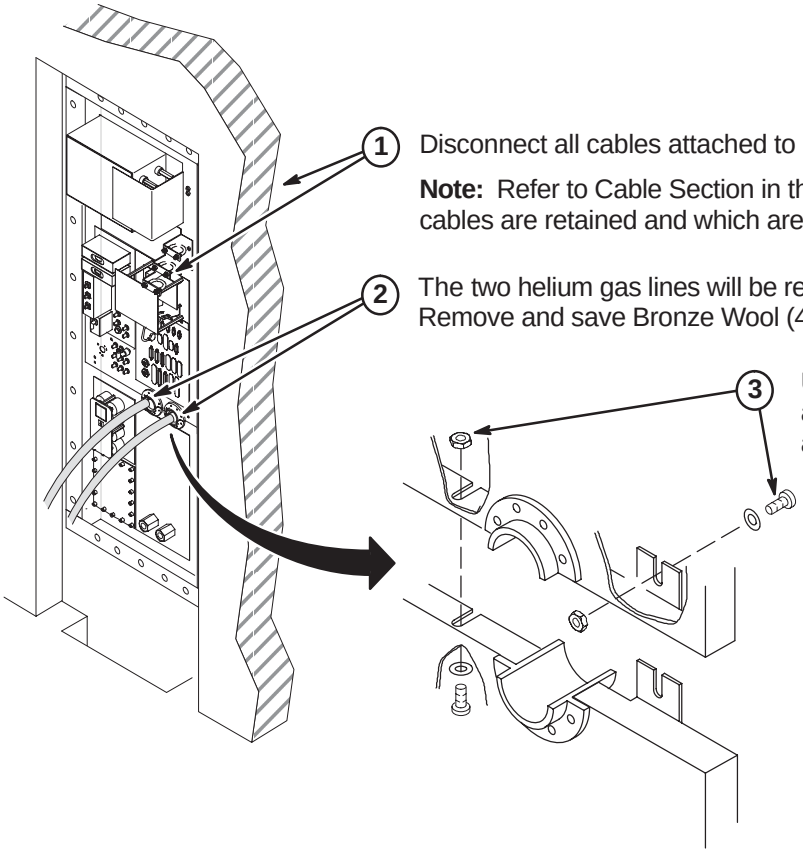
WARNING!

FERROUS MATERIAL HAZARD!

THE MATERIAL USED TO SECURE THE PENETRATION PANEL TO THE THE SCREEN ROOM MAY BE MAGNETIC. FERROUS MATERIAL WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

ALL MOUNTING MATERIALS OR RECTANGULAR FRAME THAT SECURES THE PENTRATION PANEL TO THE SCREEN ROOM ON THE MAGNET SIDE MUST BE PASSED FROM THE MAGNET ROOM TO THE EQUIPMENT ROOM THROUGH THE PENETRATION PANEL OPENING IMMEDIATELY AFTER REMOVAL OF PENETRATION PANEL.

REFER TO SECTION 5–5.



1. Disconnect all cables attached to both sides of Penetration Panel.

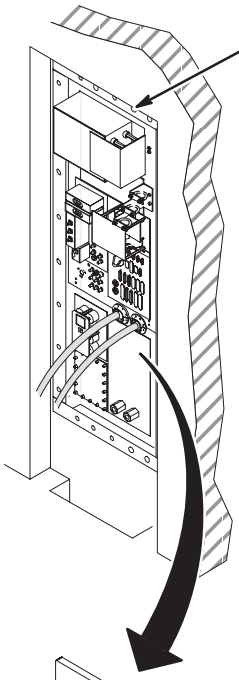
   Note: Refer to Cable Section in this manual to determine which cables are retained and which are returned to GE for recycling.

2. The two helium gas lines will be reused and MUST NOT be disconnected. Remove and save Bronze Wool (46–318068P1) from inside of J60 and J61.

3. Unfasten Nylock Lock Nuts, Flat Washers, and 10–32 x 3/8 Screws that attach the upper and lower panels around the gas lines.

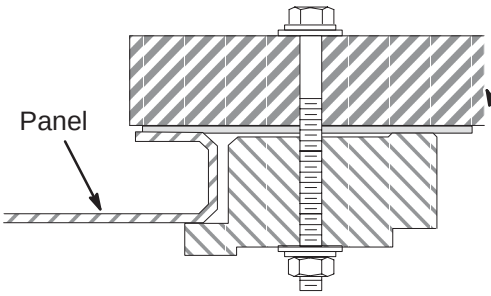
5–5 DE–INSTALL PENETRATION PANEL

At least two people are required to deinstall the Penetration Panel.

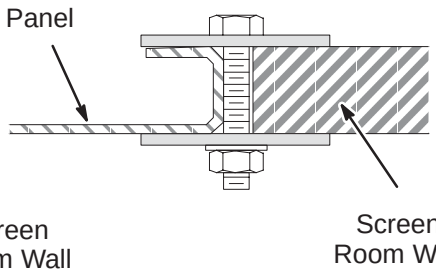


1. Deinstall Penetration Panel by removing the mounting hardware located on outside edge of panel. Two different typical methods are shown below.

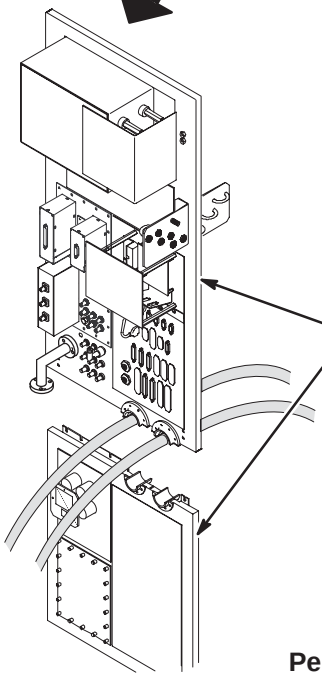
Hold Down Method



Brass Sealing Plate Method



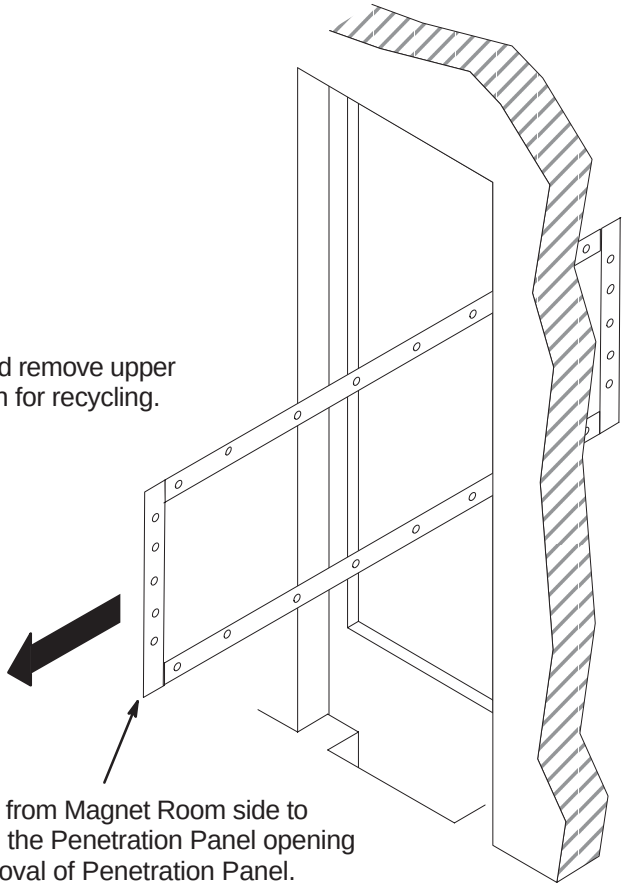
2. Save the grade 8 bolts and related mounting hardware for installing the new Penetration Panel.



3. Leave gas lines in place and remove upper and lower panels and return for recycling.

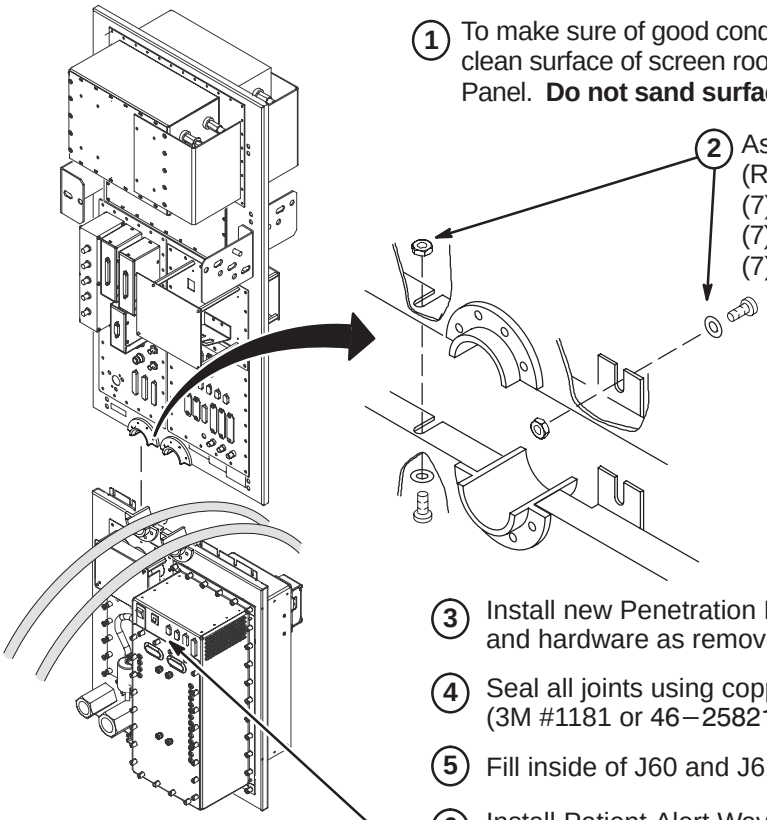
WARNING!

Penetration Panel mounting frame may be magnetic.



4. Move mounting frame from Magnet Room side to Equipment Room thru the Penetration Panel opening immediately after removal of Penetration Panel.

5–6 INSTALL EXCITE HD PENETRATION PANEL



1. To make sure of good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**
2. Assemble Penetration Panel around helium lines (Runs 621 and 622) using:
   (7) #10–32 Nylock Lock Nuts (46–208937P6)
   (7) 0.2 ID, 0.5 OD Flat Washers (1000904P425)
   (7) #10–32 x 3/8 Screws (46–208560P82)
3. Install new Penetration Panel using same method and hardware as removed penetration panel.
4. Seal all joints using copper foil tape with a conductive adhesive (3M #1181 or 46−258218P4, 5, or 6) for RF–tight integrity.
5. Fill inside of J60 and J61 guides with Bronze Wool (46–318068P1).
6. Install Patient Alert Wave Guide into “ALERT AIR LINE” hole or another available hole. If required, install Extender Box.
7. Install CAN Terminator (2303317) at J155.
8. If required, install previously removed Shim Filter and cables.

5–7 CONNECT CABLES TO PENETRATION PANEL

For information and instructions on cable and hose connections at the Penetration Panel, refer to applicable:

- Installation Flowchart in Section 1
- Coolant Hose installation in Section 12
- Cable Interconnect Map in Section 4
- Cable and System Installation in Section 7 .

Make sure to reinstall all customer added option connections.

SECTION 6 – MAGNET ENCLOSURE

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CAUTION

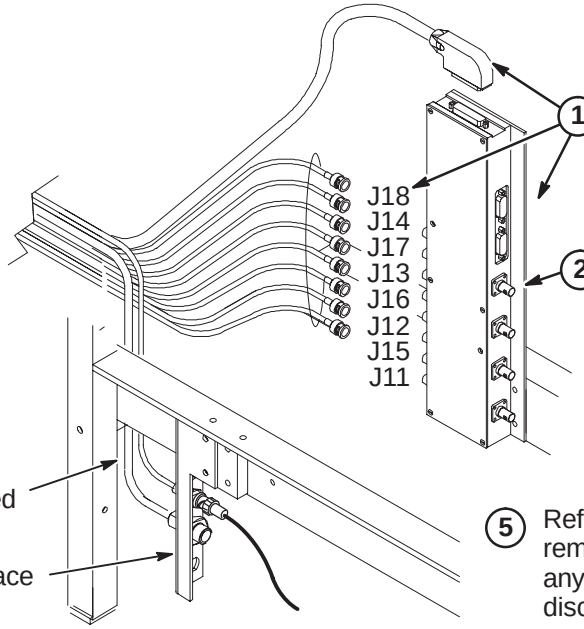
MAKE SURE POWER TO SYSTEM IS OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

6–1 INTRODUCTION

Modules on the Rear Pedestal, Bridge, LPCC and SRI will be replaced. Several cables will be removed and replaced with new cables, some cables will be relabeled and reused, and new cables will be routed and connected. Laser Warning and Hearing Loss Warning signs will also be installed.

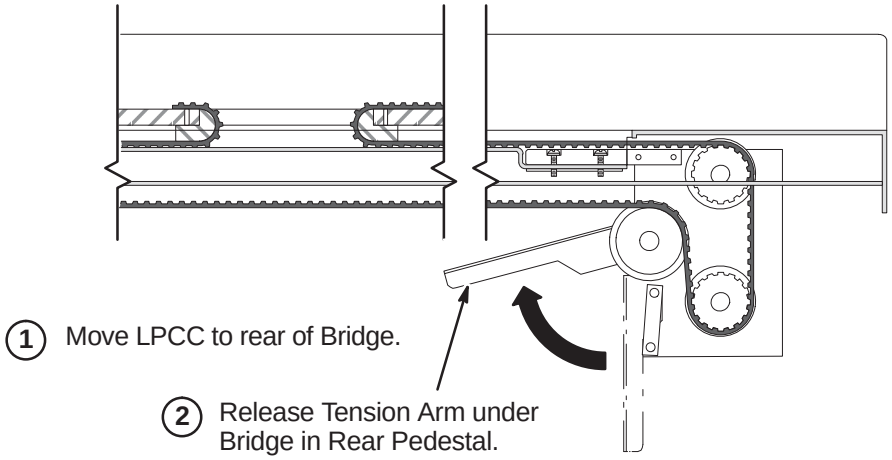
6–2 DE–INSTALL EXISTING COMPONENTS

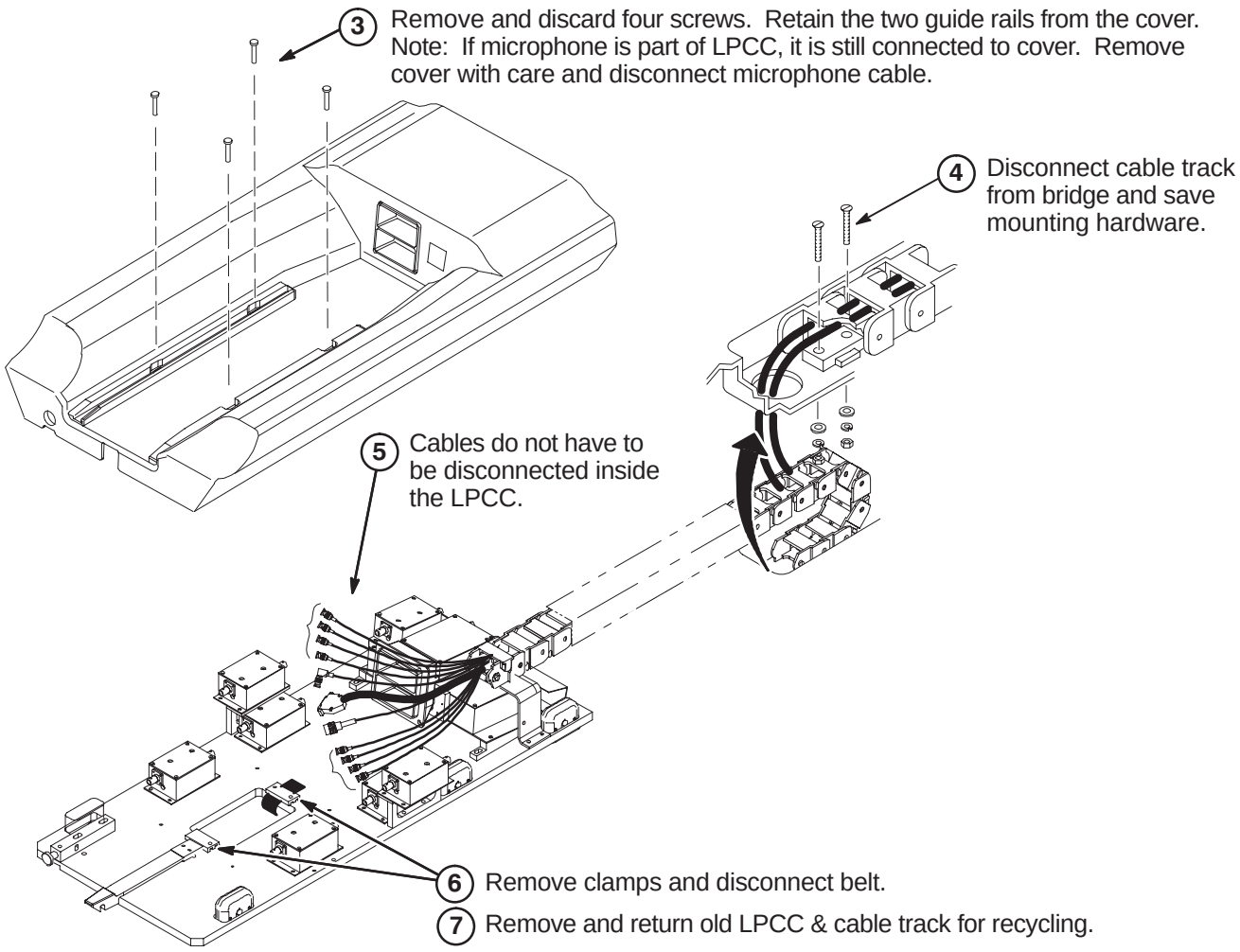
6–2–1 DE–INSTALL COMPONENTS IN REAR PEDESTAL



1. Disconnect all cables connected to Select Switch.
2. De–install Select Switch and return for recycling.
3. Disconnect all cables connected to Cable Takeup Interface.
4. De–install Cable Takeup Interface and return for recycling.
5. Refer to applicable cable removal map for information on any other cables that need to be disconnected and/or removed.

6–2–2 DE–INSTALL EXISTING LOW PROFILE CARRIAGE COVER & CABLE TRACK

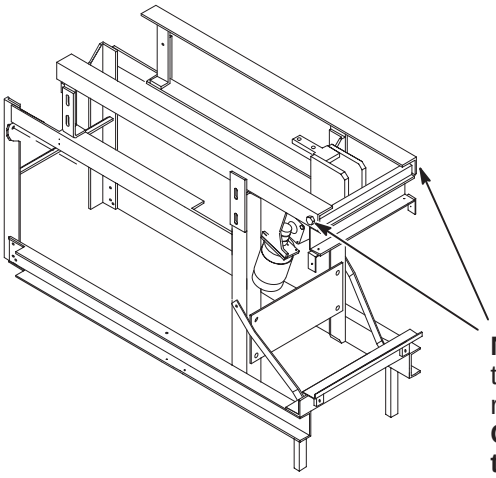


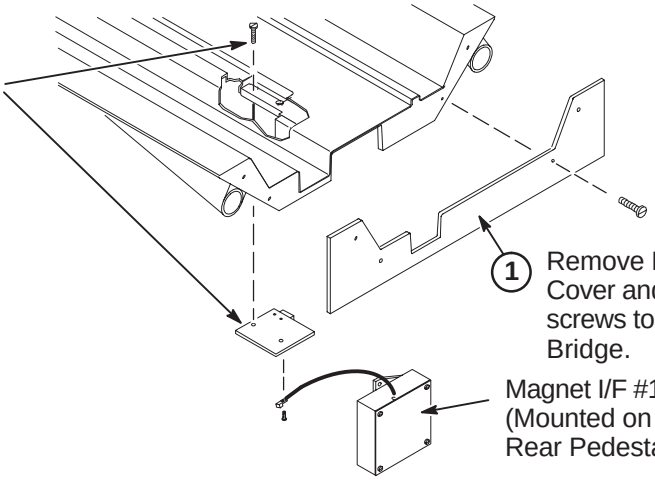


6–2–3 DE–INSTALL BRIDGE

6–2–3–1 Unfasten Bridge from Rear Pedestal

(2) Unfasten and save two 10–32 x 1–1/2 Round Head Slot Brass Screws (46–220184P47) and disconnect Bridge from existing Tongue and Rear Bridge Mount Asm.

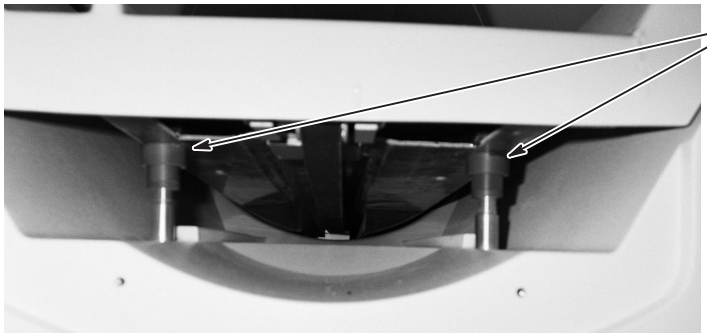




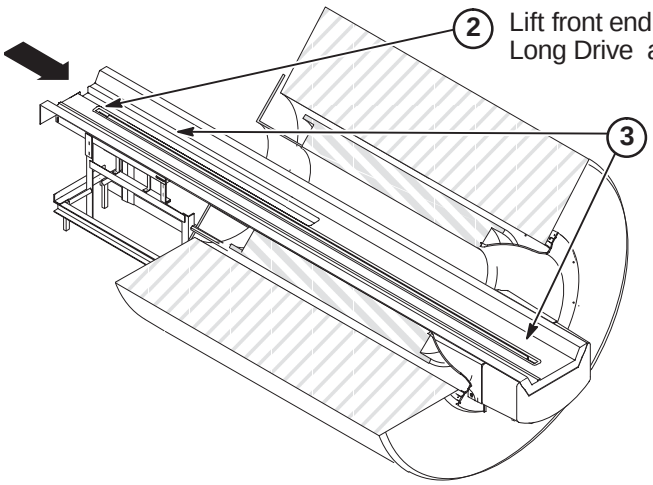
Note: If necesssary, loosen, but do not remove, the screw on each side of the Longitudinal Drive Motor Asm so the Tongue will tilt up for easier removal of Bridge.

CAUTION: Longitudinal Motor contains ferrous material. Holding this assemlby to close to magnet bore will forcibly attract it to magnet and possibly cause bodily injury or material damage.

6–2–3–2 Remove Bridge from Magnet



(1) Under the front of the Bridge, two Cap Support Cups (2318948) are epoxied to the bottom side of the Bridge. Remove these from the existing de–installed Bridge. If new Cups are not provided with upgrade, these will be reused on new Bridge.



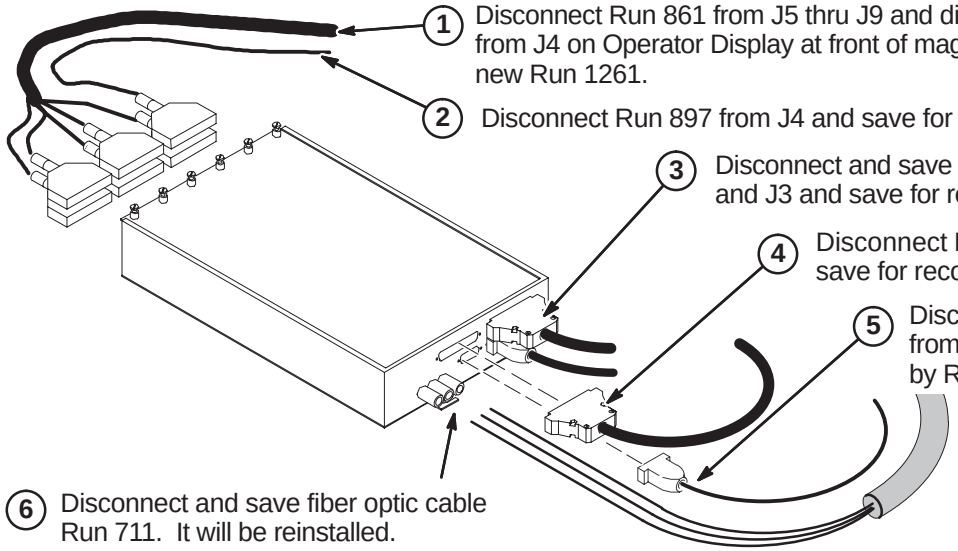
(2) Lift front end of Bridge to disengage tongue on Long Drive assembly at Rear Pedestal.

(3) While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore towards the front as it is being removed.

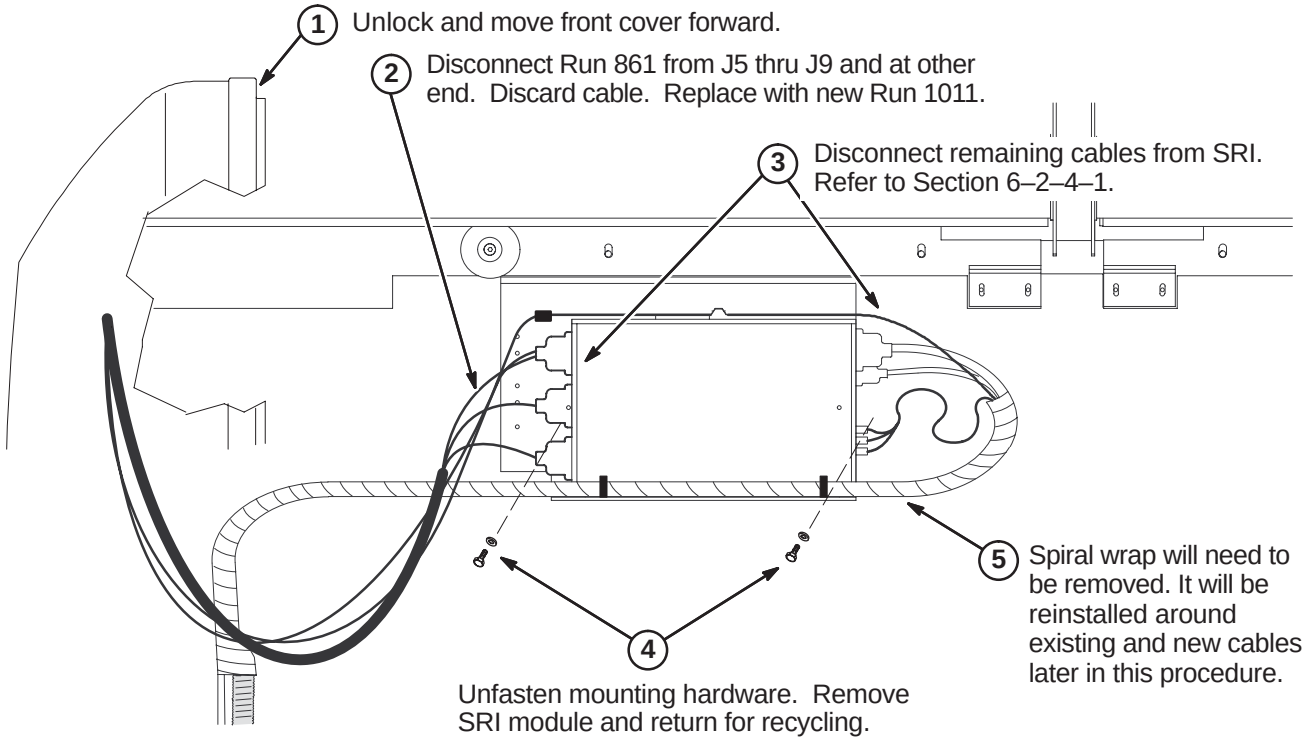
(4) Remove and save Drive Belt and Front Pulley for transfer to new Bridge.

6–2–4 DE–INSTALL SRI MODULE

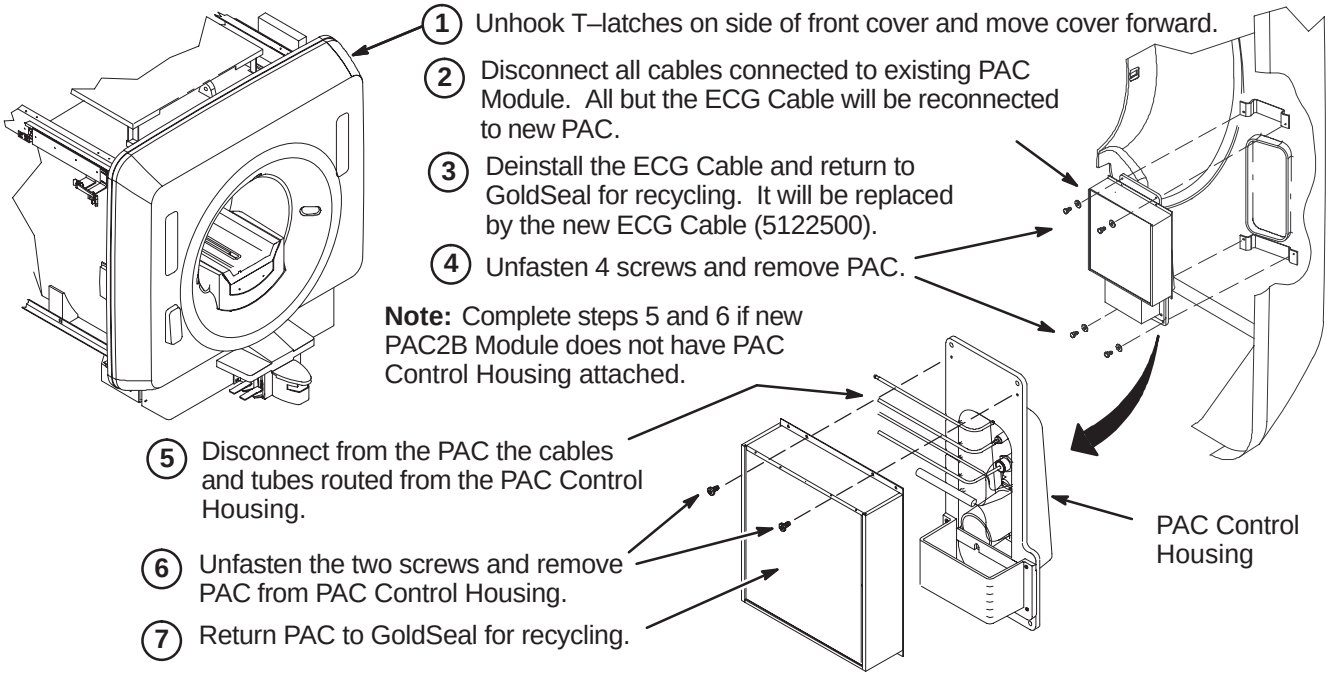
6–2–4–1 Disconnect Cables from Existing SRI Module



6–2–4–2 Remove Existing SRI Module



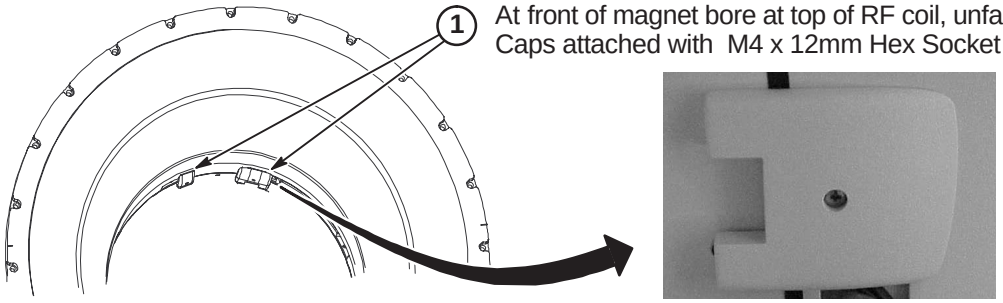
6–2–5 DE–INSTALL PAC MODULE

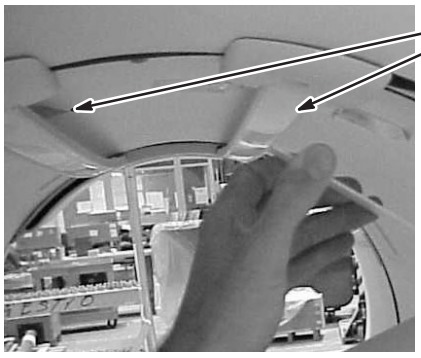


6–3 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL

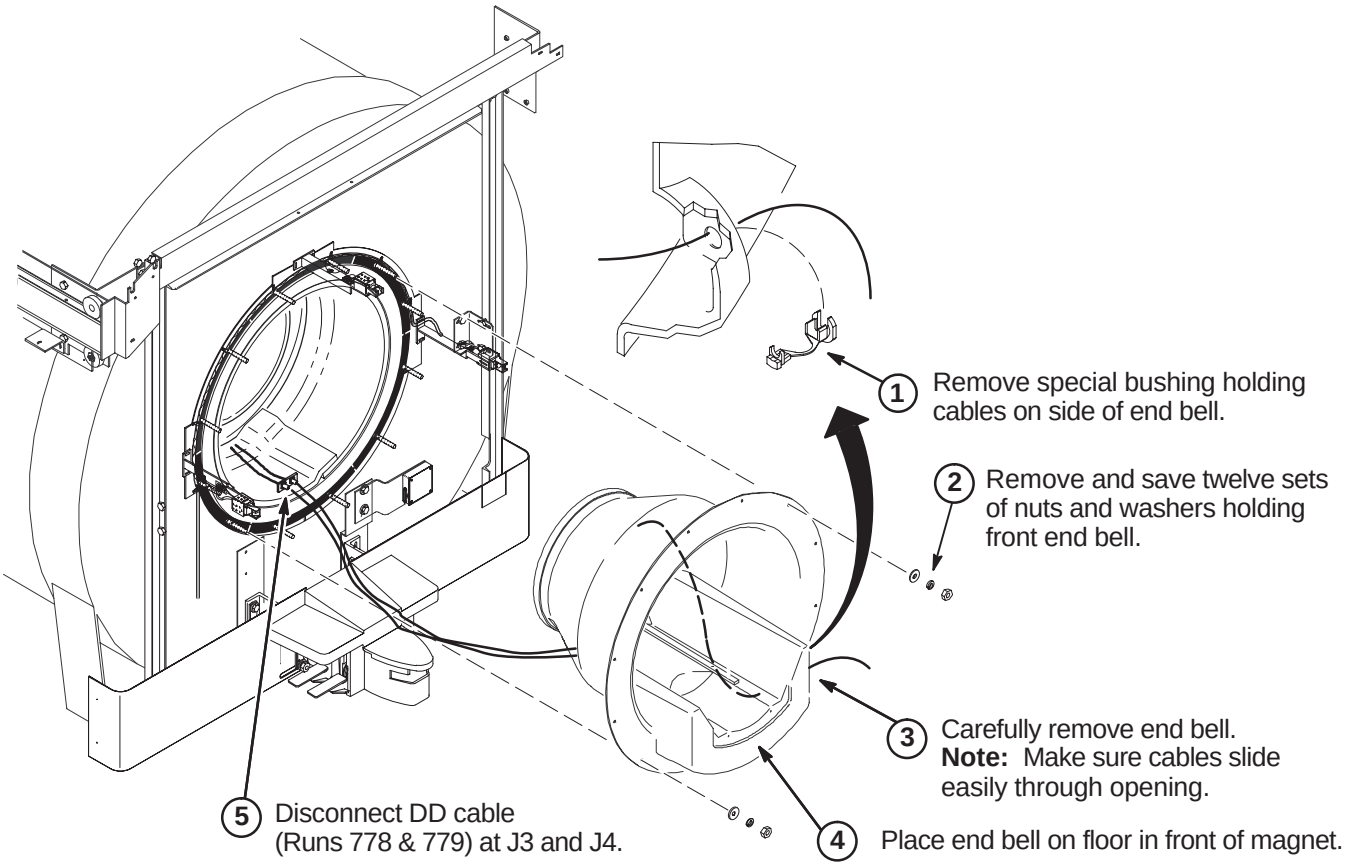
The existing Temperature Sensor Cable, Run 749, needs to be removed and must be replaced with new Dual Thermal Sensor cable, Run 1260.

6–3–1 UNFASTEN FLAT PANEL BORE LIGHTS



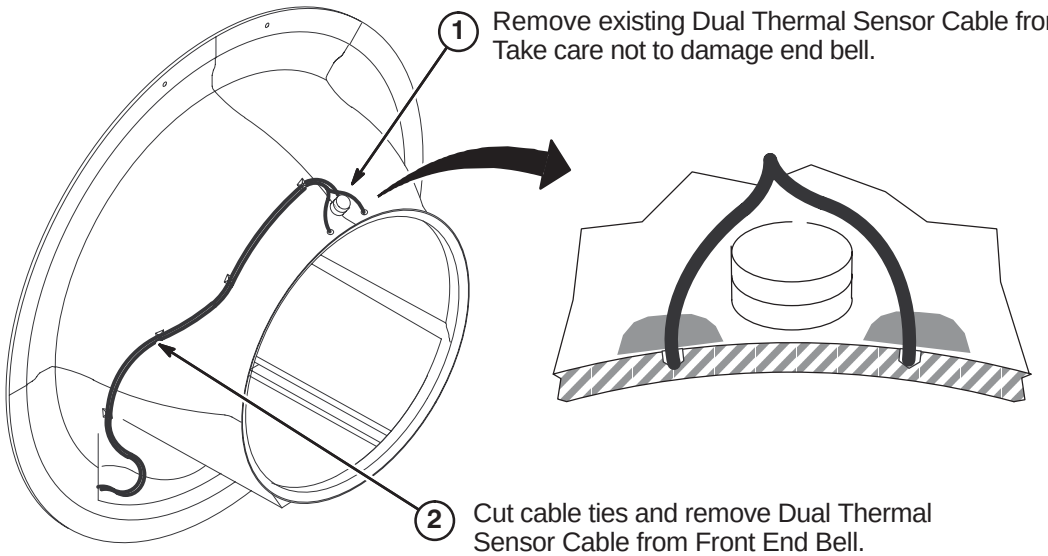


6–3–2 REMOVE FRONT END BELL



6–3–3 DE–INSTALL EXISTING RUN 749

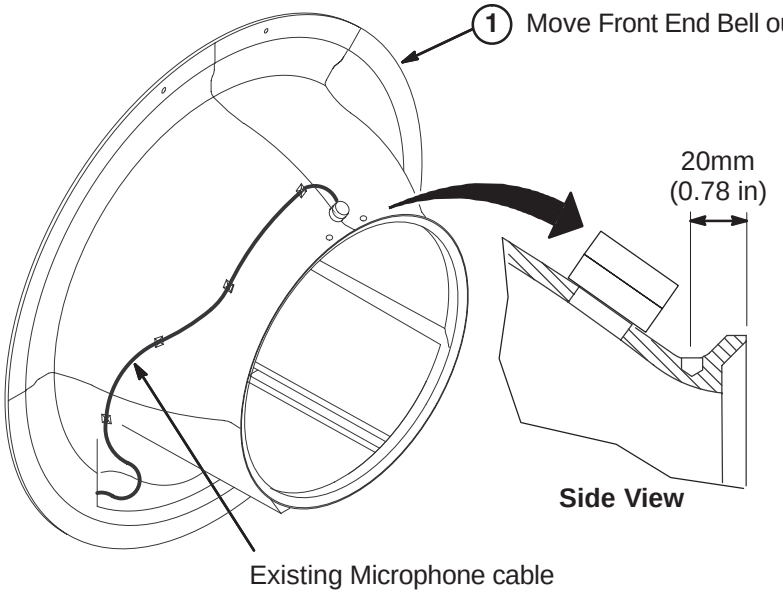
The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). EXCITE HD requires Run 1260 to be installed.

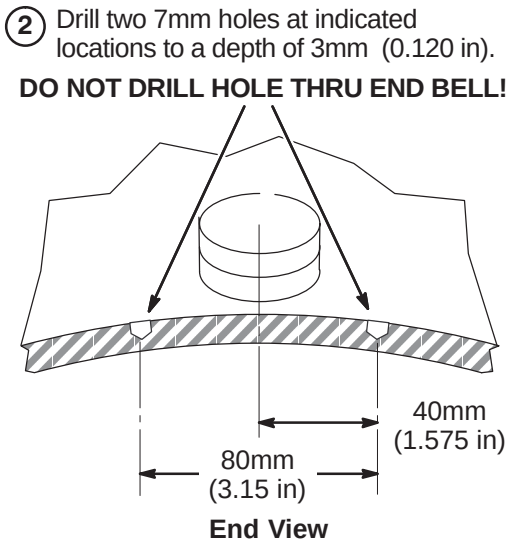


6–3–4 PREPARE END BELL FOR INSTALLATION OF DUAL THERMAL SENSOR CABLE

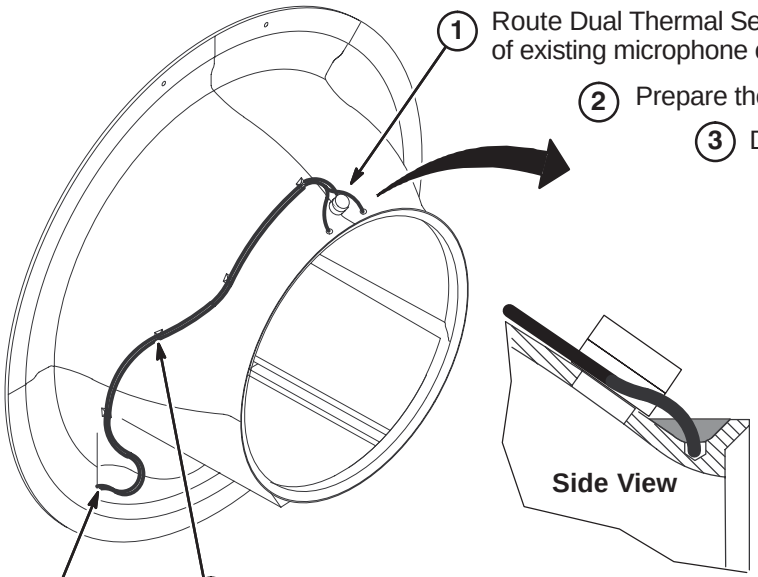
WARNING!

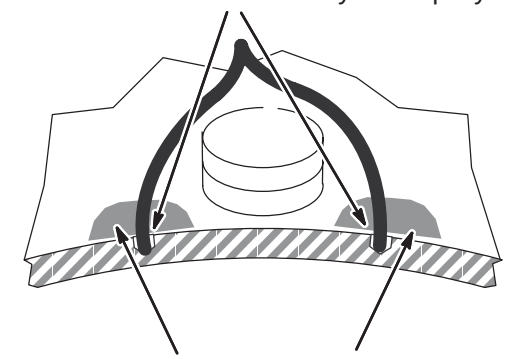
FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!





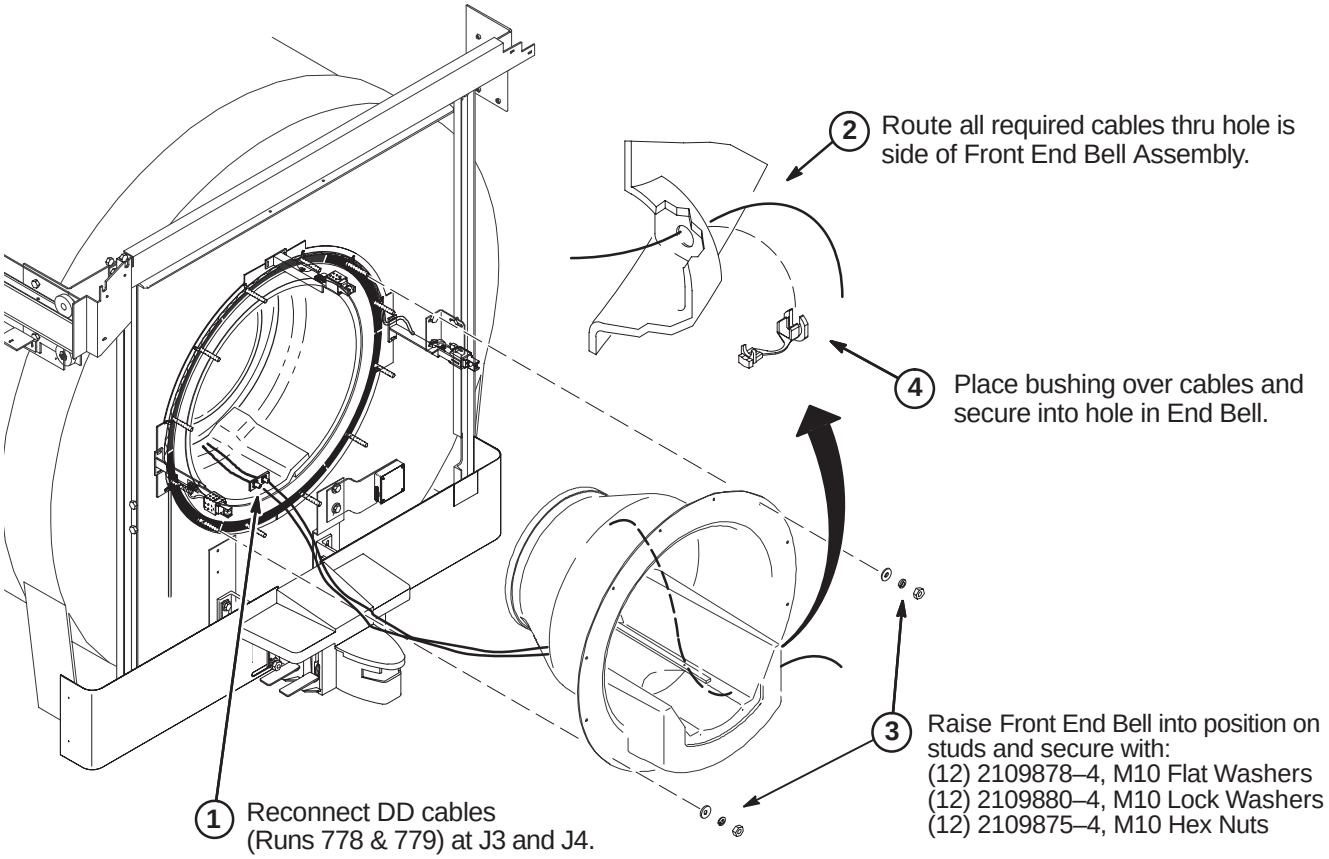
6–3–5 ATTACH DUAL THERMAL SENSOR CABLE



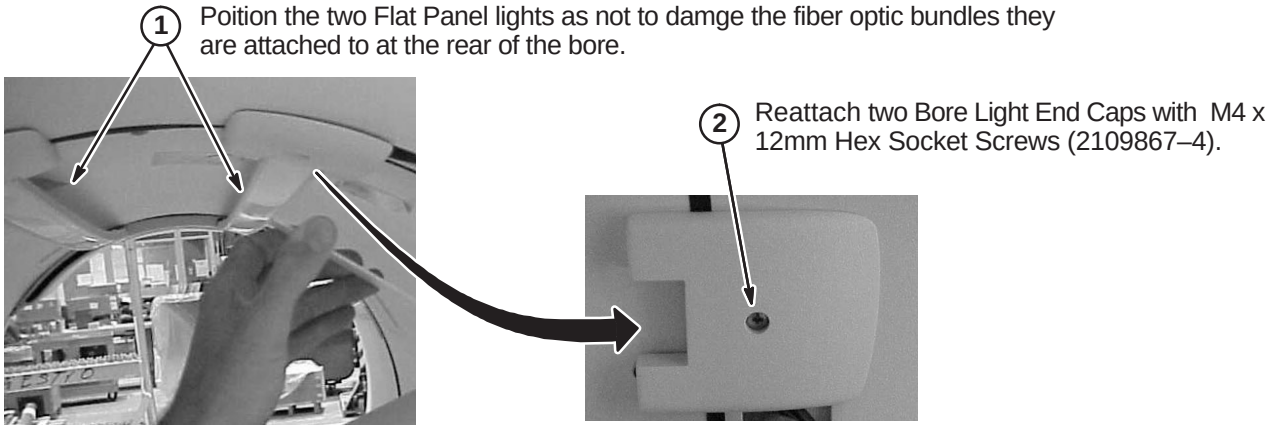


6–3–6 ATTACH FRONT END BELL TO MAGNET

Note: A good seal is required to prevent air leaks. Any air leaks between RF Tube (Gradient Coil) and End Bell joint can cause a loud humming noise during patient scanning. If possible, conduct an air leakage test prior to Bridge installation. This would require connecting power to the Blower Box and connecting 6" blower hose from Blower Box to Bulkhead.



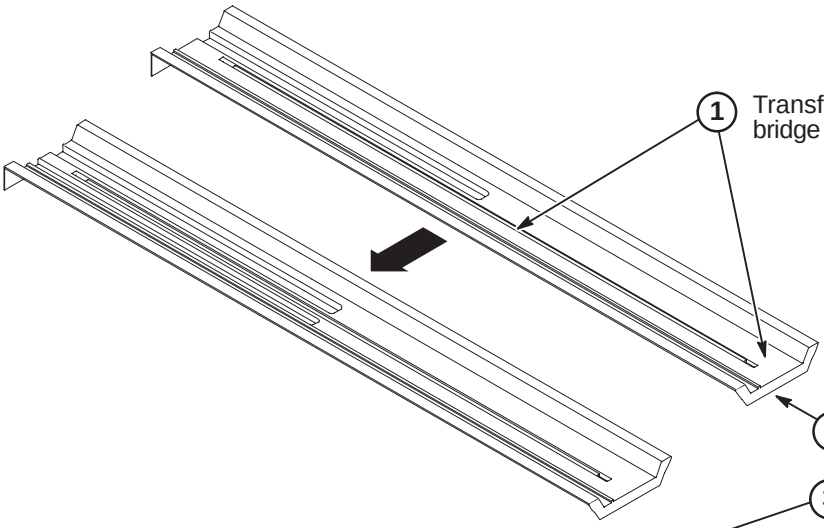
6–3–7 ATTACH BORE LIGHTS



6–4 INSTALL NEW COMPONENTS AND CABLES

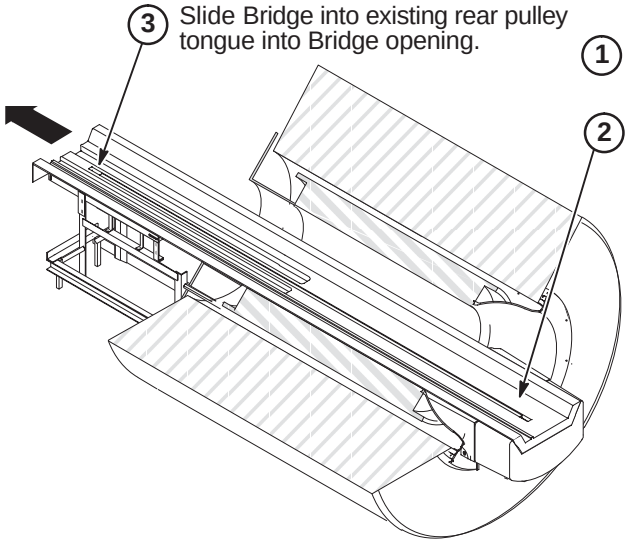
6–4–1 INSTALL NEW DUAL TRACK BRIDGE

6–4–1–1 Transfer Components to New Bridge



1. Transfer Drive Belt and Front Pulley from de–installed bridge to new EXCITE HD Bridge (5113537).
2. If still attached to bottom of Bridge, remove the two Cap Support Cups (2318948).
3. Clean the top surface of Cups and position on Bridge Cup Support and studs at front of magnet enclosure.

6–4–1–2 Install Bridge

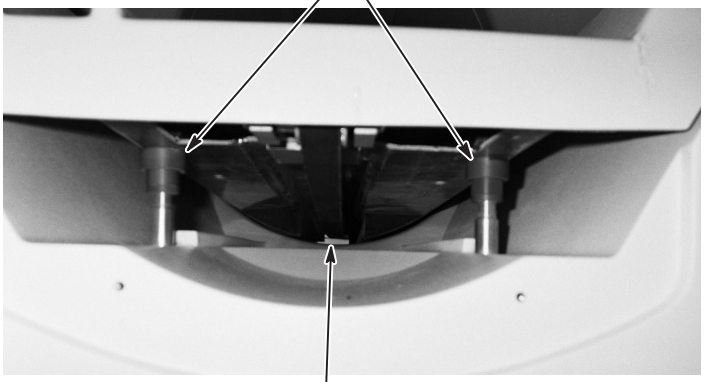


3. Slide Bridge into existing rear pulley tongue into Bridge opening.

1. Position rear end of Bridge Asm at front of Gradient Coil Bore.
2. While one person holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.

Note: Bridge should be resting on Rear Pedestal and Front Support Rods of mounting assembly.

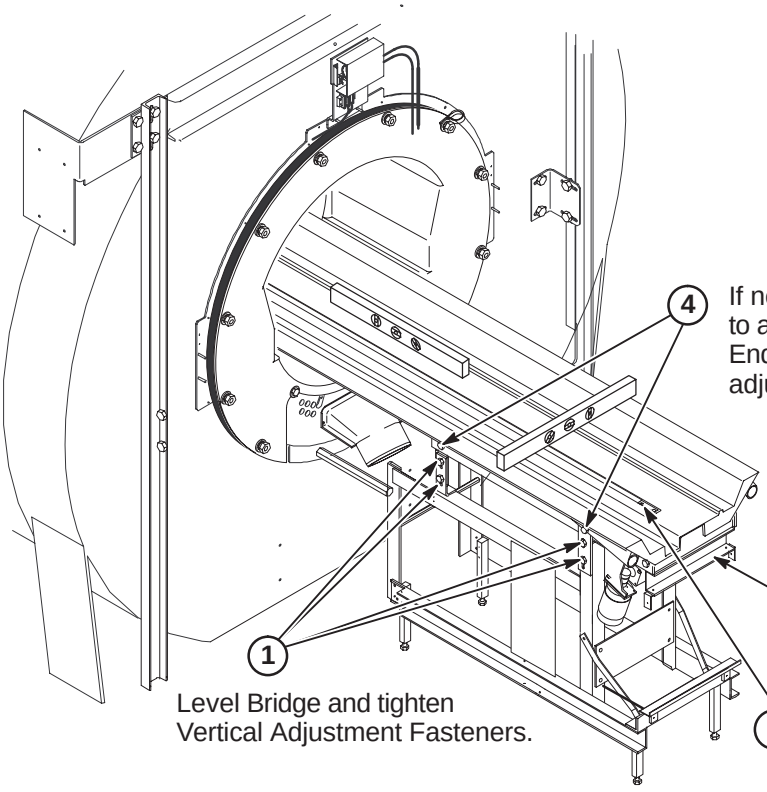
6–4–1–3 Level Front of Bridge



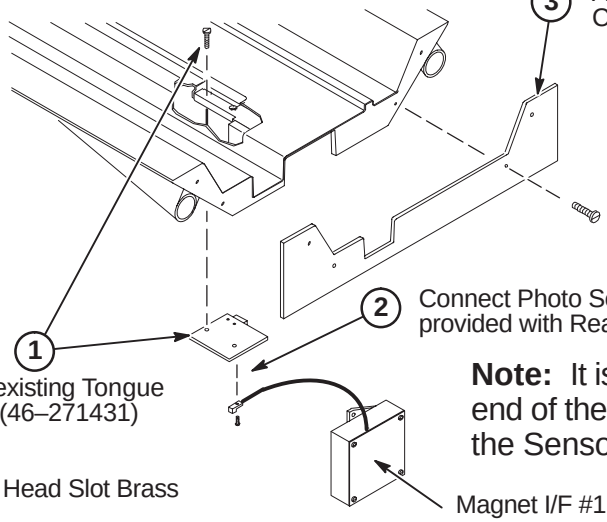
IMPORTANT!

THE BRIDGE MUST NOT COME IN CONTACT WITH THE RF COIL.
There must be an air gap of 6mm to 8mm thru entire length of bore.

6–4–1–4 Level All of Bridge



6–4–1–5 Connect Photo Sensor and Bridge End Cover



3 Attach previously removed Bridge Cover and (4) screws to end of Bridge.

2 Connect Photo Sensor with screw provided with Rear Bridge Mount Asm.

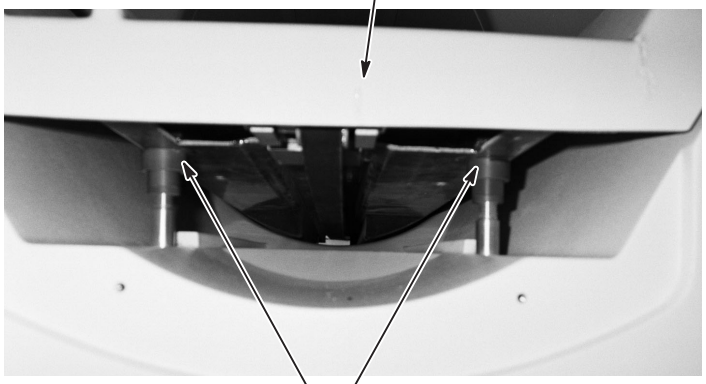
1 Attach together new Bridge, existing Tongue and Rear Bridge Mount Asm (46–271431) using:
(2) 46–220184P47
10–32 x 1–1/2 Screw, Round Head Slot Brass

Note: It is important that the Sensor end of the Sensor body fits flush with the Sensor Bracket.

Magnet I/F #1 (Mounted on Rear Pedestal)

6–4–1–6 Apply Adhesive to Front Bridge Support Caps

1 Raise the front of the Bridge high enough to apply adhesive in next step.



2 Apply epoxy adhesive to top of each Bridge Support Cap.

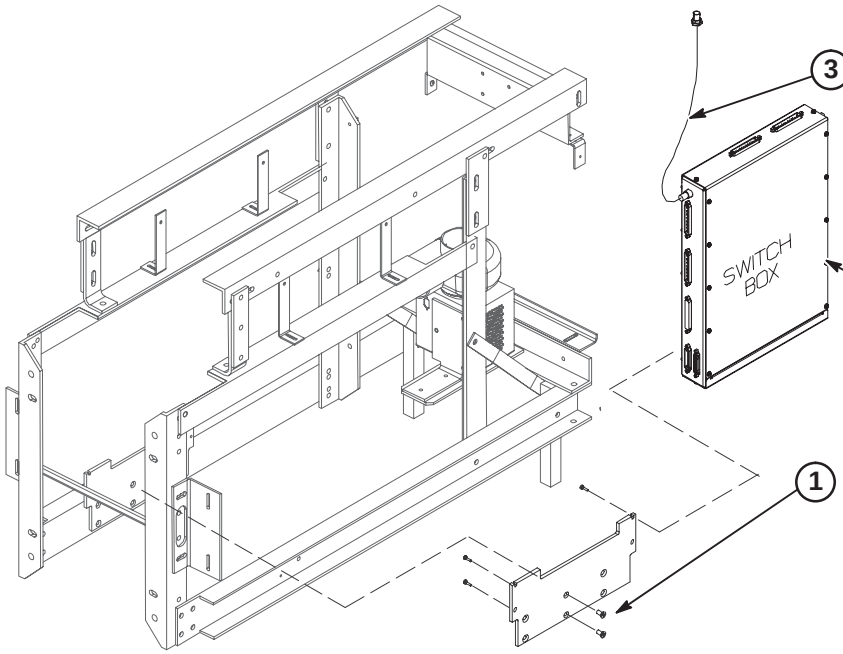
3 Lower the Bridge to rest firmly to the bridge support caps making sure the bridge does not touch the body coil or the End Bells

4 Allow adhesive to cure.

5 After adhesive has cured, adjust the bridge to the final vertical position. Make sure bridge is level.

6–4–2 INSTALL NEW COMPONENTS ON REAR PEDESTAL

6–4–2–1 Install 16 Channel Switch



3. Connect Run 1204 (5116105) as shown.

NOTE: Components and cables not shown on Rear Pedestal for clarity.

2. Attach 16 Channel Switch (5109645) to Mounting Plate using Loctite 242 (46–170686P2) and four M3 x 12mm Pan Head screws (46–312342P24).

1. Using the two center holes, install the 16 Channel Switch Mounting Plate (5110428) using Loctite 242 (46–170686P2) and:

- *If Welded Pedestal* – two M6 x 12mm Slotted Flat Head Screws (25120369–86)
- *If Bolted Pedestal* – two 10–32 x 1/2 Slotted Flat Screws (46–208913P12)

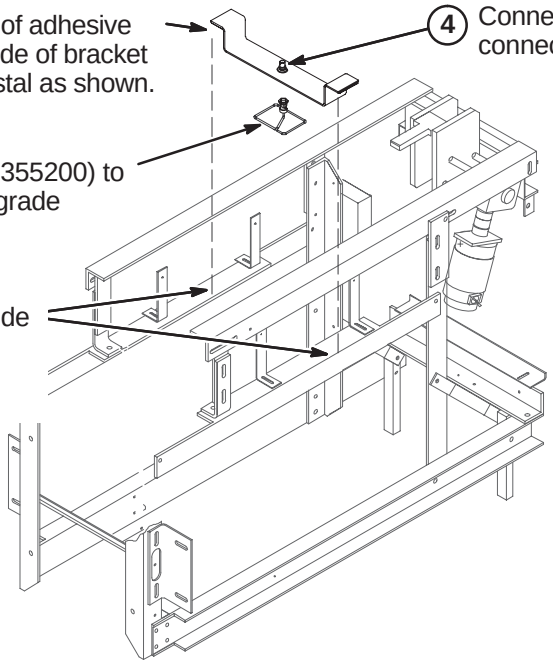
6–4–2–2 Install UTNS Antenna

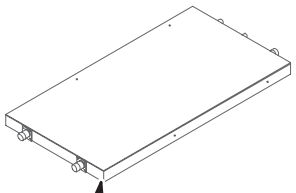
3. Peel of liner from bottom of adhesive backed Velcro on each side of bracket and attach to Rear Pedestal as shown.

4. Connect Run 1204 to UTNS Antenna connector at top of bracket.

2. Attach UTNS Antenna (2355200) to bottom side of UTNS Upgrade bracket (2384700).

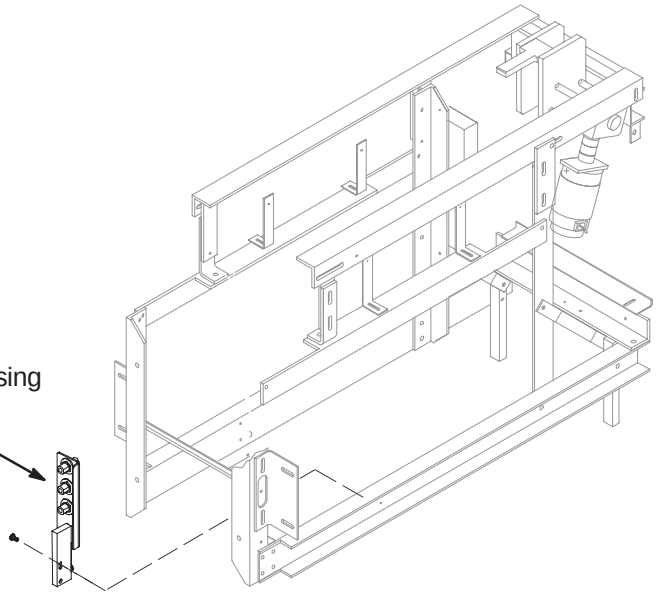
1. Make sure top surfaces of side braces are clean and dry for attachment of adhesive backed Velcro.





NOTE:
For clarity, Hybrid Splitter not shown in installed position.

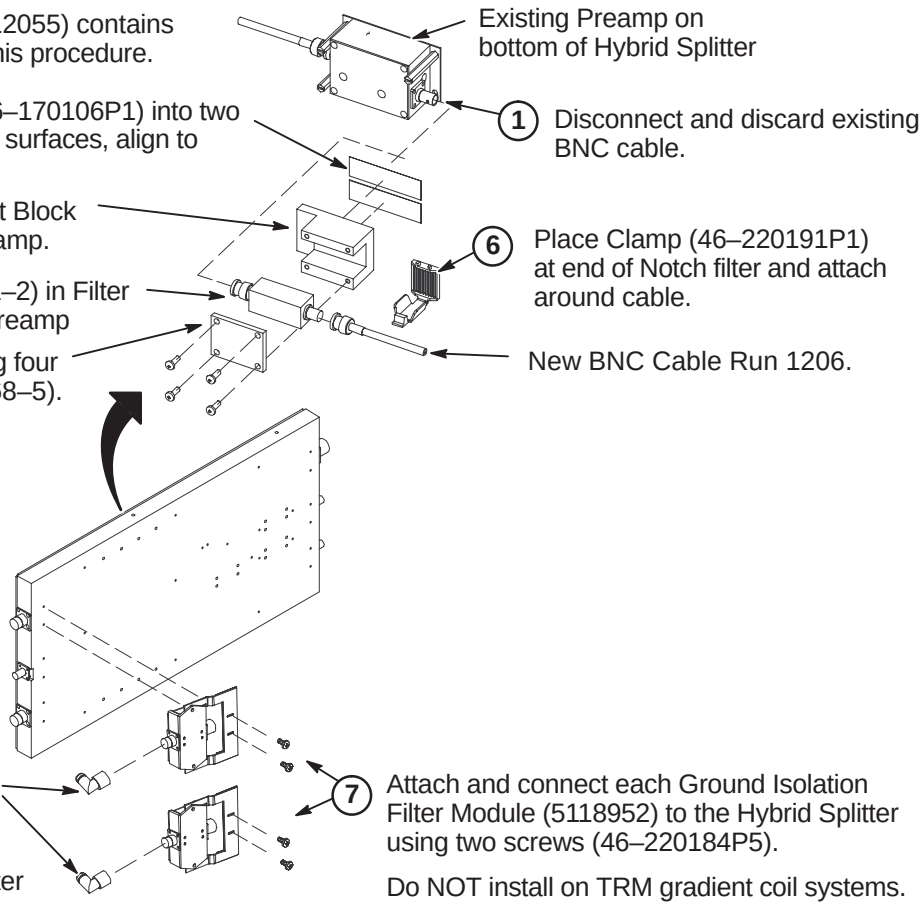
6–4–2–3 Install Bulkhead Interface



(1) Install new Bulkhead Interface (5113250) using Loctite 242 (46–170686P2) and two M6 x 12mm Pan Head Screws (2109873–24).

6–4–2–4 Install Filter Mount Asm and Isolation Filters to Body Hybrid Splitter

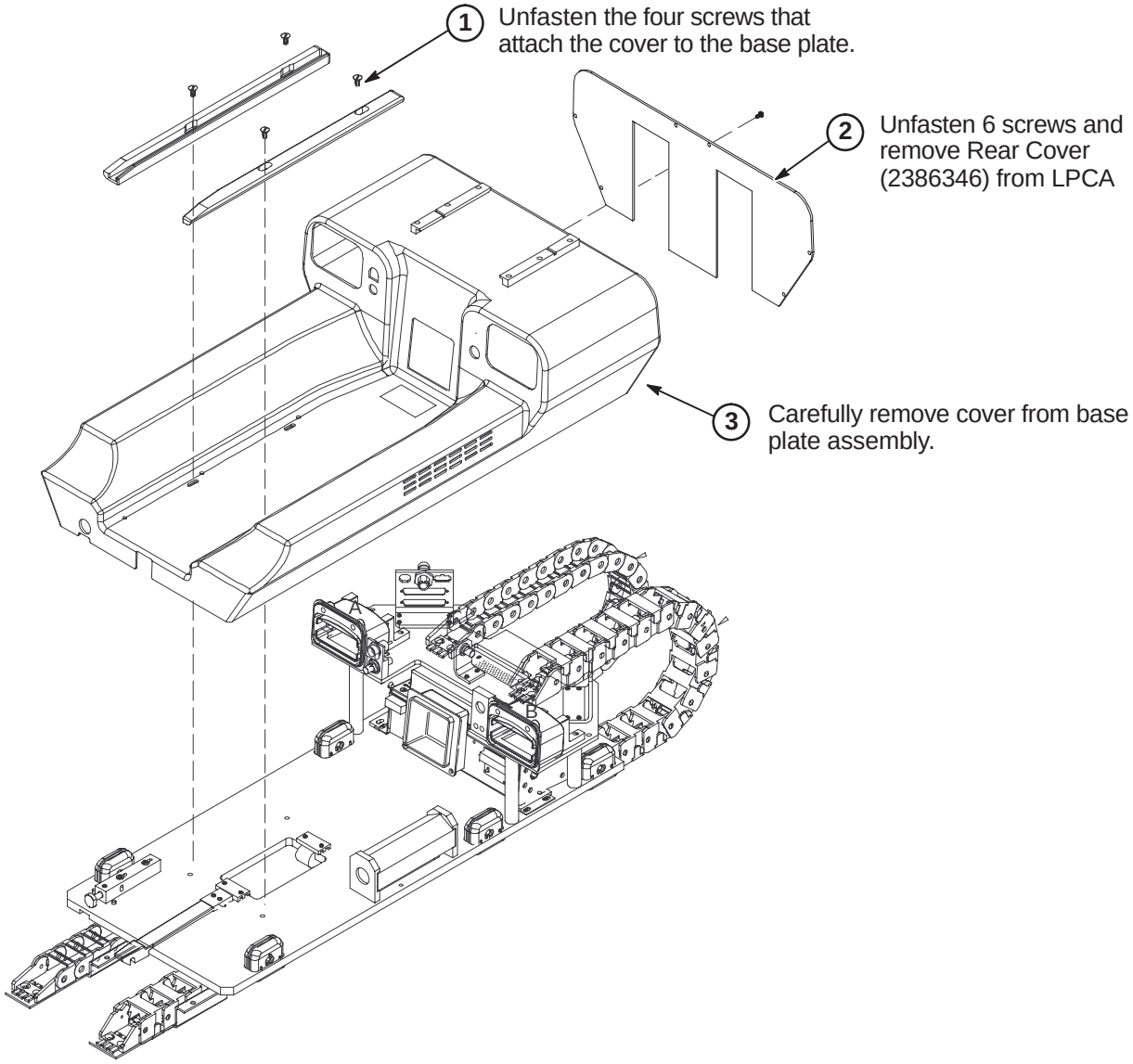
NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.



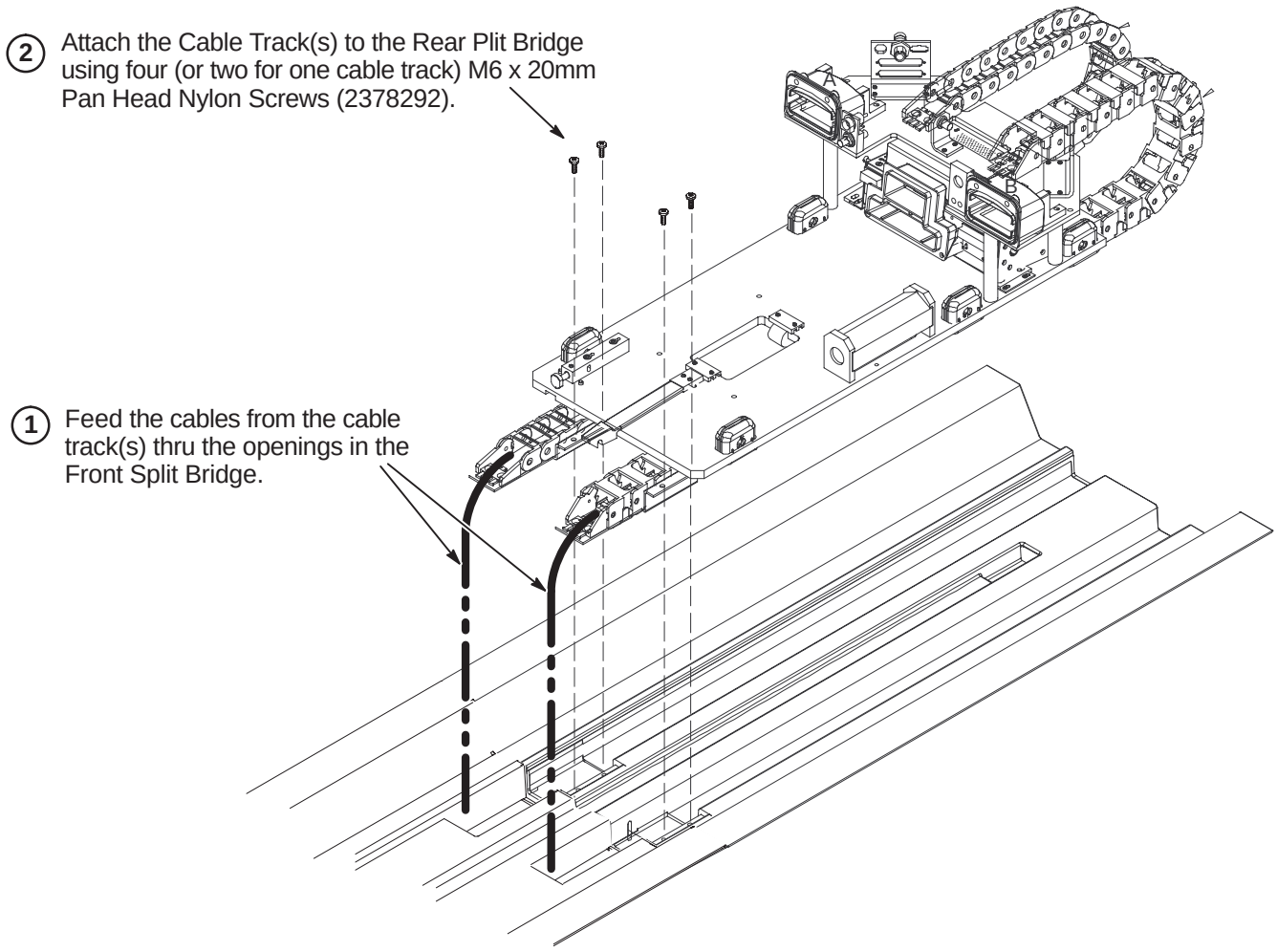
6–4–3 INSTALL LPCA AND CABLE TRACKS

The WideOpen and TwinSpeed style enclosures will be upgraded to an LPCA with a dual cable track. The S–Series and Cx magnet enclosures will install a new LPCA with only one cable track.

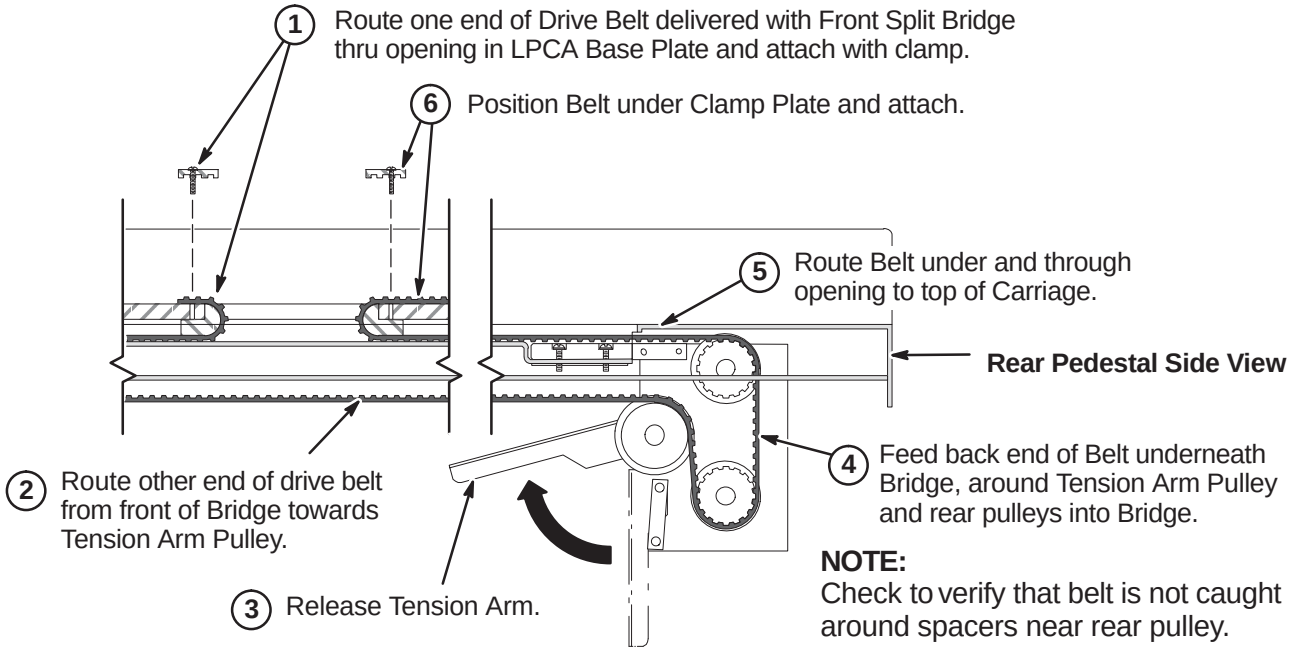
6–4–3–1 Remove LPCA Cover



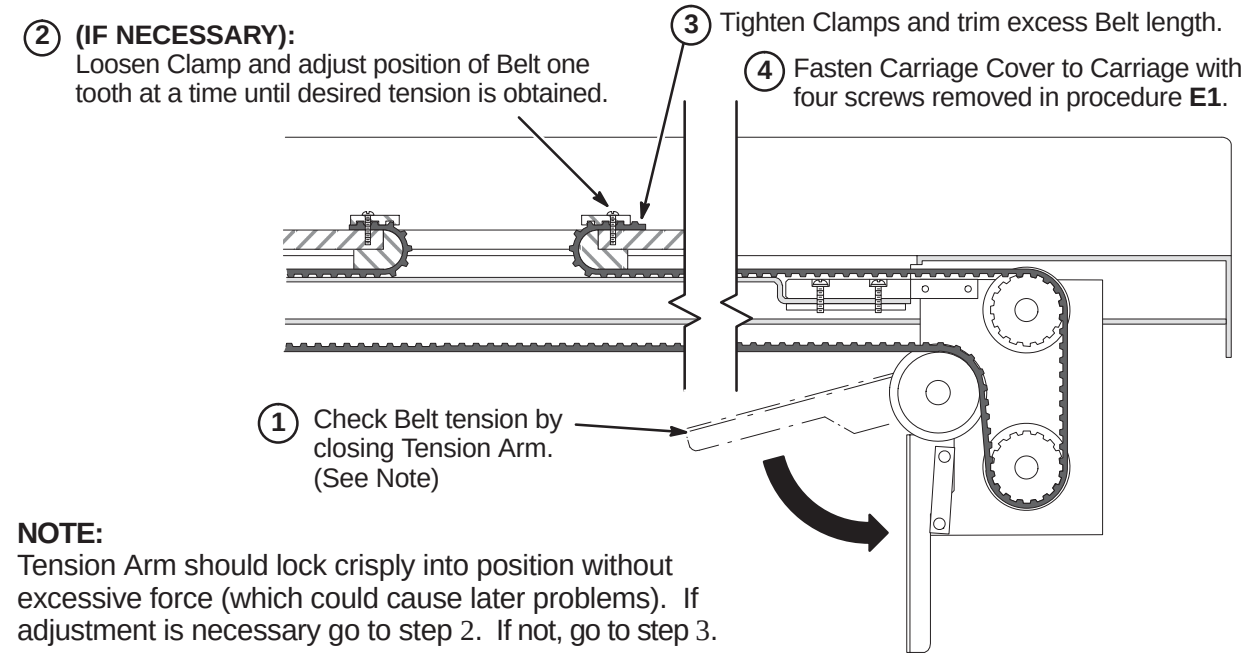
6–4–3–2 Attach Cable Tracks to Bridge



6–4–3–3 Route and Attach Drive Belt



6–4–3–4 Final Adjustment of Drive Belt

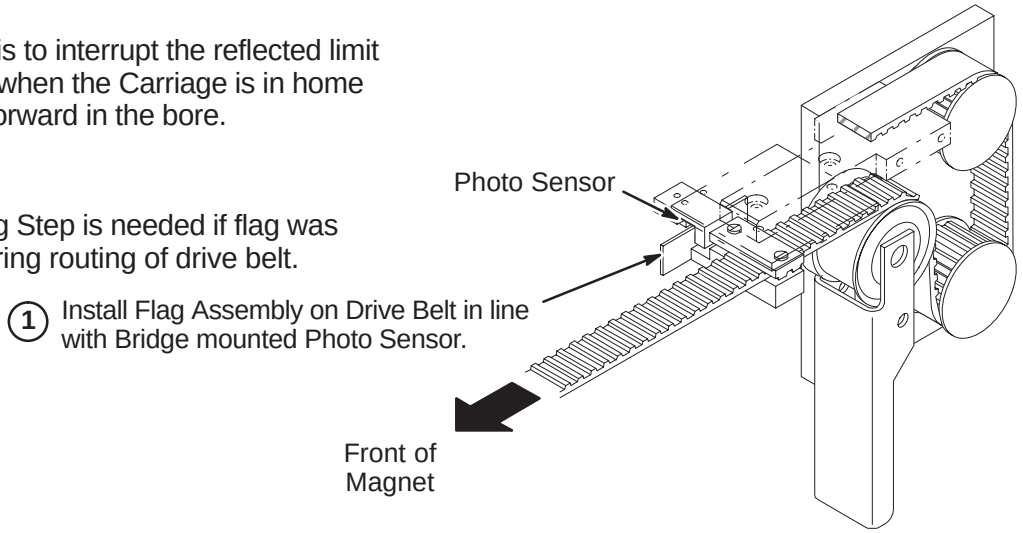


NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

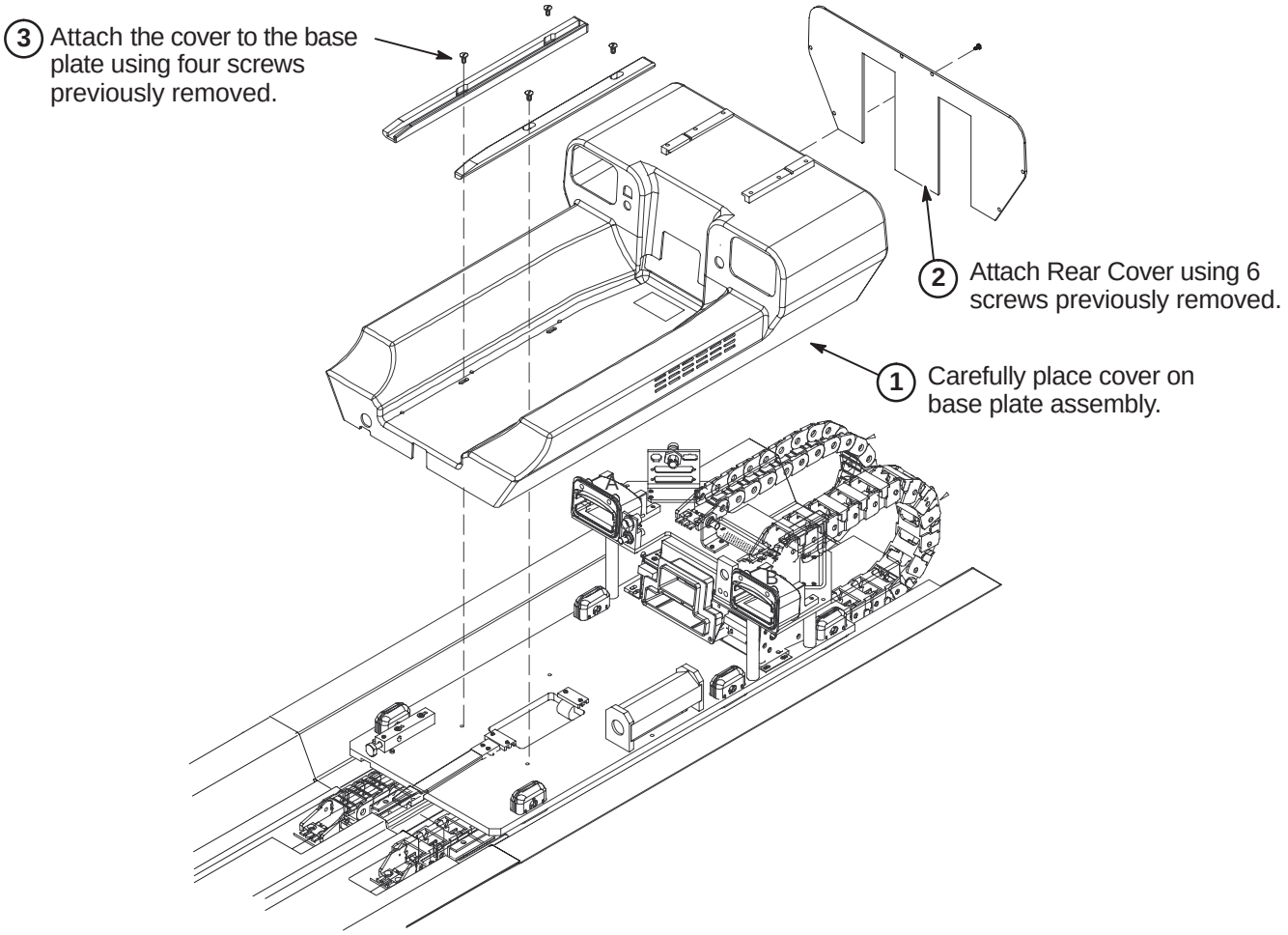
6–4–3–5 Installation of Sensor Flag

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.



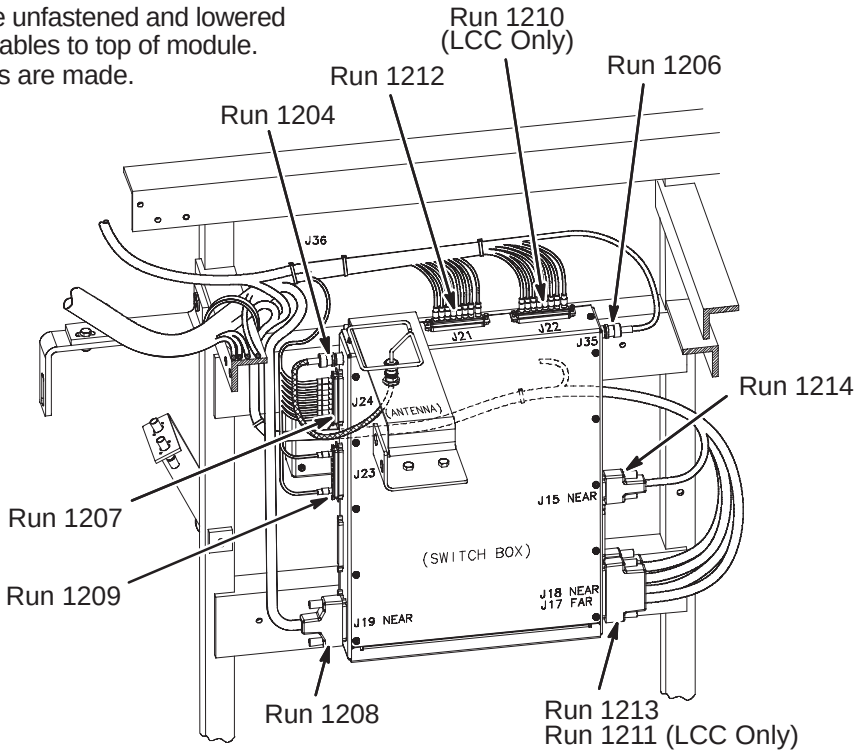
6–4–3–6 Installation of LPCA Cover



6–4–3–7 Connect Cables from Cable Track to 16 Channel Switch Module

(1) 16 Channnel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re–mount module after connections are made.

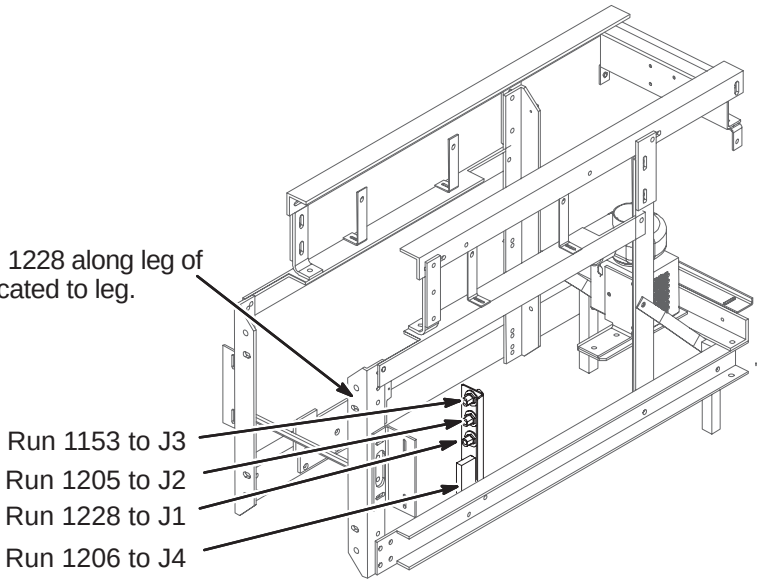
(2) Connect cable Runs as indicated.



6–4–3–8 Connect Cables from Cable Track to Bulkhead Interface

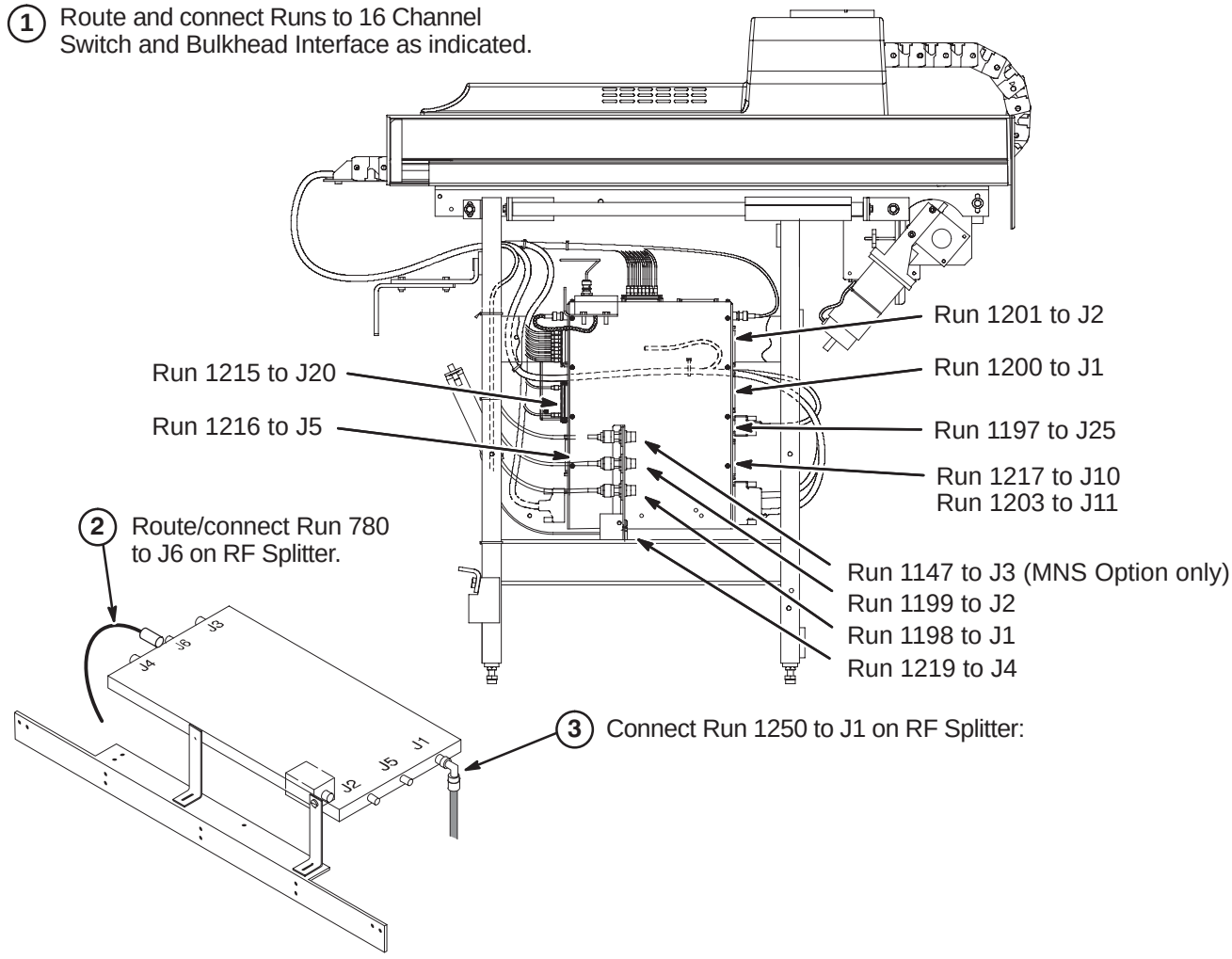
(1) Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty–wrap as indicated to leg.

(2) Connect Runs to Bulkhead Interface as indicated.

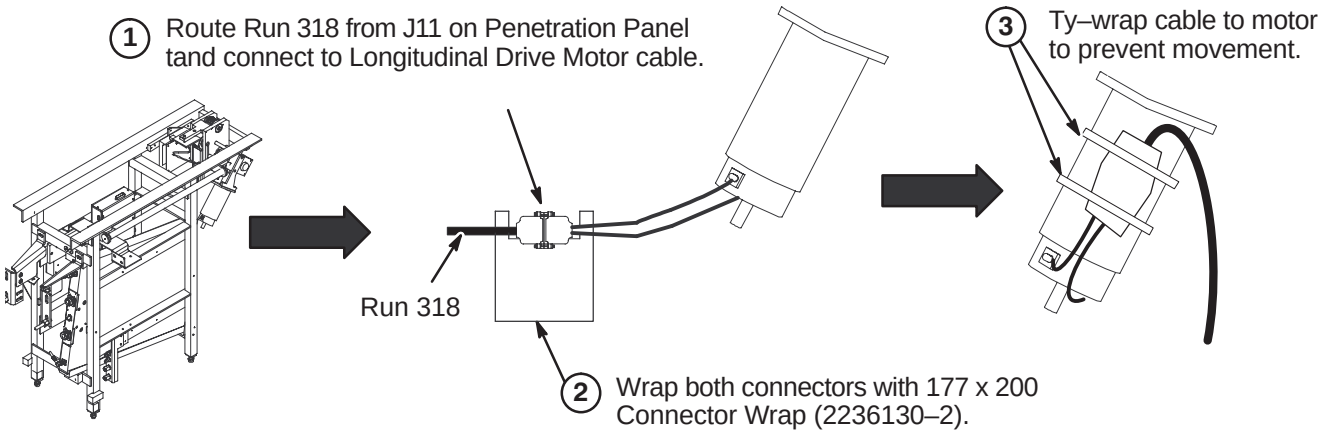


6–4–4 CONNECT REMAINING CABLES TO REAR PEDESTAL

6–4–4–1 Cable Connections to 16 Channel Switch and Bulkhead Interface

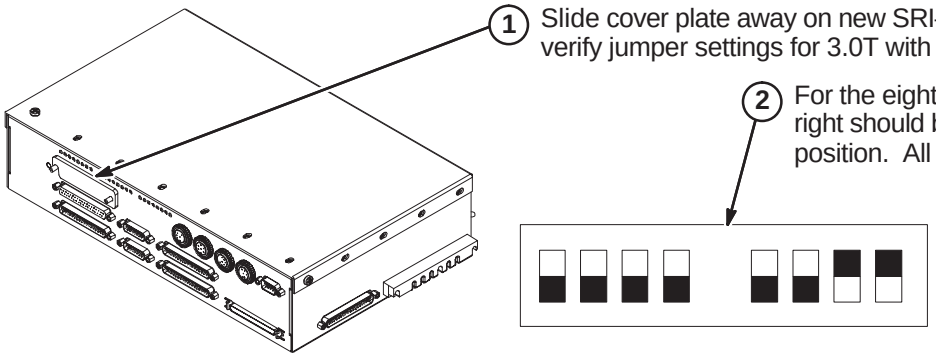


6–4–4–2 Route/Connect Run 318 to Longitudinal Drive Motor



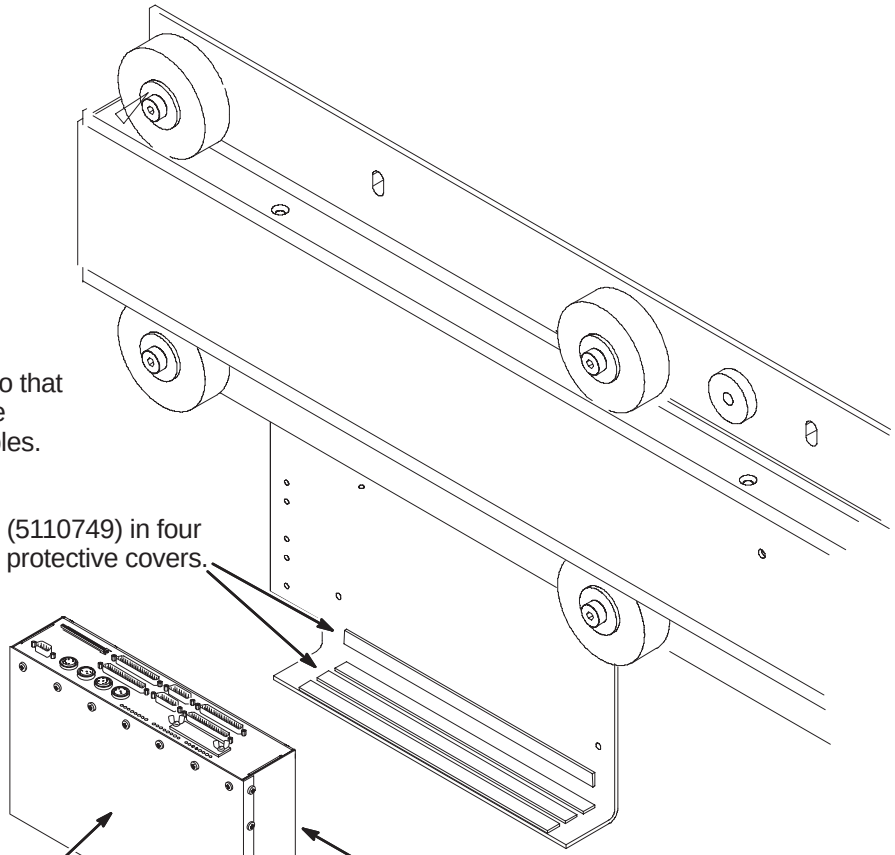
6–4–5 INSTALL SRI–3 MODULE

6–4–5–1 Verify Jumper Positions for SRI–3



1. Slide cover plate away on new SRI–3 and verify jumper settings for 3.0T with Coil ID.
2. For the eight jumpers, the two jumpers on the right should be in the “ON” or located in the up position. All others should be “OFF” or down.

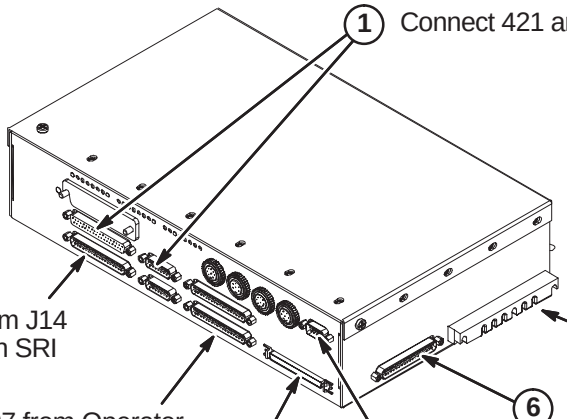
6–4–5–2 Position SRI–3 in L Bracket



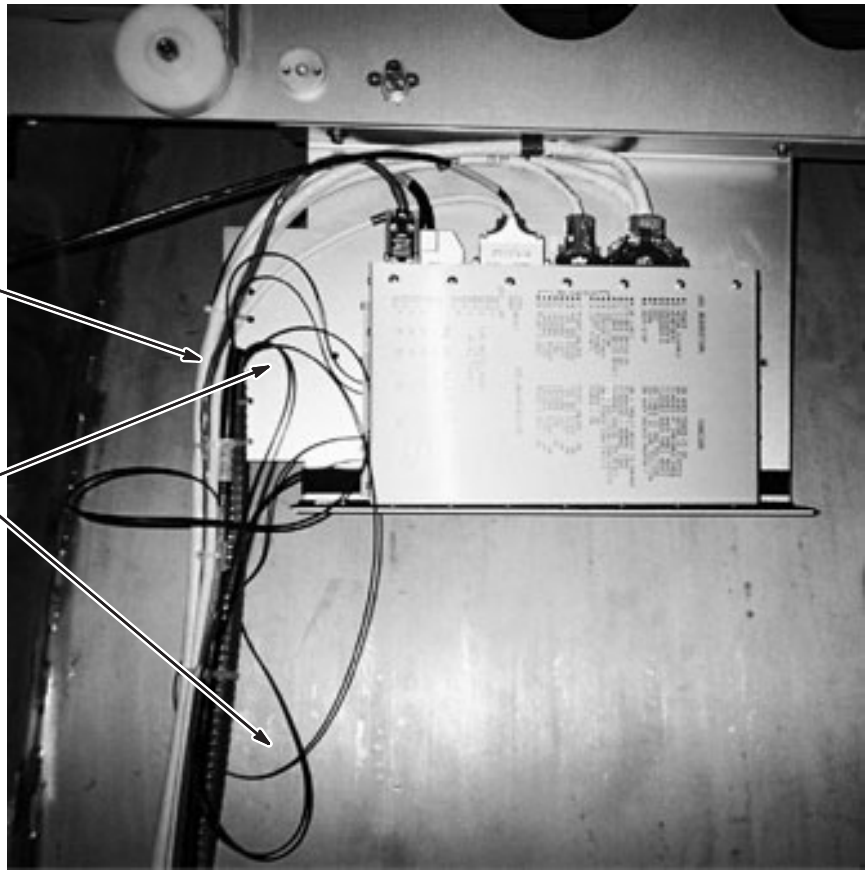
Note:
SRI–3 Module to be installed so that it can be easily pulled off for the installation and removal of cables.

1. Install Loop Fastener strips (5110749) in four places as shown. Remove protective covers.
2. On back of SRI–3 Module, remove protective cover from Velcro at the four locations.
3. Press the SRI–3 Module on bracket and press into place against the adhesive of the velcro.

6–4–5–3 Route and Connect Cables to SRI–3



1. Connect 421 and 422 to J3 and J2.
2. Connect Run 715 from J14 on Pen Panel to J1 on SRI Module.
3. Connect Run 897 from Operator Display to J4.
4. Route Run 1261 from Operator Display and connect to J5.
5. Route Run 1260, Dual Temperature Sensor, from Front End Bell and connect to J10.
6. Route Run 1215 from J20 on 16 Channel Switch and connect to J20.
7. Connect fiber optic cable Run 711 to J11, J12, and J13.



8. Route all cables and fiber optic cables as shown.

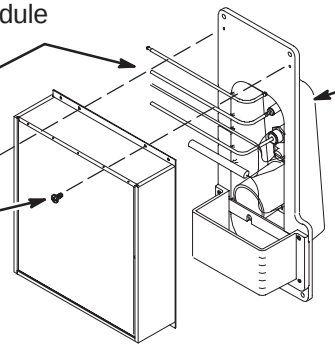
Note:
Fiber optic cables loops must not be smaller than two inch (50mm) radius.

9. Reapply spiral wrap around all cables and fiber optics.

6–4–6 INSTALL PAC MODULE

6–4–6–1 Connect Run 712, 716, & Pneumatic Patient Alert Tubing

Note: Complete steps 1 and 2 if new PAC2B Module did not have PAC Control Housing attached.



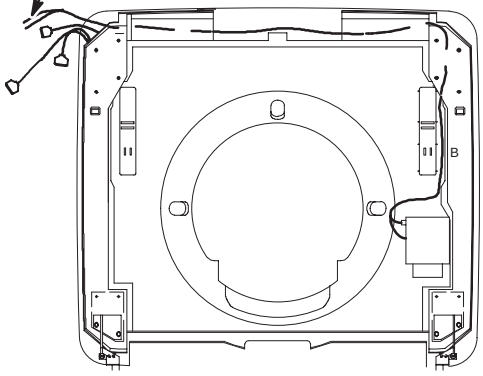
PAC Control Housing

(2) Connect cables and tubes from the existing PAC Control Housing to new PAC2B module.

(1) Attach new PAC2B to PAC Control Housing using two previously removed fasteners.

(3) Mount new PAC2B Asm to Front Cover using flat washers and screws previously removed.

(4) Route Run 712 and Pneumatic Patient Alert Tubing along route of existing cables to PAC Asm location. Ty–wrap as necessary.

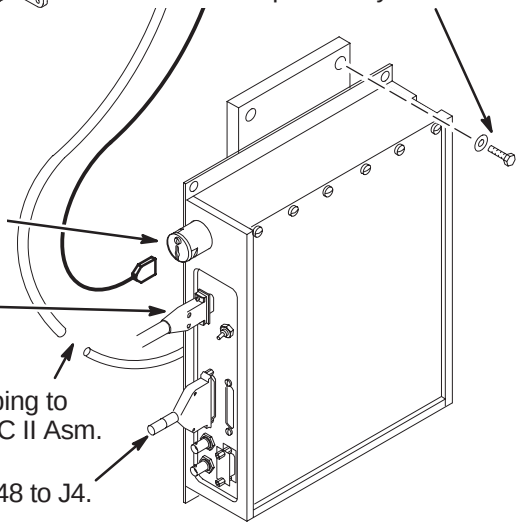


(5) See next Step for Run 712 connection instructions.

(6) Connect cable Run 716 to J2.

(7) Connect Pneumatic Tubing to Tubing extension on PAC II Asm.

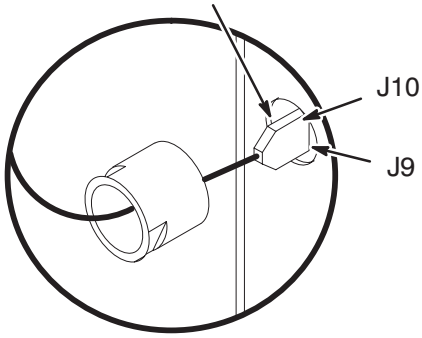
(8) Connect Run 848 to J4.

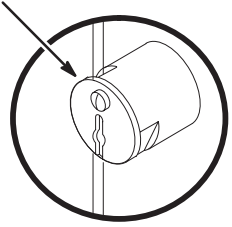


6–4–6–2 Connect Run 712

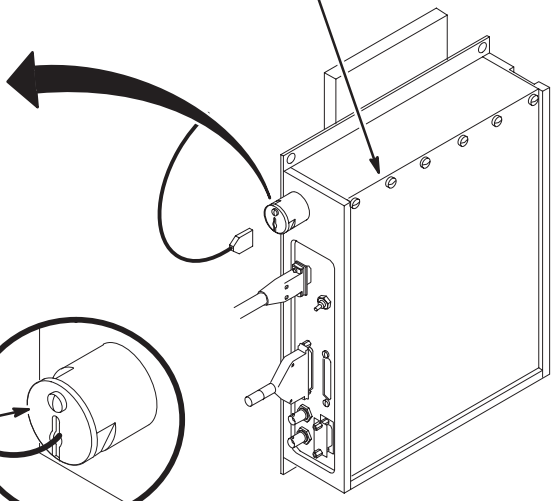
(1) Remove shield cover by removing one screw and unscrew shield from module.

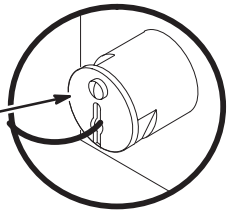
(2) Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.





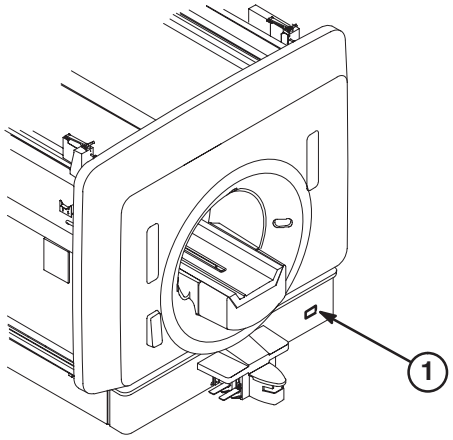
Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.





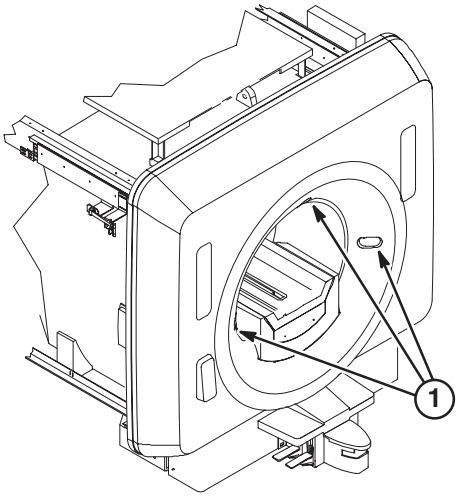
(3) Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.

6–5 INSTALL RATING PLATE ON FRONT TRIM COVER



1 Place Rating Plate (46–302200P12) in this area of Front Skirt

6–6 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS



A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.

1 For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.

French

German

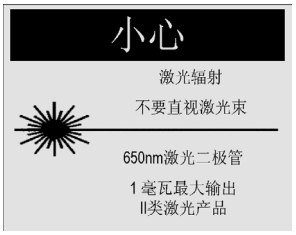
Portuguese

Spanish

Italian

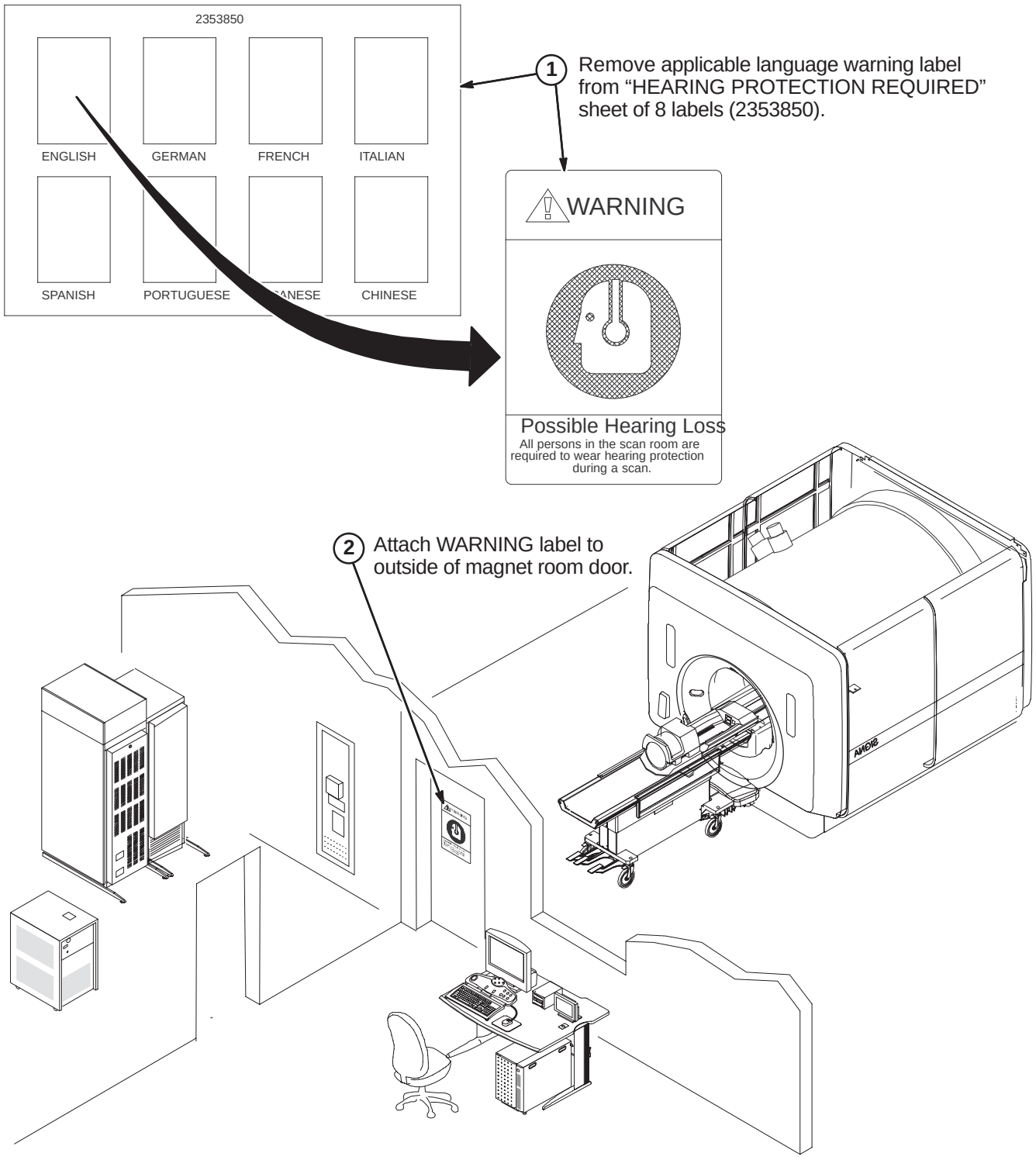
Swedish

Japanese



Chinese

6–7 INSTALL HEARING LOSS WARNING SIGN



SECTION 7 – CRADLE RELEASE UPGRADE

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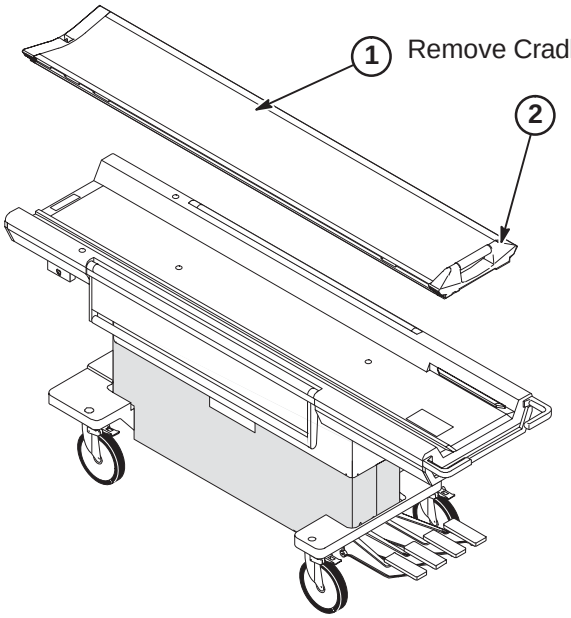
7–1–1 PREPARE CRADLE FOR UPGRADE 7–2
7–1–2 UPGRADE PROCEDURE 7–2

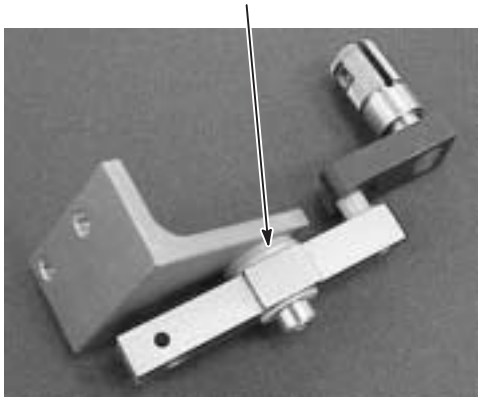
7–1 PATIENT TABLE CRADLE RELEASE UPGRADE

This upgrade is performed to improve the release mechanism and improve image quality.

7–1–1 PREPARE CRADLE FOR UPGRADE

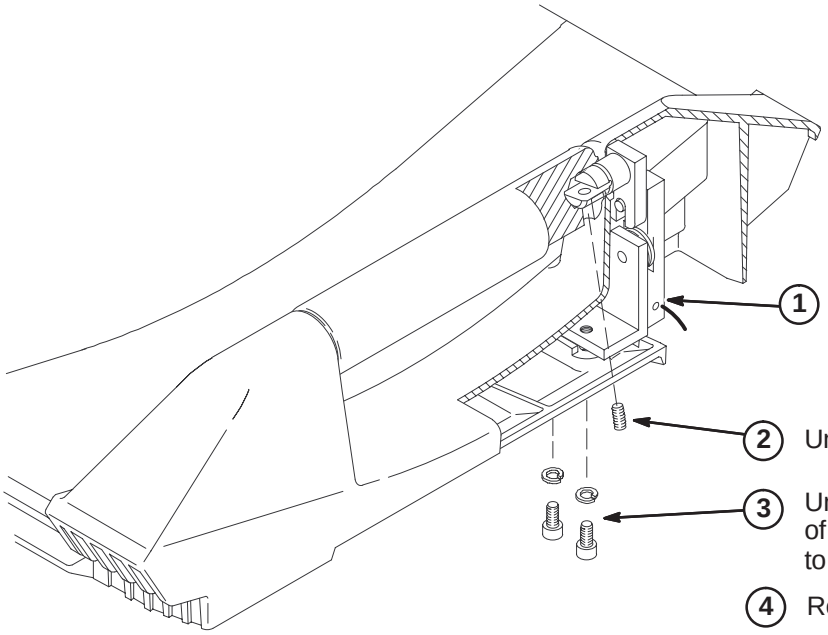


1. Remove Cradle from patient table.
2. Position Cradle to access release mechanism located under this corner of cradle.
3. The new mechnism is designed with two washers and two bushings inserted into the lever. The lever has also been widened in the center portion.



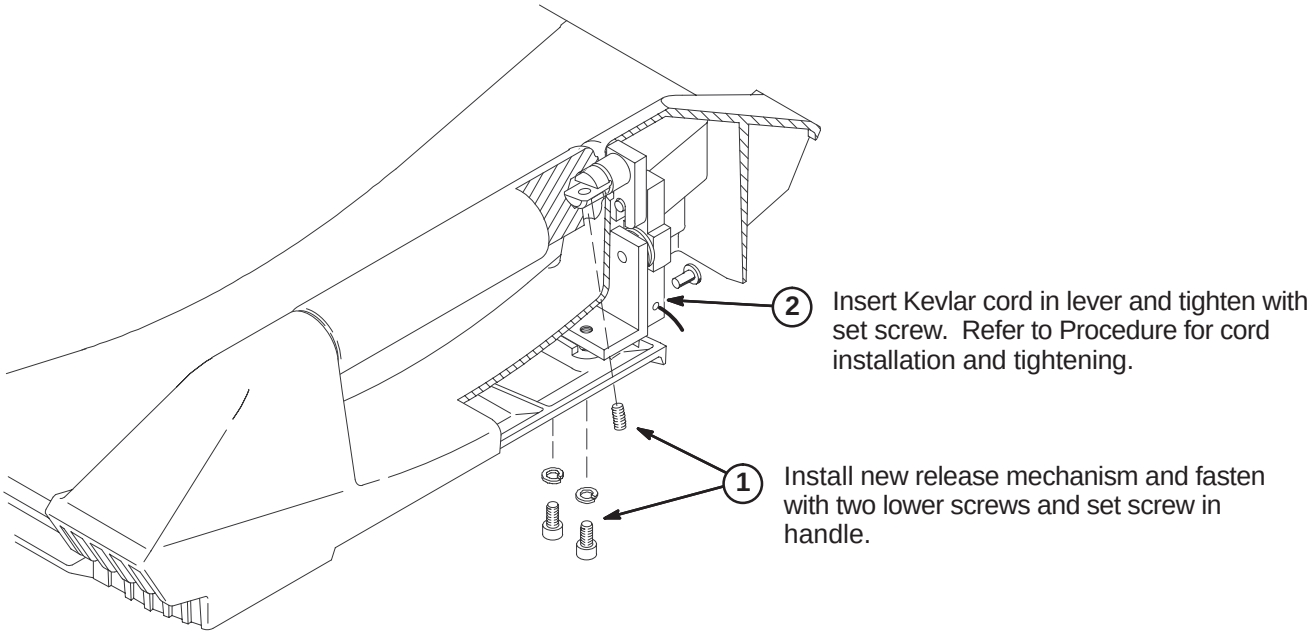
7–1–2 UPGRADE PROCEDURE

7–1–2–A Remove Existing Release Mechanism

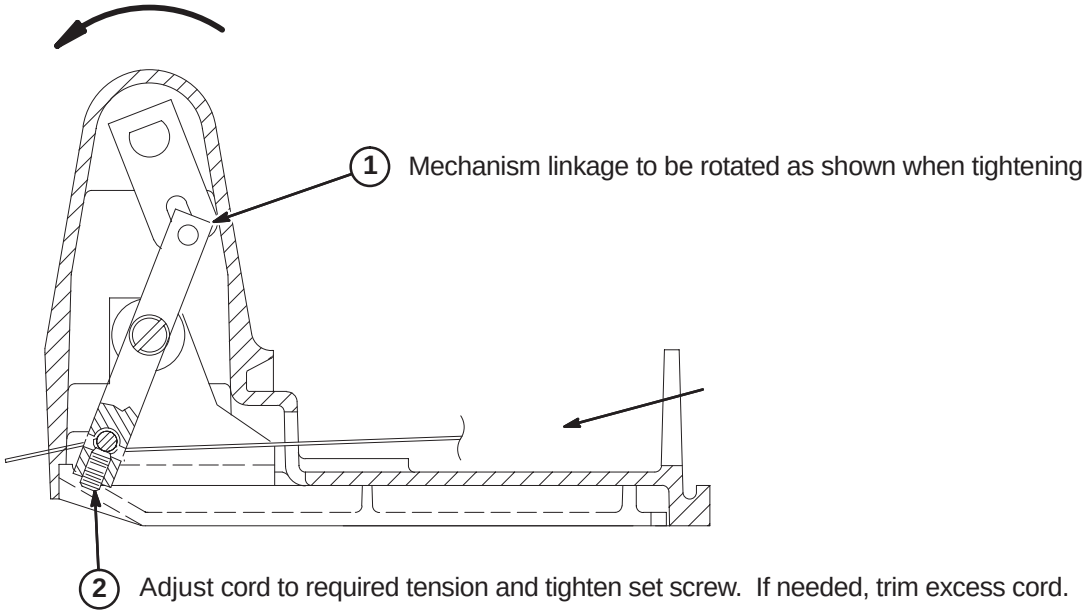


1. Unfasten set screw at bottom of lever and remove Kevlar cord from lever. Retain set screw and flat head rivet for transfer to new mechanism if required.
2. Unfasten set screw at bottom of cradle handle.
3. Unfasten two screws and washers at bottom of cradle that attach the existing mechanism to cradle.
4. Remove old release mechanism.

7–1–2–B Install New Release Mechanism



7–1–2–C Attach Cord



SECTION 8 – OPERATOR WORKSPACE

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8–2–2 DE–INSTALL POWER MONITOR PC AND PERIPHERALS 8–5
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8–1 INTRODUCTION

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

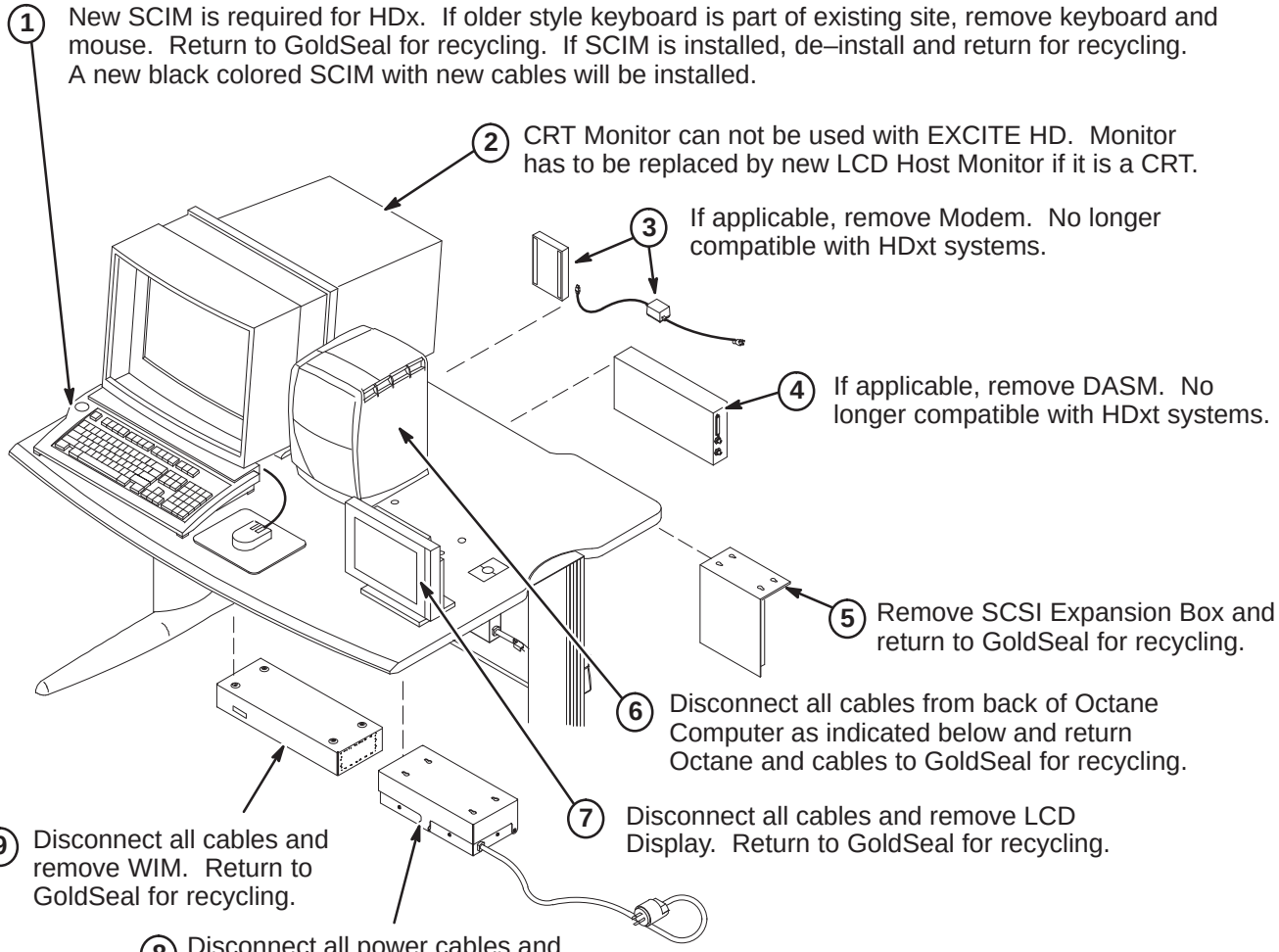
8–1–1 DE–INSTALLATION INFORMATION

The equipment de–installed for the HDx upgrade depends on the existing site configuration. The general guideline for removal of equipment is:

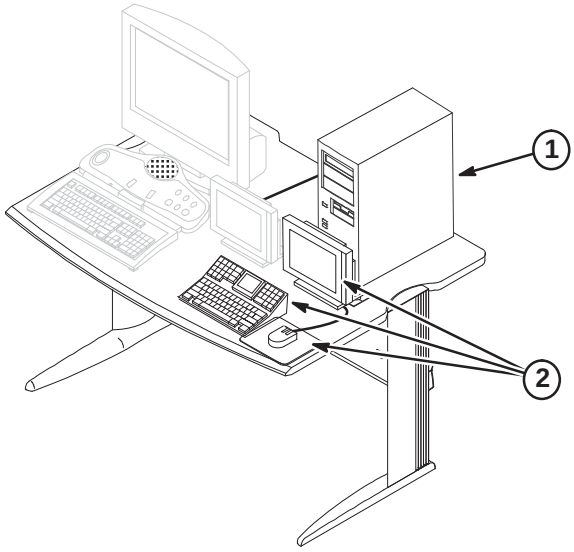
- DAT and/or MOD drives are not supported in HDxt. Refer to Section 8–3 for legacy media support.
- Confirm that all cables are disconnected from OperatorWorkspace and Computer.
- Perform the following procedure for return of the OW modules and Computer.
- Table will remain as part of upgrade.

8–2 DE–INSTALL OPERATOR WORKSPACE

8–2–1 DE–INSTALL OW TABLE MODULES

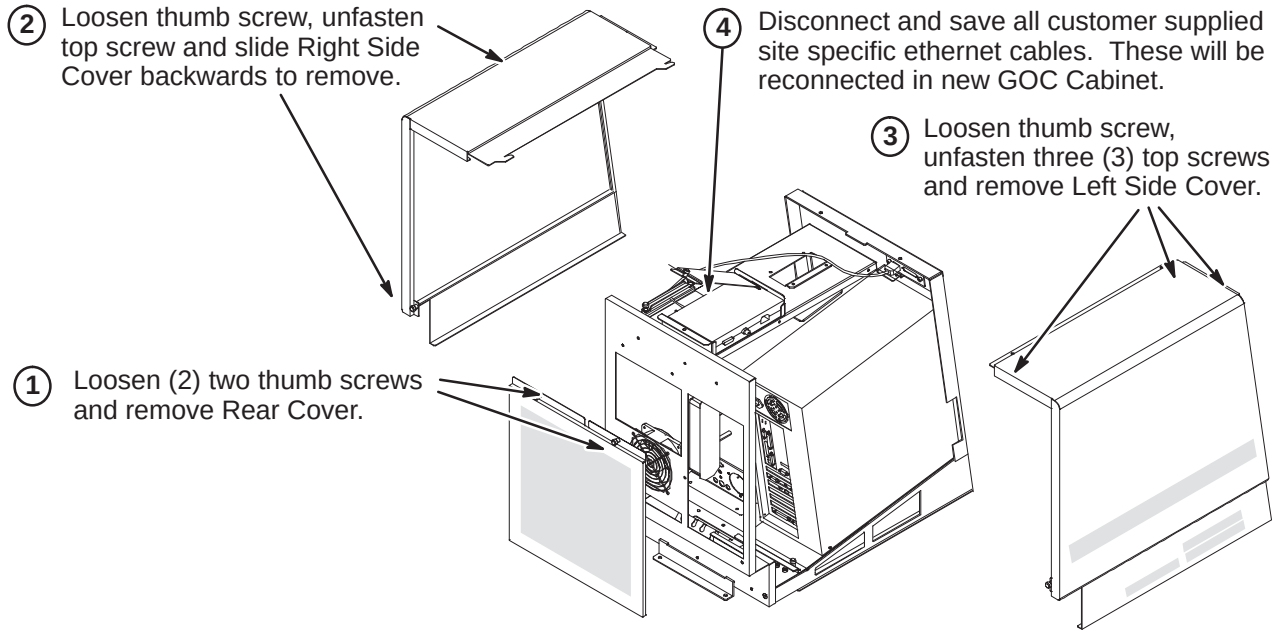


8–2–2 DE–INSTALL POWER MONITOR PC AND PERIPHERALS



1. Disconnect all cables from back of Power Monitor PC and return to GoldSeal for recycling.
2. Remove Monitor, Keyboard, and Mouse and return for recycling.

8–2–3 DE–INSTALL OCTANE DUPLEX COMPUTER CABINET



2. Loosen thumb screw, unfasten top screw and slide Right Side Cover backwards to remove.
4. Disconnect and save all customer supplied site specific ethernet cables. These will be reconnected in new GOC Cabinet.
3. Loosen thumb screw, unfasten three (3) top screws and remove Left Side Cover.
1. Loosen (2) two thumb screws and remove Rear Cover.

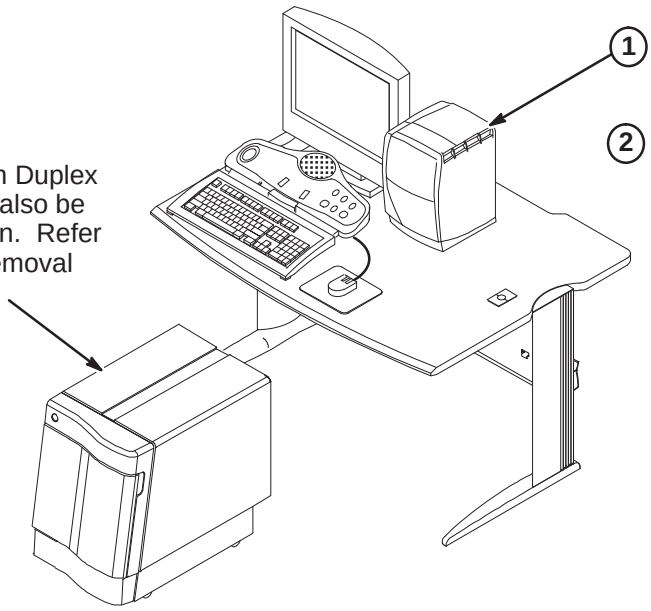
5. Refer to Section 14–3–1 for instructions on transferring Legacy Archive modules to EXCITE HD.
6. The remaining Duplex cables will not be used on the new GOC Cabinet and must be returned for recycling to Gold Seal.
7. The Duplex must be returned for recycling to Gold Seal. Duplex can be inserted onto skid that was used to ship the GOC by removing the white foam on the bottom that supported the GOC.

8–3 LEGACY ARCHIVE MEDIA SUPPORT

HDx series does NOT support Legacy Archive Media, such as DAT and Pioneer MOD Archive Devices. Complete the applicable procedure for transfer of Legacy Archive Media for use with HDxt.

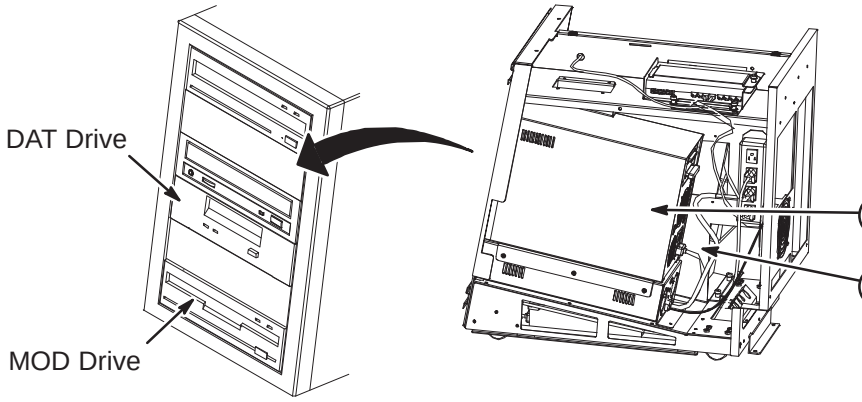
The Octane computer and legacy Media Devices will be retained as part of the upgrade. This upgrade would require a Monitor (M1000NT) and Keyboard (M1000MN).

8–3–1 DE–INSTALL OCTANE COMPUTER



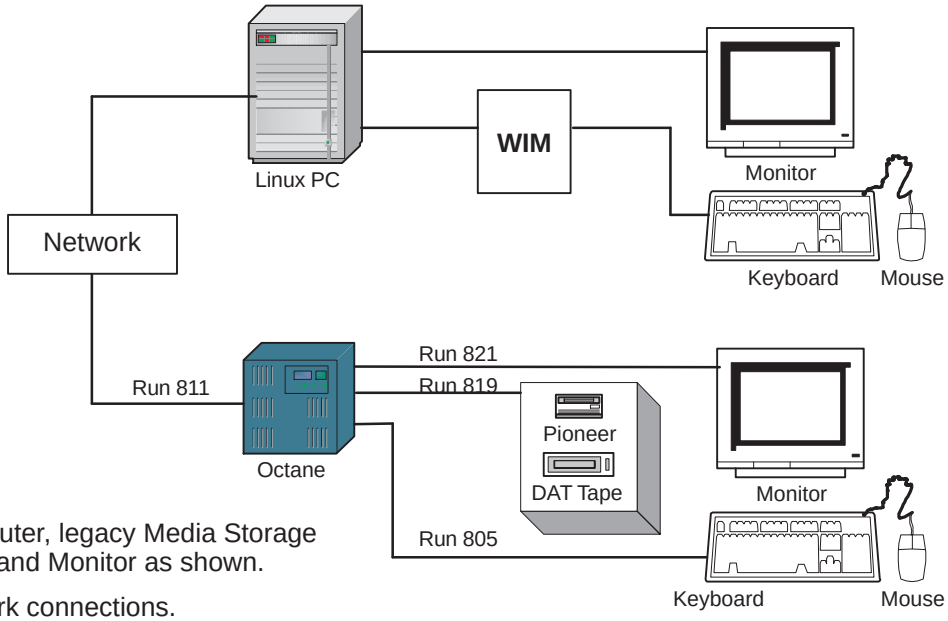
1. Move Octane Computer to suitable location determined by customer.
2. Retain:
   Ethernet cable Run 811
   Video cable Run 821
   Keyboard Cable Run 805
   and Power Cord.
3. SCSI Expansion Box in Duplex Computer Cabinet will also be moved to same location. Refer to Section 8–3–2 for removal instructions.

8–3–2 DE–INSTALL SCSI EXPANSION BOX AND CABLES



1. Remove SCSI Expansion Box from Duplex Computer Cabinet.
2. Retain data cable Run 819 and SCSI Power Cord.

8–3–3 HARDWARE CONFIGURATION INTERCONNECT



1. Connect the Octane Computer, legacy Media Storage Device, Keyboard, Mouse and Monitor as shown.
2. Connect power and Network connections.
3. After Octane computer is powered up and switched ON:
   1. Login as 'root'
   2. cd /export/home
   3. touch **.SHOW_MODE**
   4. Logout
   5. Login as 'signa'
4. The 'Octane' system will now boot up in **SHOW_MODE** which will enable to further setup the DICOM Networking to the different hosts to transfer images.

8–4 OPERATOR WORKSPACE UPGRADE

8–4–1 OPERATOR WORKSPACE TABLE INSTALLATION (If Needed)

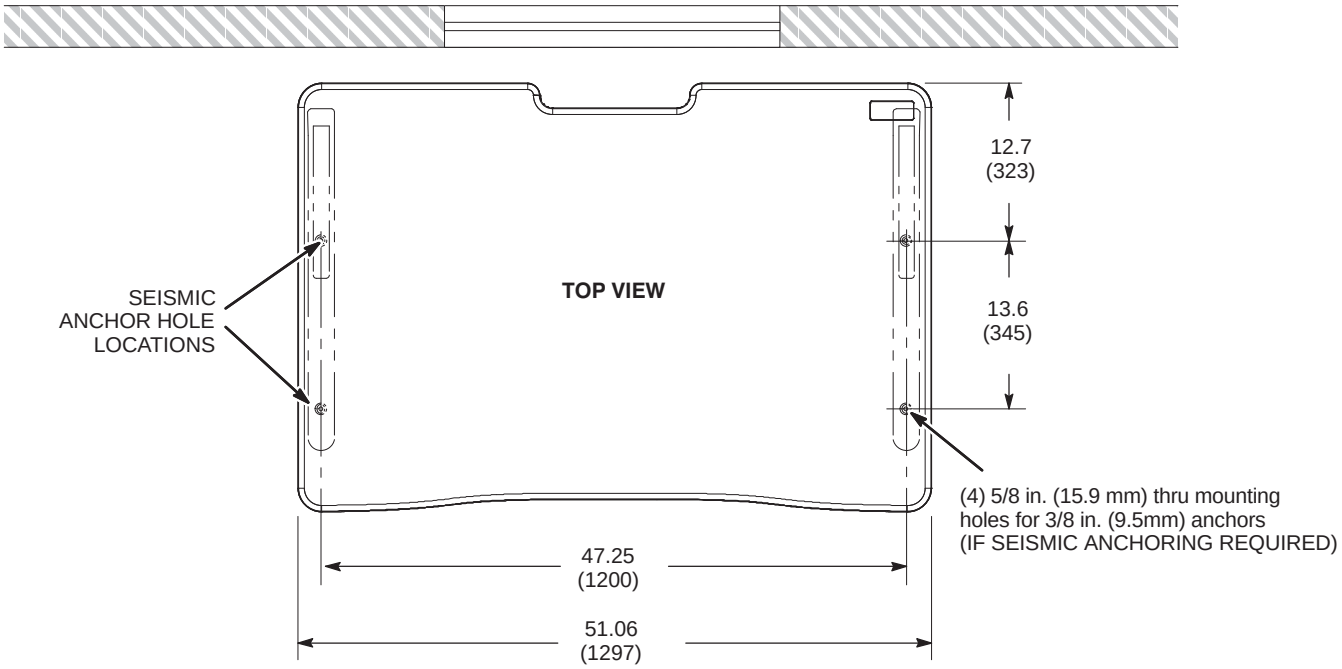
Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table.

1. Open palletized M1000MW box that contains the Operator Workspace table. Use furnished *Operator's Workspace Table Assembly Instructions* (5189137–50) to assemble the model 5189137 table.
2. Move table into designated position. Refer to room layout plan and/or consult with Site customer for exact location of Operator Workspace Table. Typical location with table in front of Magnet Room viewing window is shown below.

Note

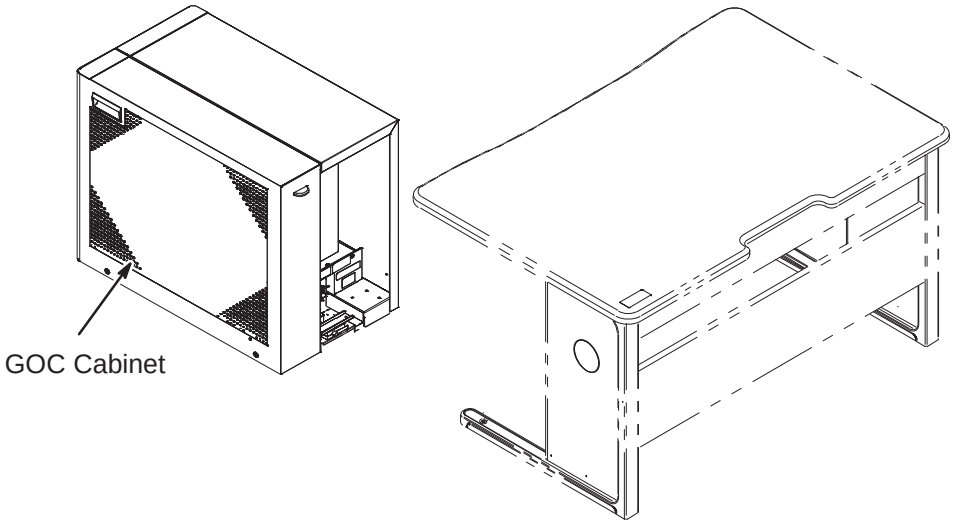
The design of the new model 5189137 table does not require anti–tip anchors. This table meets the ANSA/BIFMA anti–tip standards.

3. If seismic anchoring is required, refer to site architectural drawings and procure and install specified anchors.



8–4–2 POSITION COMPUTER CABINET

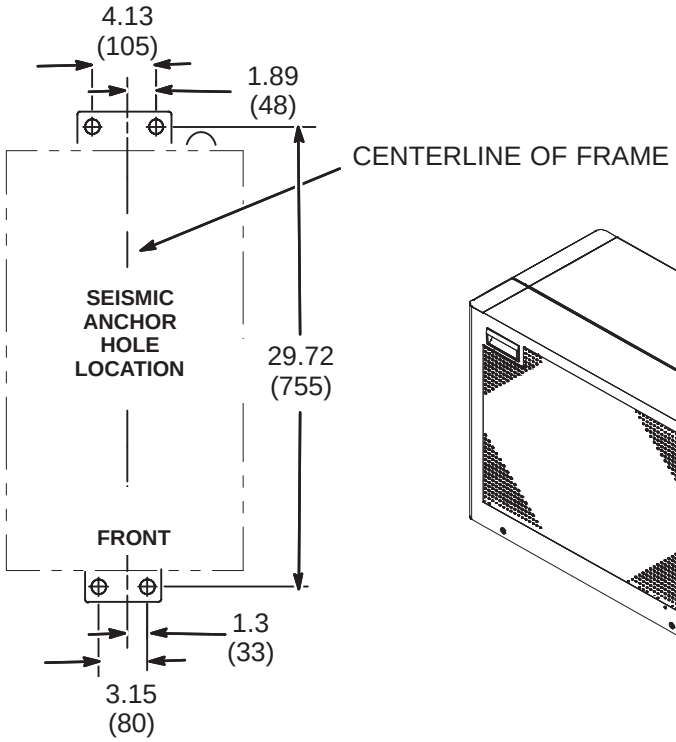
Consult with Site customer for exact location of Global Operator Console (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



8–4–3 SEISMIC ANCHORING (IF REQUIRED)

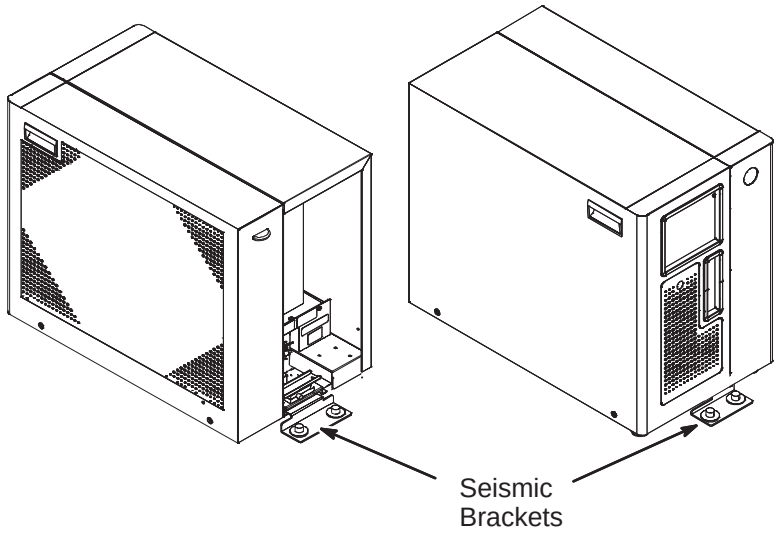
GOC Seismic Mount kit is shipped with GOC. Find kit inside of plastic bag.

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre–installation manual. Illustration below indicates hole location for installing brackets.



NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



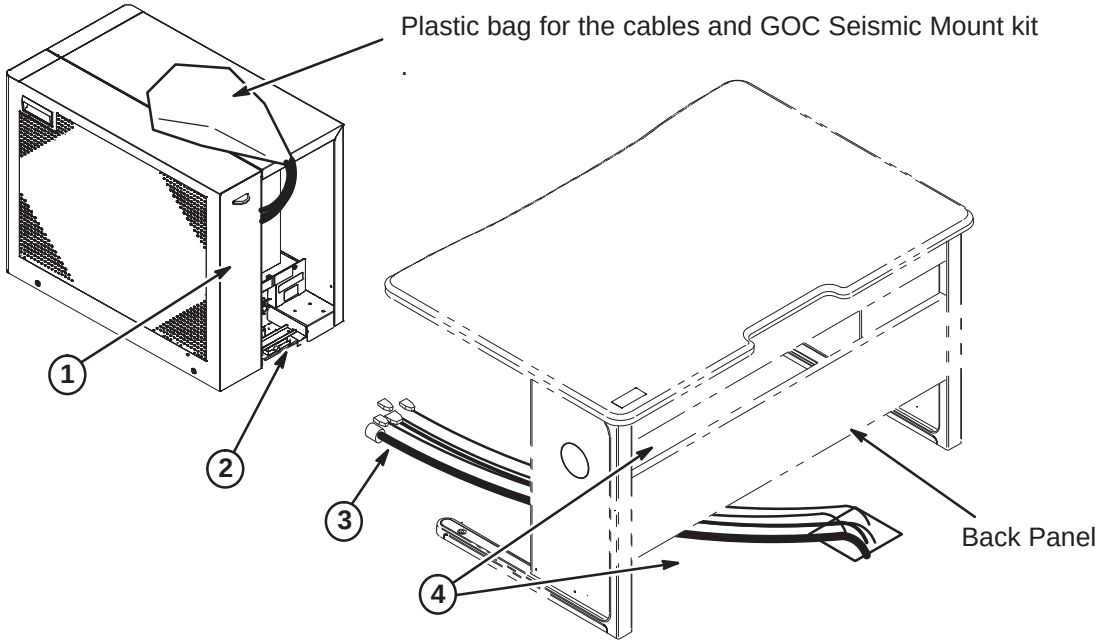
8–4–4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Two power cords
- One video ouput cable
- Keyboard output cable
- SCIM output cable
- Modem output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

1. Runs 1064, 1070, E0015, E3045, E3046, E3047 and E3048 are pre–connected to computer and are routed at top of opening.
   (These cables are inside of plastic bag)
2. Runs 1381, 1085, 1255, 1371, power cord for the Patient Alert system, and any other option cables will be routed thru lower opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.
4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the Back Panel. The cables coming out of the lower opening will be routed below the Back Panel.



8–4–5 GOC CABINET COVER REMOVAL

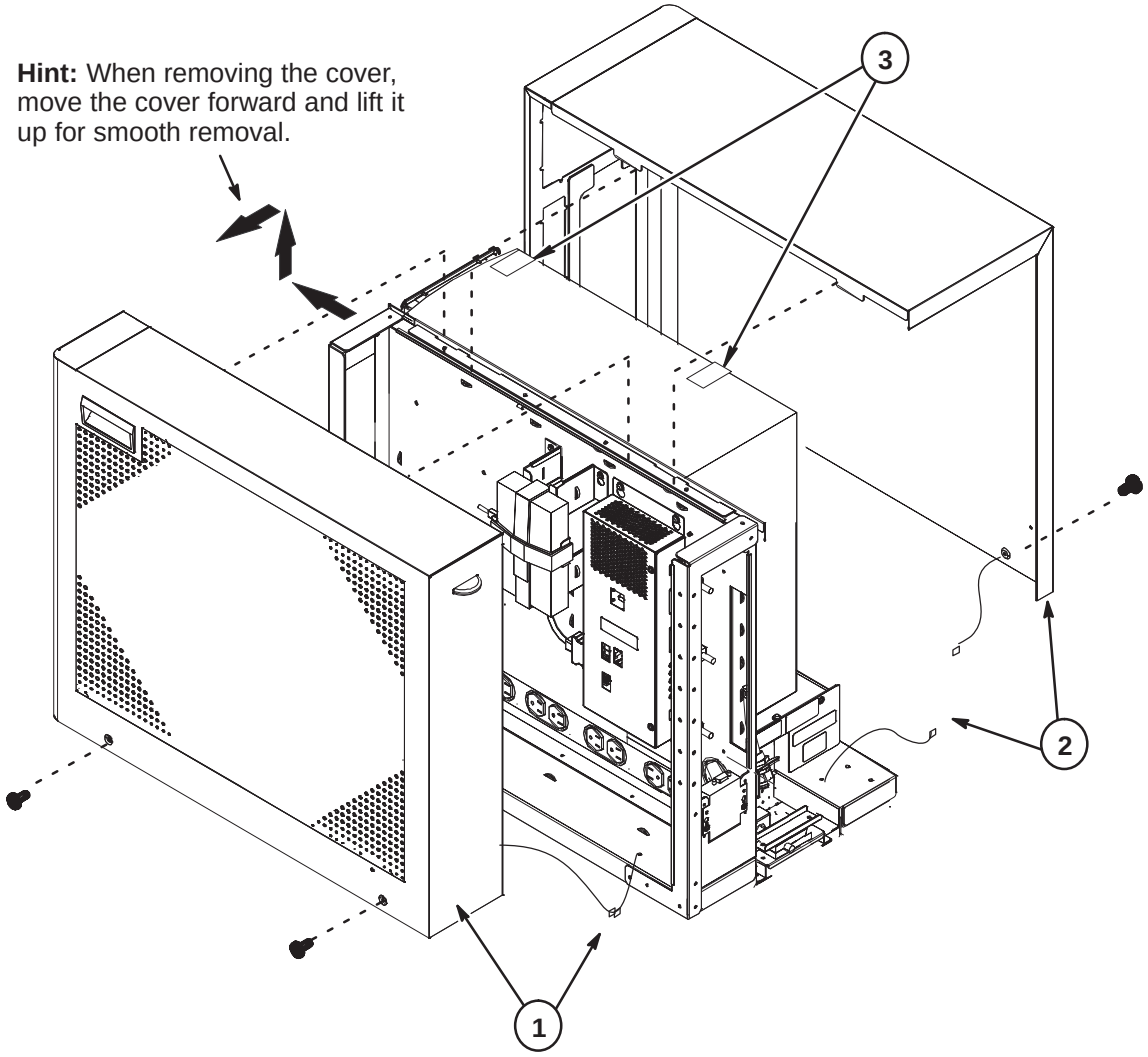
The Computer Cabinet covers need to be removed to access the cable connection points.

1. For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
2. For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

Note

A ground wire is connected to each side cover. Use care when removing side covers.

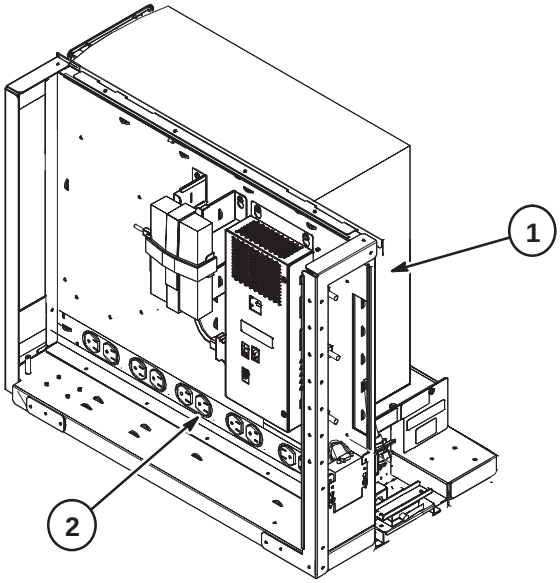
3. Peel the two shipping tapes off from PC.



8–4–6 INSTALL MODEM CABLE TO GOC CABINET (If Applicable)

If Modem is part of system installation or upgrade, it will be placed on Operator Table or GOC.

1. Modem cable to the Linux PC is provided. Connect Run E3045 to Modem.
2. Modem power cord must be connected to J7 on GOC PDU Module. Route this cable to the rear top of the GOC Cabinet.



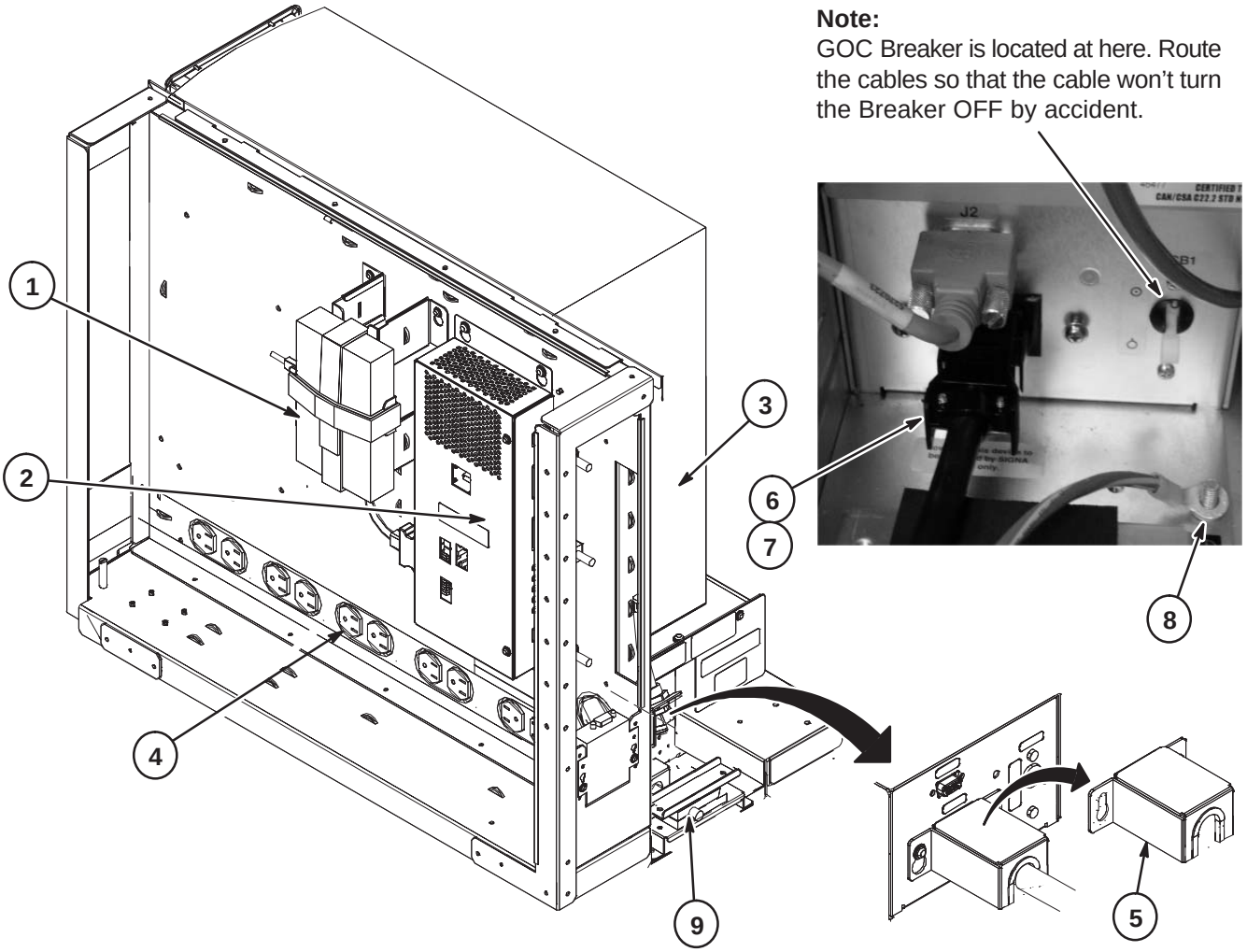
8–4–7 GOC CABLE CONNECTIONS

Refer to cable map for information on cables to be connected thru lower area of rear opening.

1. Route and connect Ethernet cables to Lan Hub.

Note Use ports 3, 7 or 8 for connecting site ethernet cables such as AW. Don not use any other ports for site ethernet cables.

2. Connect Run 1085 to J7 and Run 1255 to J18 on GOC AA.
3. Refer to Section 8–2–9 for all cables connected to Linux PC computer.
4. Refer to Section 8–2–8 for power cable connections to GOC PDU. Make sure Patient Alert power is connected to J3. If Modem is installed, power cable is connected to J7.
5. Before connecting GOC power cable Run 1381 to J1, loosen screws and remove cover.
6. Route and connect GOC power cable Run 1381 to J1 on PDU.
7. Confirm that GOC Power Cable is firmly connected to J1 by pulling the cable once. Re–install cover removed in Step 5 after connecting cable and tighten screws.
8. Connect Run 1382 to GND. Make sure nut securing GND cable is firmly tightened to GND stud.
9. Tighten clamp to secure cables in place.

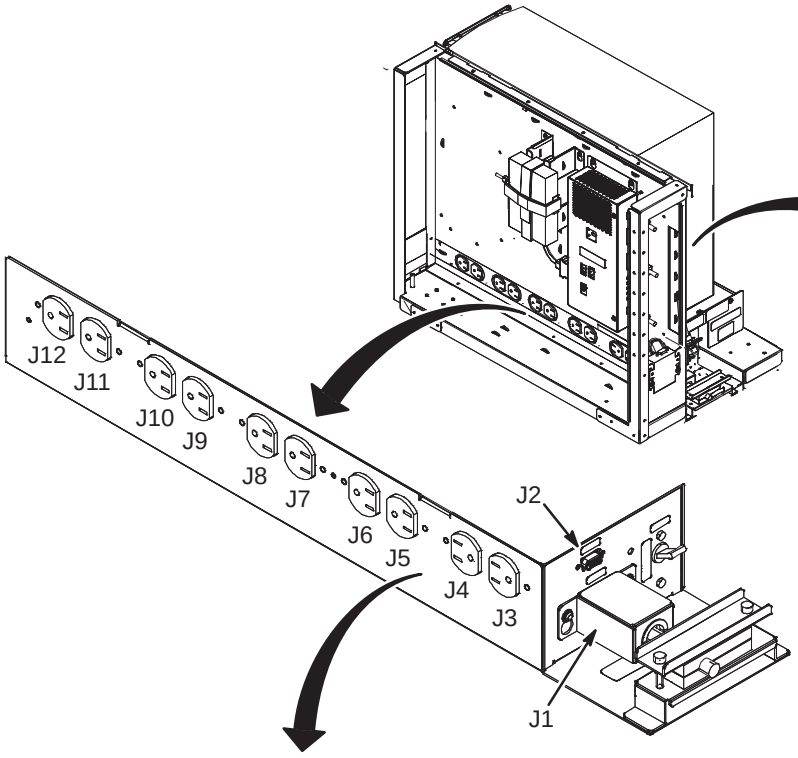


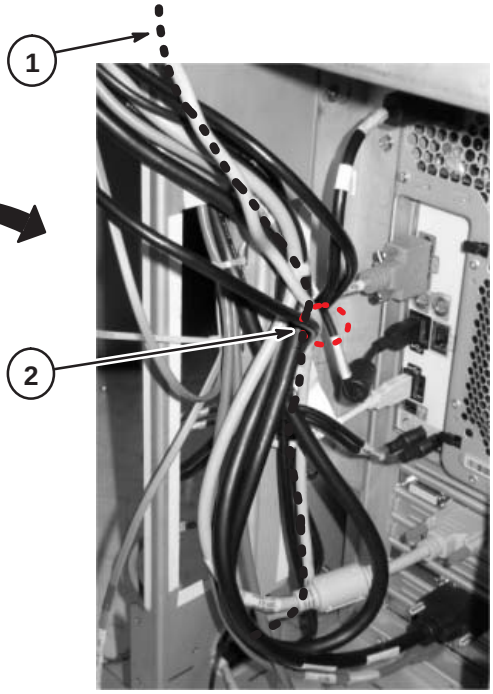
8–4–8 POWER CABLE CONNECTIONS

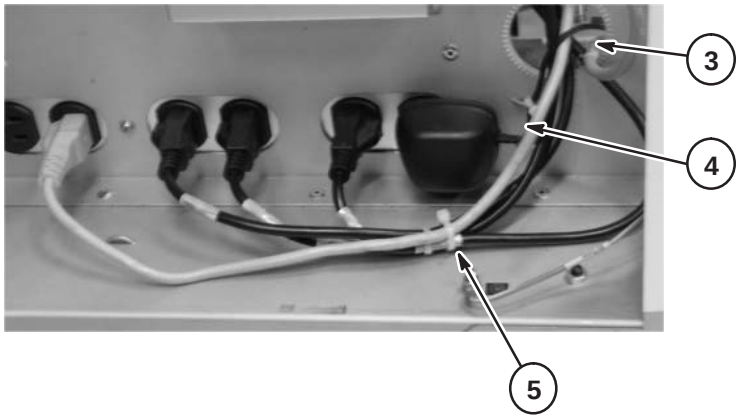
The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.

Also refer to cable map for information on power cable conections to OW1–PDM.

1. If Modem is installed, route power cable and connect to J7.
2. Tie–wrap point for Modem power cable.
3. Route power cables connected to J3 thru J7 through access hole.
4. route and connect Patient Alert power cable to J3 and tie–wrap at this point.
5. Tie–wrap power cables J4 to J7 at this point.





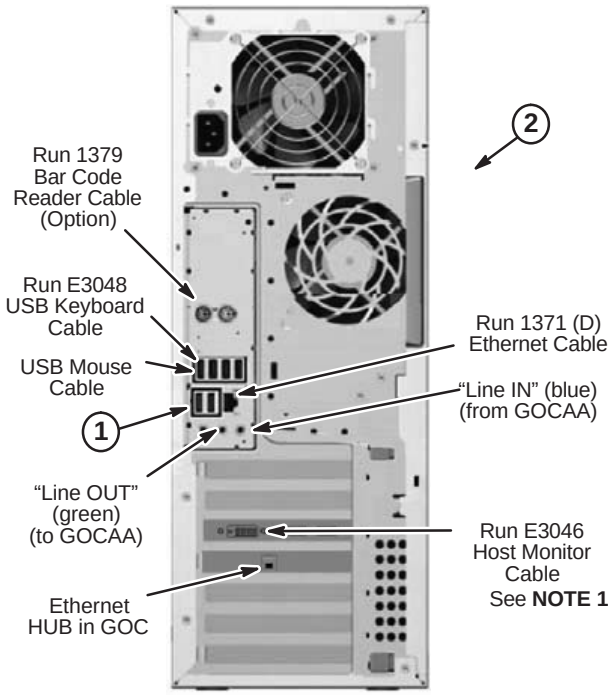


POSITION	NAME
J1	GOC Input
J2	GOCAA
J3	Patient Alert
J4	Host Computer
J5	Host LCD
J6	SCSI Tower
J7	Modem
J8	
J9	LAN Hub
J10	
J11	
J12	One Wire Hub

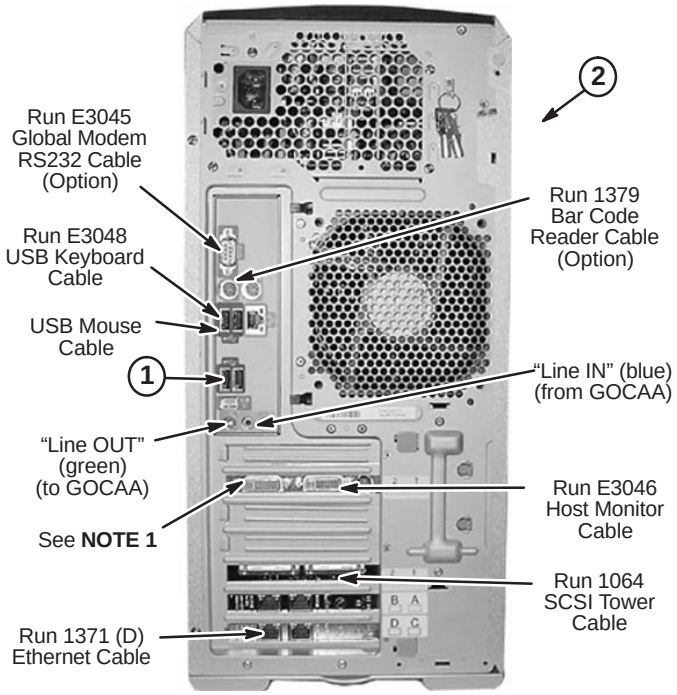
8–4–9 LINUX PC CABLE CONNECTIONS

The connection points of cables routed to the Linux PC are described in the following two sections. All cables shown are preconnected except for Run 1371 and USB cable for USB–Serial DB9 Adaptor which has to be routed and connected at site. Refer to the applicable PC for your site.

1. Refer to Section 8–2–10 for installing the USB to 1–Wire Adapter (5307784) to USB port.
2. Connect remaining cables as marked at rear of computer.



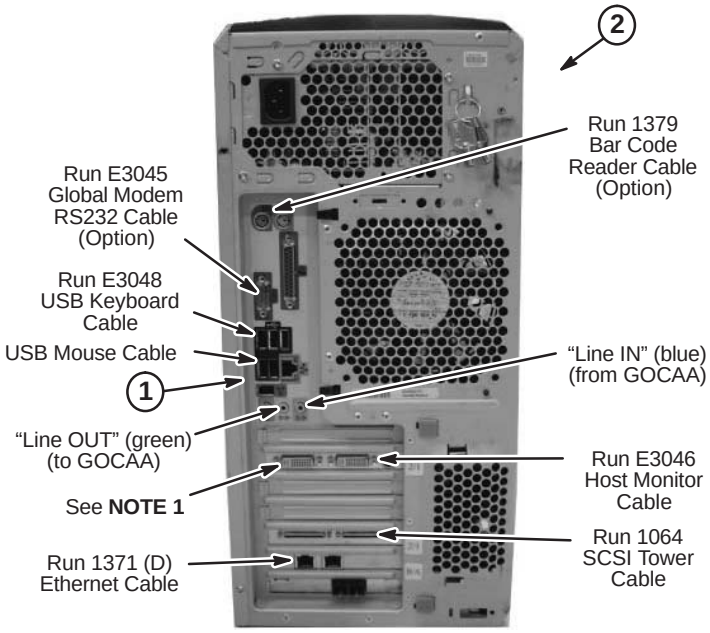
Z400
See **NOTE 2**



XW9300

NOTE 1:
Do not connect any additional monitors prior to loading operating system software. Doing so will cause boot–up to stop.

NOTE 2:
SCSI Tower and DASM are not supported for Z400PC.



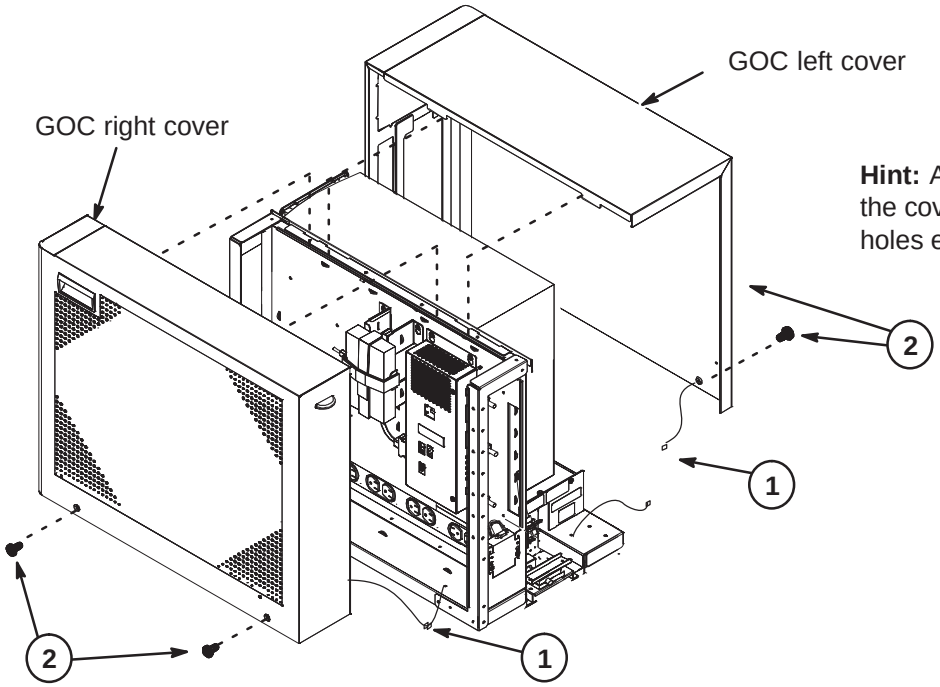
XW8400

8–4–10 INSTALL TEMPERATURE AND HUMIDITY SENSOR COMPONENTS

The components of the Temperature and Humidity Sensor Kit (5147136–2) will be installed on the back of the GOC cabinet and on the back of the System Cabinet with one cable routed from the GOC and connected to it.

8–4–10–1 RESTORE GOC COVERS

1. Connect Ground cables and install both side covers by inserting tabs at top of covers into slots.
2. Install screws that were removed in earlier procedure



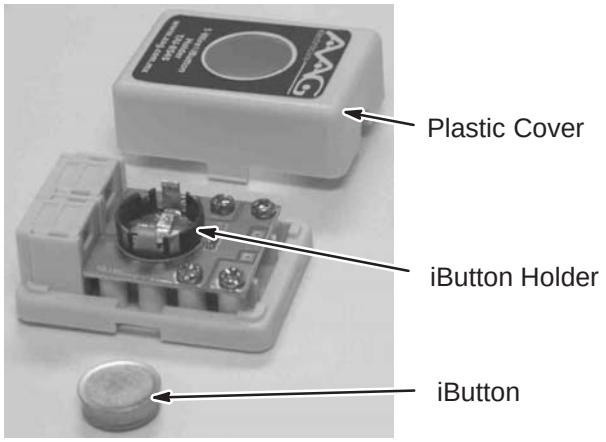
Hint: After restoring the covers, push the covers backward to find the screw holes easily.

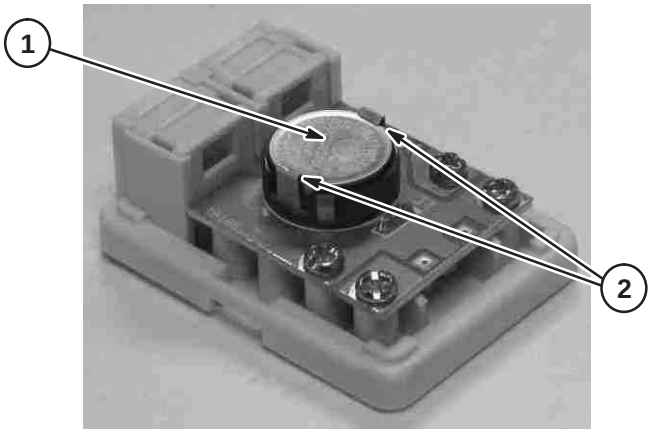
8–4–10–2 INSTALL iBUTTON INTO HOLDER

Install the iButton into the holder in the orientation shown in the illustration.

Note: The pictures below show the plastic cover removed although this is not necessary to install the iButton**.**

1. Install the iButton in the holder so that the solid part of the case is exposed.
2. Make sure the “fingers” latch over the top of the iButton and secure it in the holder.
3. If removed, reinstall the Plastic Cover.



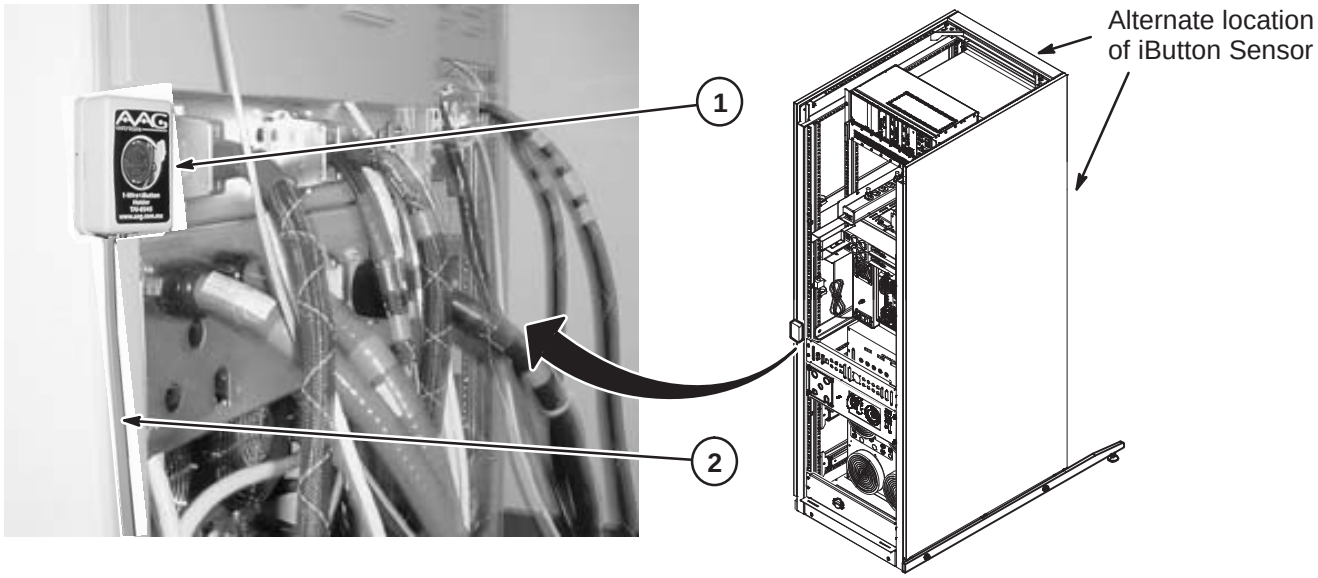


8–4–10–3 INSTALL TEMPERATURE & HUMIDITY SENSOR ON SYSTEM CABINET

Note: iButton Holders are unique to each location. Make sure to install 5141857–4 to System Cabinet and 5141857–5 to GOC.

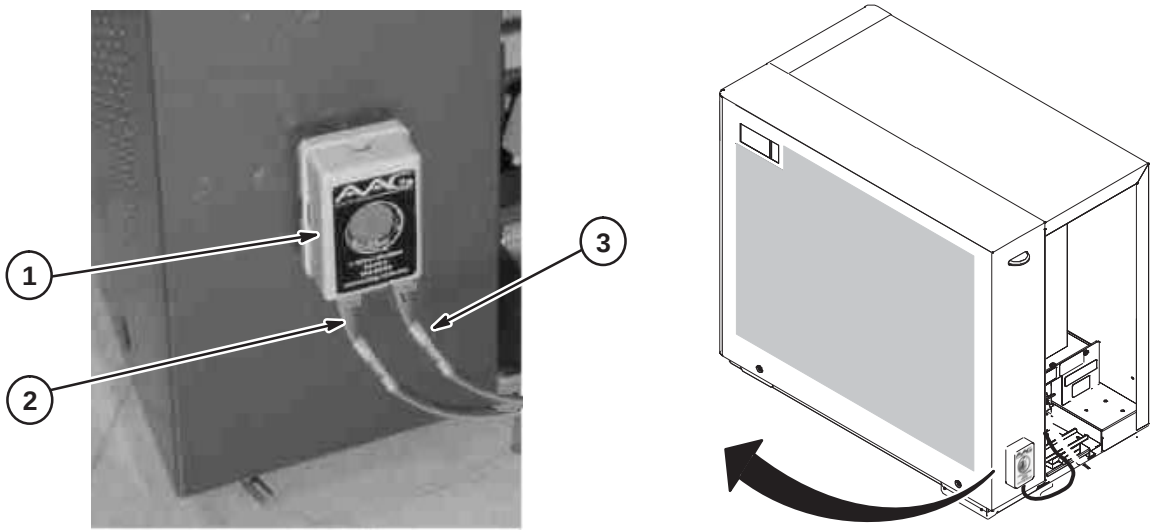
Alternate Location: For Mobile sites, or other sites where forced air may blow directly on the sensor which may cause false readings, the iButton Sensor can be located at the front or top of the System Cabinet.

1. Attach iButton Sensor assembly (Holder, 5141856, and Humidity iButton (5141857–4), to back of System/RFS Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779–2).
2. Connect cable Run 1384 (5143794–2) to Sensor. Route and to iButton at rear of GOC.



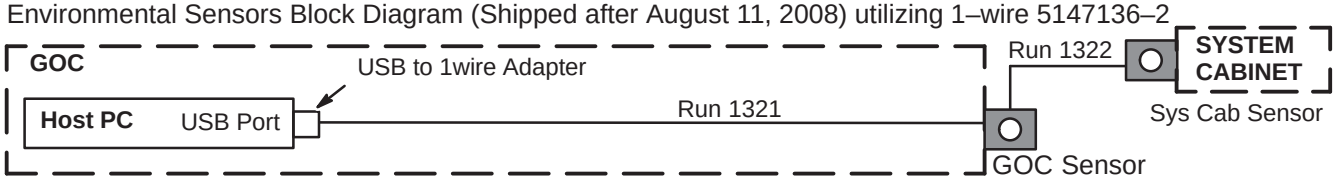
8–4–10–4 INSTALL TEMPERATURE & HUMIDITY SENSOR ON GOC

1. Attach iButton Sensor assembly (Holder, 5141856, and iButton 5141857–5), to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779–2).
2. Route and connect Run 1322 (5143794) from System cabinet to Sensor at back of GOC.
3. Connect cable Run 1321 from Sensor to USB to 1–Wire Adapter (5307784) installed in any open USB port at rear of PC. Refer to step 1 in Section 8–2–9.

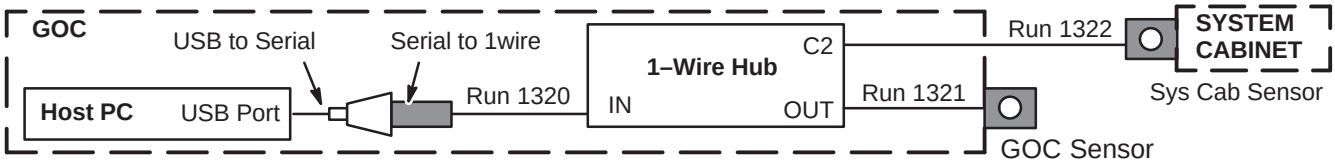


8–4–10–5 TEMPERATURE & HUMIDITY SENSOR WIRING SCHEMATICS

Environmental Sensors Block Diagram (Shipped after August 11, 2008) utilizing 1–wire 5147136–2



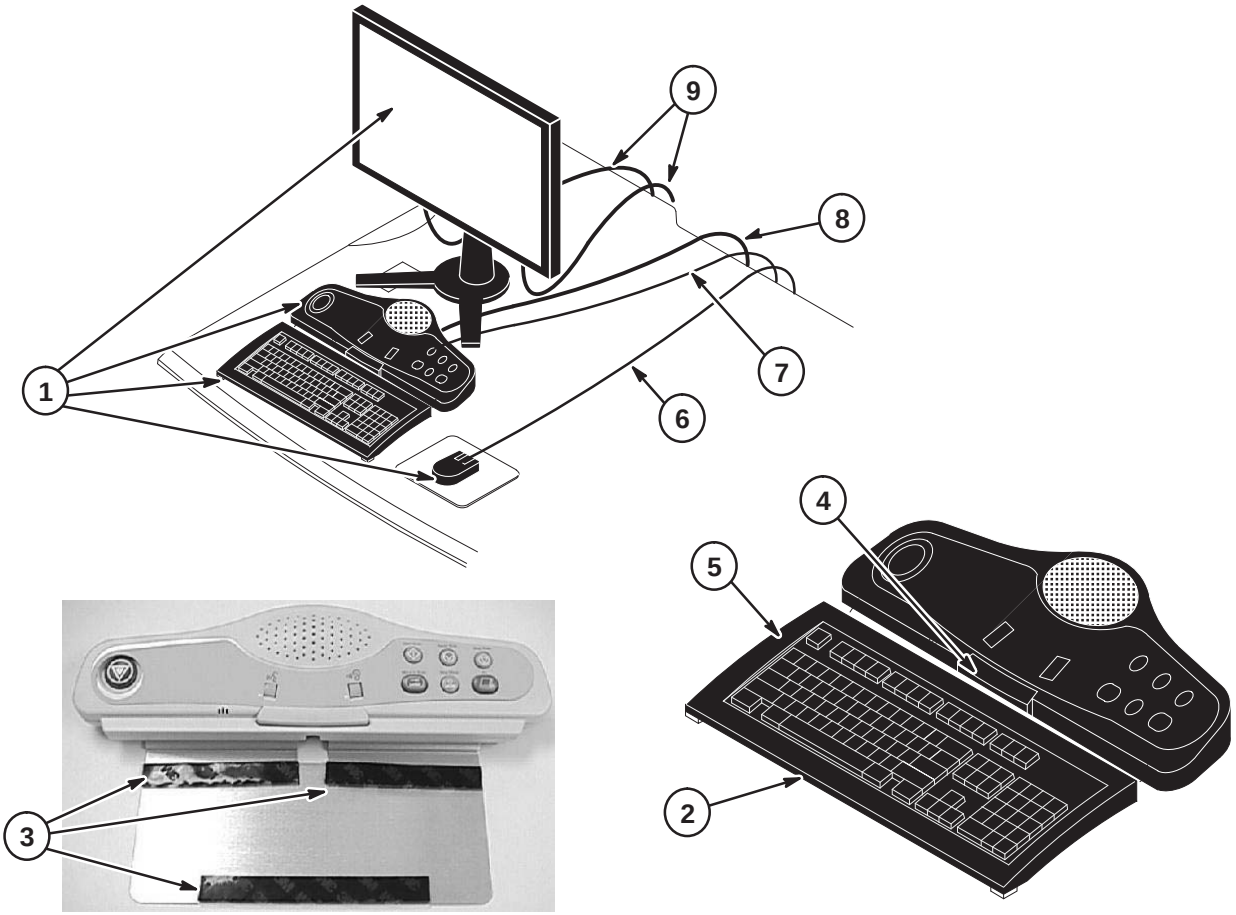
Environmental Sensors Block Diagram (Shipped before August 11, 2008) utilizing 1–wire 5147136



8–4–11 INSTALL MONITOR AND SCIM & KEYBOARD CABLES

Consult with Site customer for exact location of SCIM/Keyboard and Monitor on Operator Workspace (OW) Table.

1. Position Host Display Monitor, SCIM, Mouse, and keyboard on table.
2. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
3. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface.
4. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
5. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.
6. Connect Mouse cable to USB port at rear of computer in GOC.
7. Connect Run E3048 to USB Keyboard cable. Route and connect other end to computer in GOC.
8. Connect Run E3047 to SCIM. Route and connect other end to computer in GOC.
9. Route power cord Run E0015 and Run E3046 from GOC and connect to Display Monitor.



Note

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

8–4–12 INSTALL SCSI TOWER and MODEM (Option)

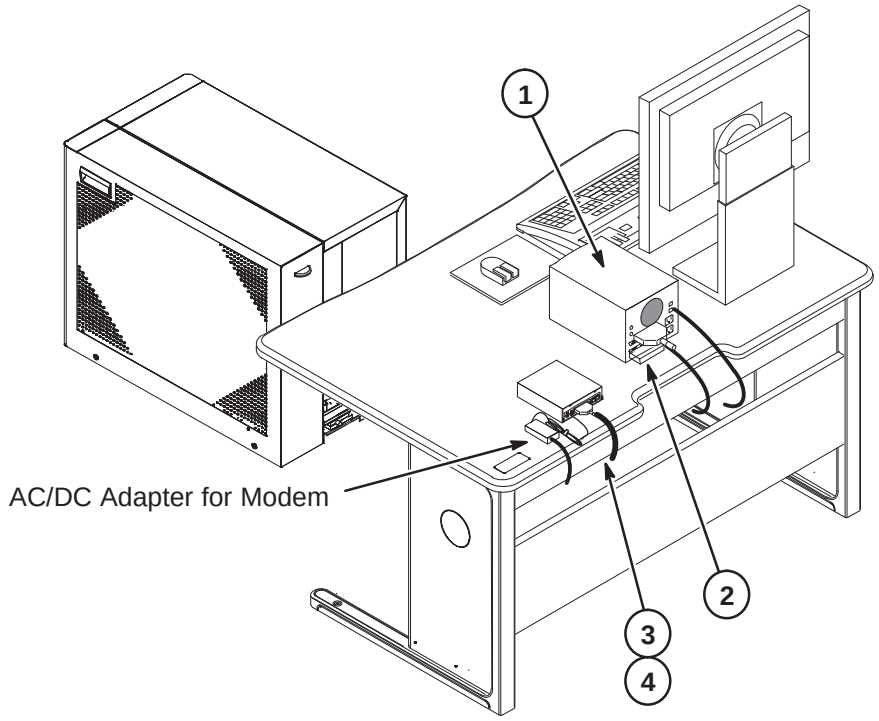
Note: SCSI Tower is an option with release 15.x or HDxt.

Note: SCSI Tower and Modem are not compatible with Z400 computer. The Z400 computer will be delivered and installed at sites around the 4th quarter of 2010.

1. 64 and 1070 from GOC Cabinet.
2. Install terminator on back of SCSI Tower.

The following steps are for the Modem option:

3. If applicable, place the Modem on the OW table and connect Runs E3045 and power cable from GOC Cabinet.
4. Connect customer supplied telephone line.



8–4–13 FINAL POSITIONING OF GOC CABINET

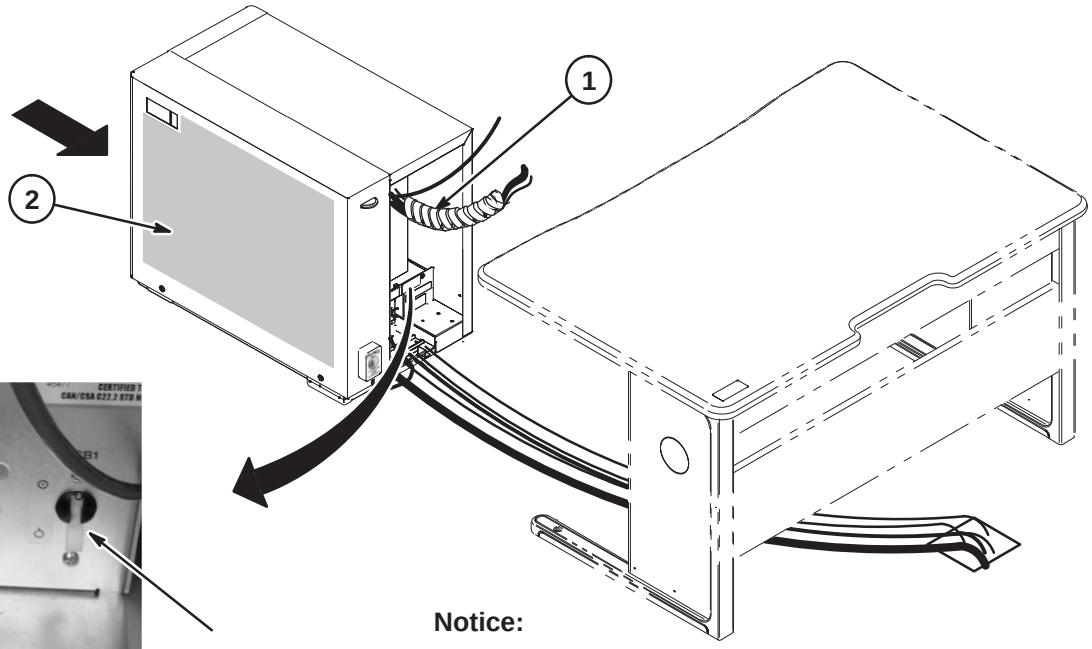
1. Use the spiral tube to bind the following cabes. (Length of spiral tube is about 1 m) Spiral tube is included in the plastic bag which is shipped with GOC.

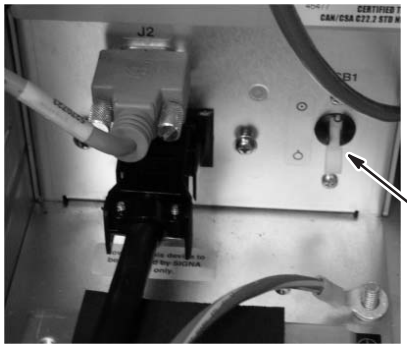
Notice

Do Not include Modem power cable in Spiral Tube.

- Three power cords (LCD Monitor ,SCSI Tower(if applicable))
- One video cable
- Keyboard cable
- SCIM cable
- SCSI Tower cable (if applicable)
- Modem cable (if applicable)

2. Place cabinet in final position as determined by customer.





Notice:

GOC circuit breaker location. To avoid turning the Breaker OFF by accident, do not place the cables inside of GOC Cabinet near the circuit breaker.

8–4–14 POWER ON INFORMATION

IMPORTANT

When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note

When powering up the GOC, there is a Main breaker located in the rear of the GOC.

SECTION 9 – HFD/PDU CABINET

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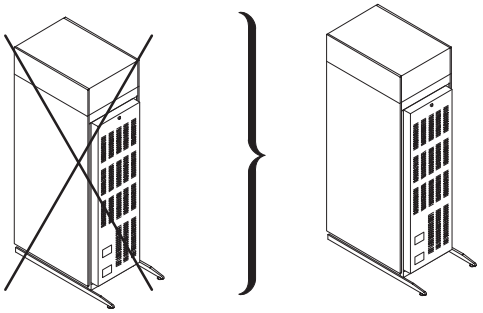
9–1 OVERVIEW 9–2
9–2 POSITION HFD/PDU CABINET 9–3
9–3 INSTALL FAN ASSEMBLY AND ATTACH CABINET IDENTIFICATION MAGNET 9–4
9–4 INPUT VOLTAGE SELECTION 9–5
9–5 CIRCUIT BREAKER DIP SWITCH SETTINGS 9–5
9–6 CABLE CONNECTIONS TO HFD/PDU CABINET 9–6

9–1 OVERVIEW

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

The HFD/PDU Cabinet will replace the ACGD/PDU cabinet shown below.



ACGD/PDU CABINET (MR3)

HFD/PDU CABINET (MR3)

The ACGD/PDU cabinet will be returned for recycling. Refer to applicable cable removal map and applicable column in cable removal Table 4–1 for disconnection and removal of designated cables for system being de–installed.

Positioning of cabinets is critical for installing cables and post–installation system operation. Refer to architectural site layouts for:

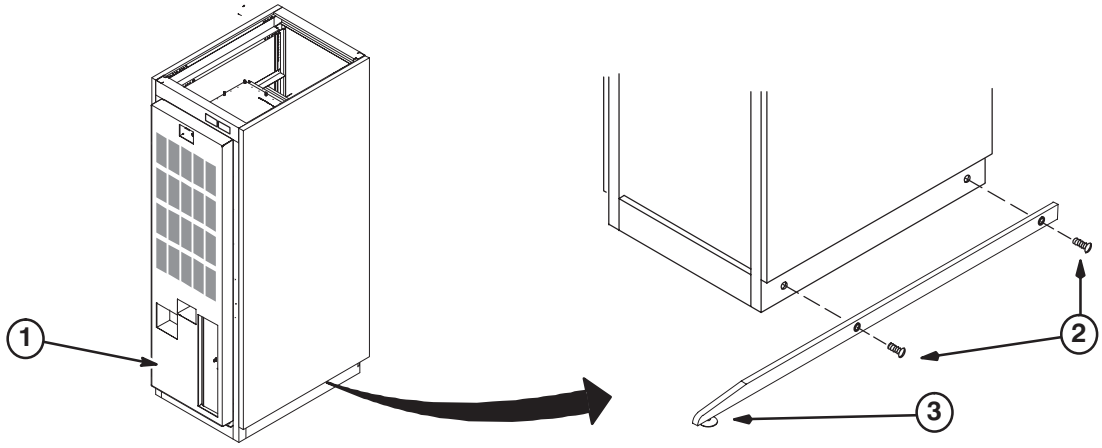
- Location of all cabinets and related equipment.
- For seismic anchoring (where required)

9–2 POSITION HFD/PDU CABINET

WARNING!

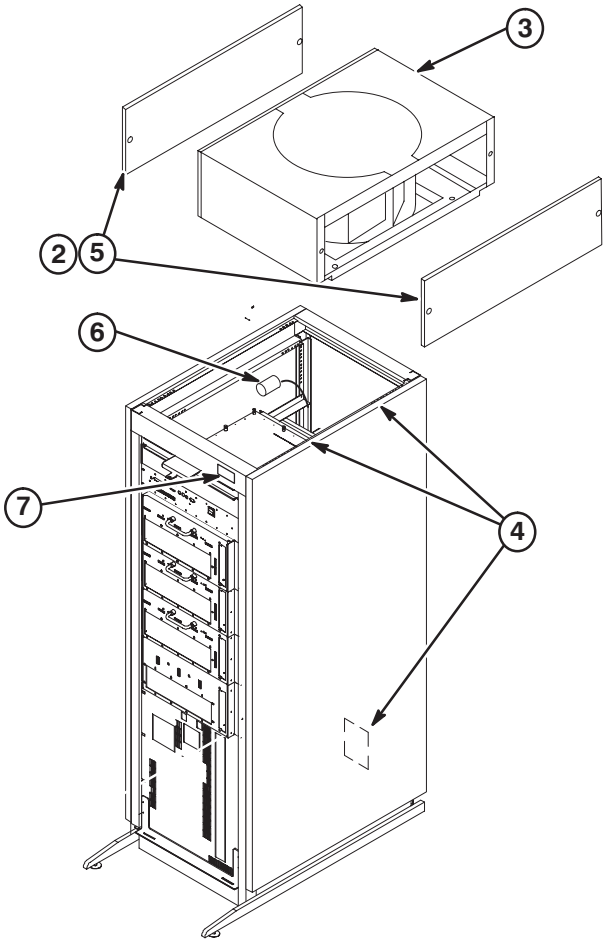
DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

1. Move cabinet to final position as indicated by site layout drawings.
2. Attach the anti–tip bars to cabinet. (Anti–tip bars are shipped separately)
3. Adjust leveling pads to level cabinet.



9–3 INSTALL FAN ASSEMBLY AND ATTACH CABINET IDENTIFICATION LABEL

1. Remove Fan Assembly from shipping container.
2. Remove two side trim panels (two screws from each).
3. Requiring two people, place the Fan Assembly on top of cabinet with the cable connector facing the rear of the cabinet.
4. Secure Fan Assembly in place with (4) four screws and washers, two on each side of cabinet. The fasteners are provided in a plastic bag located in PDU module. Rear PDU covers will need to be removed for access to plastic bags with fasteners.
5. Install two side trim panels with previously removed screws.
6. Connect cable from SGA Power Supply to the fan connector.
7. Make sure the “GRADIENT/PDU” cabinet magnet, part of the Cabinet ID Label Sheet (5126718), is attached to the front of cabinet as shown.

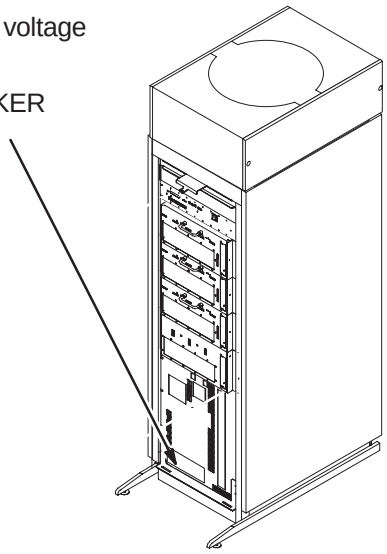


DANGER!!

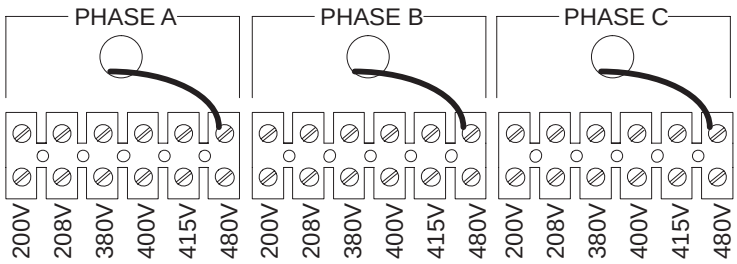
LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

9–4 INPUT VOLTAGE SELECTION

1. Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
2. Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
3. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



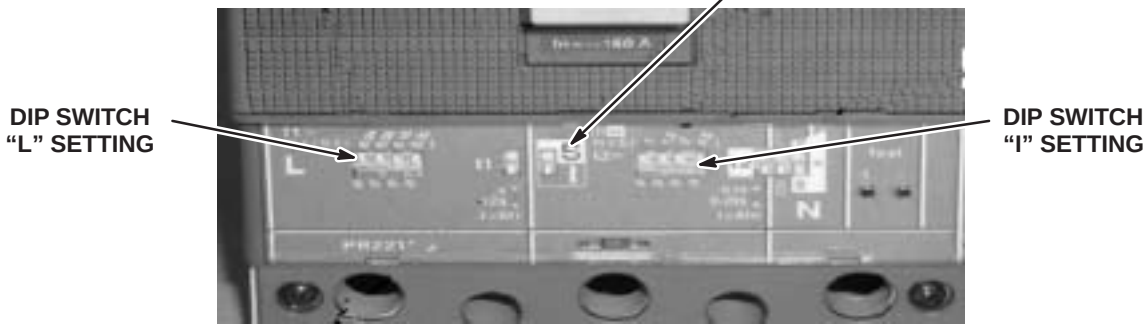
(SHOWN WIRED FOR 480V)



9–5 CIRCUIT BREAKER DIP SWITCH SETTINGS

1. Table below shows the position of the DIP switches for each input voltage selection. Set DIP switches according to input voltage. The white portion indicates the switch.

2. The “S/I” switch must be in the “I”, down position



INPUT VOLTAGE	200	208	380	400	415	480
"L" SWITCH SETTINGS						
"I" SWITCH SETTINGS						

9–6 CABLE CONNECTIONS TO HFD/PDU CABINET

Refer to applicable cable “added” map in Section 4 for information on connecting new and retained cables to cabinet. The gradient cables will be reconnected to HFD/PDU Cabinet. The other ends of the retained gradient cables will be connected to the new Penetration Panel.

SECTION 10 – RFS CABINET

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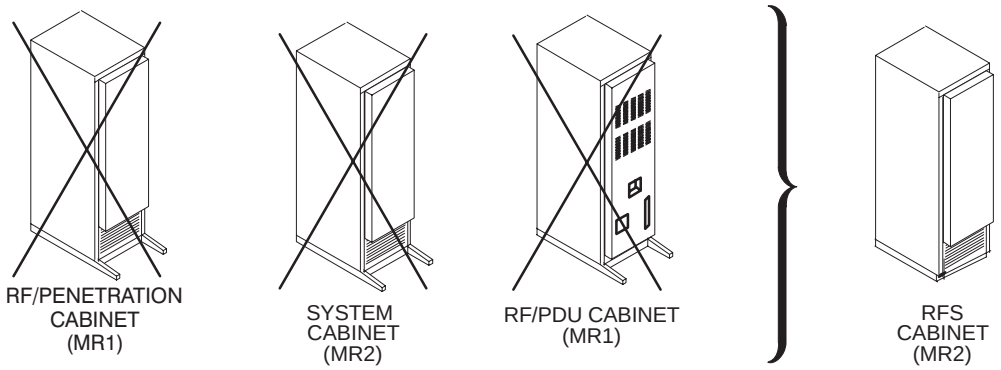
10–3 CABLE CONNECTIONS TO RFS CABINET 10–3

10–1 OVERVIEW

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

The RFS Cabinet will replace one or more of the cabinets shown below.



All de–installed cabinets are to be returned for recycling. Refer to applicable cable removal map and applicable column in cable removal Table 4–1 for disconnection and removal of designated cables for system being de–installed.

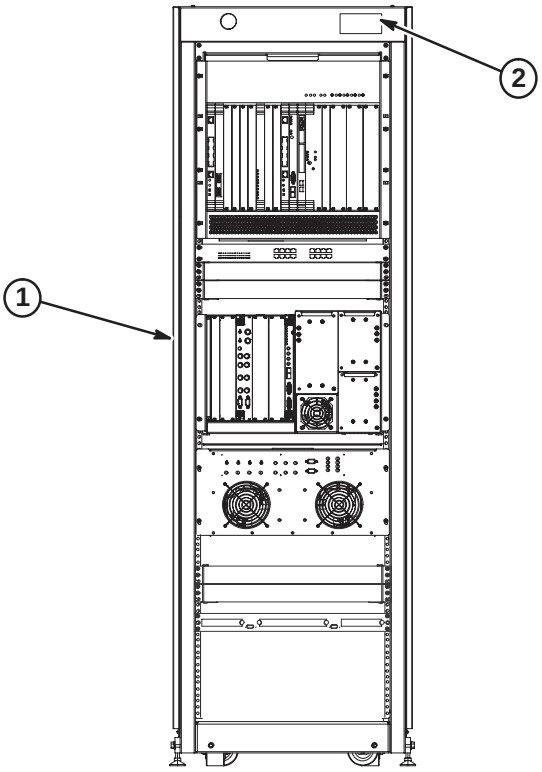
Positioning of cabinets is critical for installing cables and post–installation system operation. Refer to architectural site layouts for:

- Location of all cabinets and related equipment.
- For seismic anchoring (where required)

10–2 POSITION RFS CABINET

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

1. Position RF/System cabinet to final position as indicated by site layout drawings.
2. From the Cabinet ID Label Sheet (5126718), locate the ”RF/SYSTEMS CABINET” magnet and attach to front of cabinet as shown.



10–3 CABLE CONNECTIONS TO RFS CABINET

Refer to applicable cable “added” map in Section 4 for information on connecting new and retained cables to cabinet.

SECTION 11 – ACCESSORIES CABINET

SECTION **PAGE**

RF AMPLIFIER CABINET

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11–2 REMOVE MODULES FROM ACCESSORIES CABINET 11–2
11–3 REAR VIEW OF ACCESSORIES CABINET BEFORE UPGRADE 11–3
11–4 REAR VIEW OF ACCESSORIES CABINET AFTER REMOVAL OF MODULES & CABLES . . 11–3
11–5 INSTALL SHIM POWER SUPPLY 11–4

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

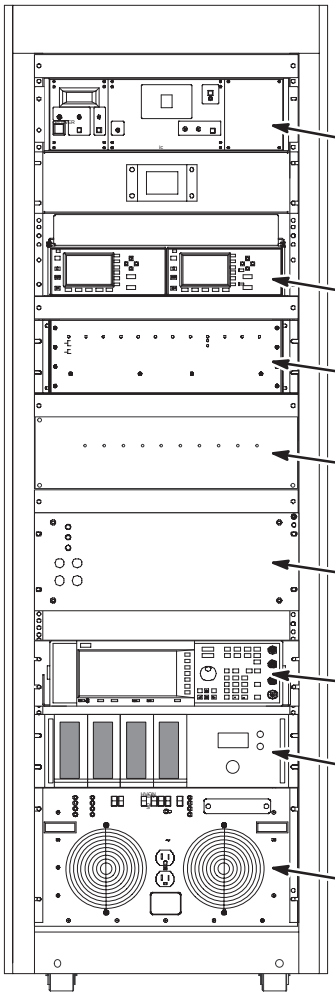
11–1 INTRODUCTION

The Accessories Cabinet will be modified for the upgrade to EXCITE HD. Refer to Section 4 for additional information and instructions on disconnecting, removing, and/or relabeling of cables when upgrading to EXCITE HD.

11–2 REMOVE MODULES FROM ACCESSORIES CABINET

The existing Accessories Cabinet pre–upgrade set–up may differ slightly from site to site. Spectroscopy cables may or may not be present. Spectroscopy cables are not shown in the System cable map in Section 4. Refer to 2.5kW MNS Option cable map on page 4–14.

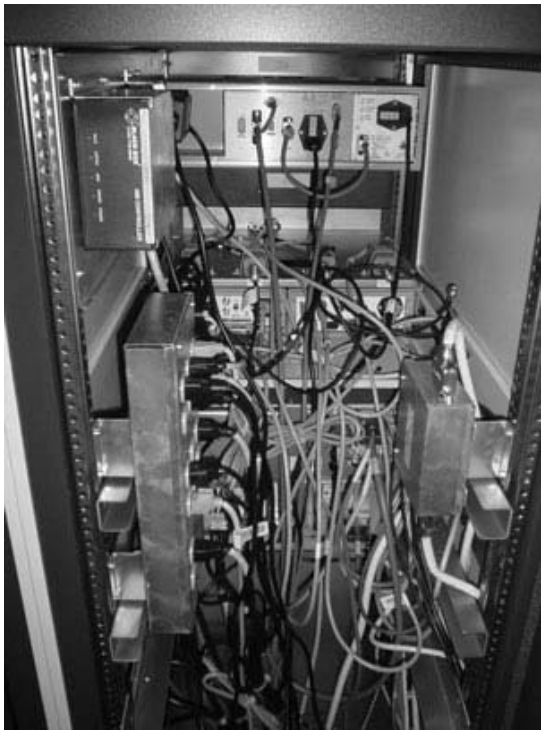
The illustration below indicates what modules are to be removed or retained.



Note: All removed modules and cables are to be returned for recycling.

1. Location of Helium (Cryogen) Meter. Retain if already installed in cabinet. (If not installed in the Accessories Cabinet, retrieve from currect location for transfer to Accessories Cabinet)
2. Remove Power Monitor and associated cables.
3. Remove 3T Control Module and associated cables.
4. Remove DRSM and associated cables.
5. If part of existing cabinet, remove Up/Down Converter module and associated cables.
6. If part of existing cabinet, remove HP Signal Generator module and associated cables.
7. High Order Shim Power Supply. Retain if already installed. (If not installed in the Accessories Cabinet, retrieve from currect location for transfer to Accessories Cabinet)
8. Remove SSM and associated cables.
9. Remove blank panels and save mounting hardware for later use.

11–3 REAR VIEW OF ACCESSORIES CABINET BEFORE UPGRADE

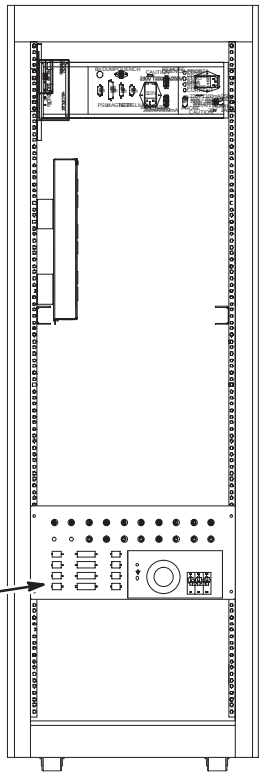


11–4 REAR VIEW OF ACCESSORIES CABINET AFTER REMOVAL OF MODULES AND CABLES

NOTE: Refer to Section 4 for cable routing and connection information.

1. Make sure Power Monitor mounting plate and brackets are removed.

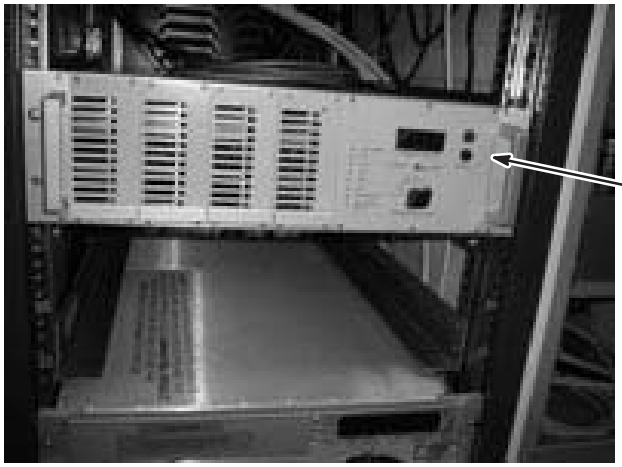
2. Disconnect external cables from interface panel as indicated by “affected” cable map in cable section.



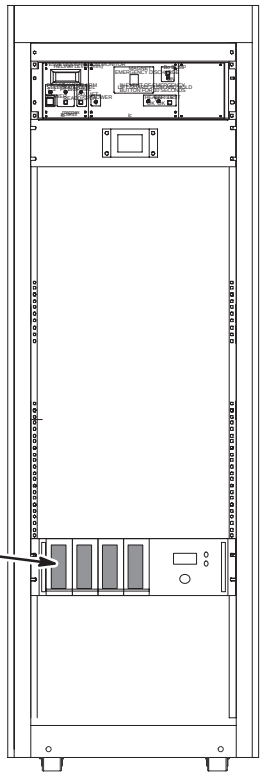
11–5 INSTALL SHIM POWER SUPPLY

The Shim Power Supply is already part of the system. For this upgrade it needs to be located in the lower part of the Accessories Cabinet.

The cables at the rear of the HOS remain the same before and after the ugprade except for one cable, Run 568, connected to J1 Comm 1. This cable will be replaced by Run 1380 which is part of the Signa Single Term Server Upgrade for RRI HOS Kit (M3335SA). Refer to cable map for routing and connection.



Install Shim Power Supply in this area.



SECTION 12 – RF AMPLIFIER CABINET

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RF AMPLIFIER CABINET

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12–2–3 INSTALL DUAL DIRECTIONAL MOUNTING PLATE 12–3
12–2–4 INSTALL HEAD AND BODY DUAL DIRECTIONAL COUPLERS 12–4
12–2–5 CONNECT CABLES TO DUAL DIRECTIONAL COUPLERS 12–5
12–3 25/35Kw RF AMPLIFER INSTALLATION 12–6

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 2 FOR POWER OFF PROCEDURES.

12–1 INTRODUCTION

Depending on site configuration, either the existing 25kW RF Amplifer will be modified or a new 25/35kW RF Amplifier will be installed.

If the site has a 25kW RF Amplifier Cabinet, it will be modified for the upgrade to EXCITE.

- Refer to Section 12–2 for step–by–step upgrade instructions.
- Refer to Section 4 for additional information and instructions on disconnecting, removing, and/or relabeling of cables when upgrading to EXCITE.

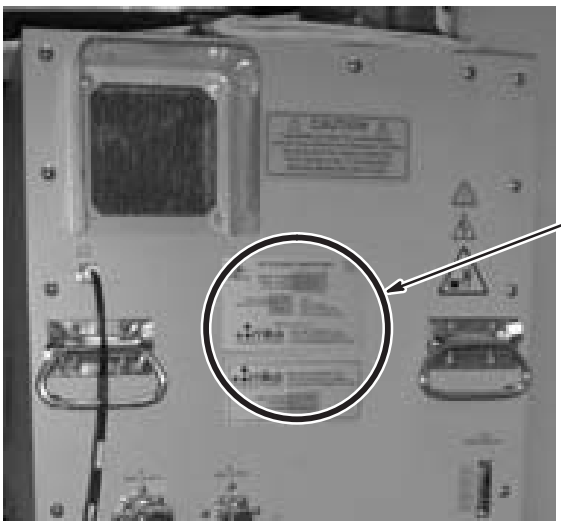
If the site is upgrading from a HEAD only system to HEAD and BODY system, a new 25/35kW RF Amplifier Cabinet will be installed.

- Refer to Section 12–3 for installation instructions.
- Refer to Section 4 for additional information and instructions on disconnecting, removing, and/or relabeling of cables when upgrading to EXCITE.

12–2 25Kw RF AMPLIFER UPGRADE INSTRUCTIONS

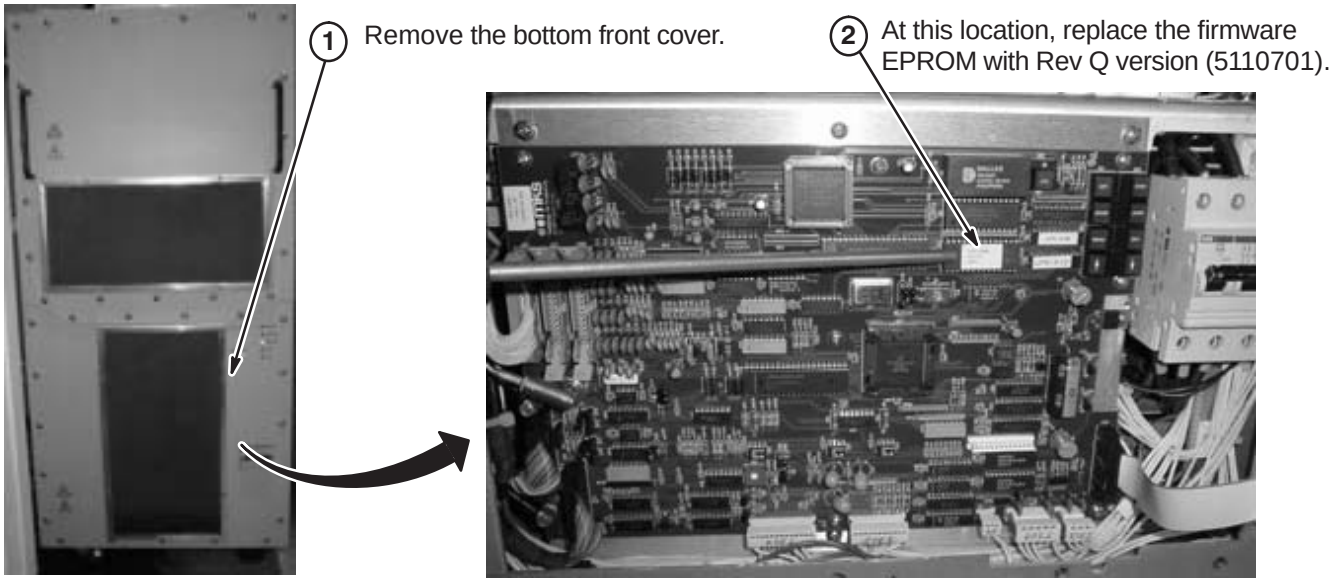
Complete the following procedures to modify an existing 25kW RF Amplifer for upgrading to EXCITE.

12–2–1 DETERMINE IF RF AMPLIFIER NEEDS FIRMWARE UPGRADE



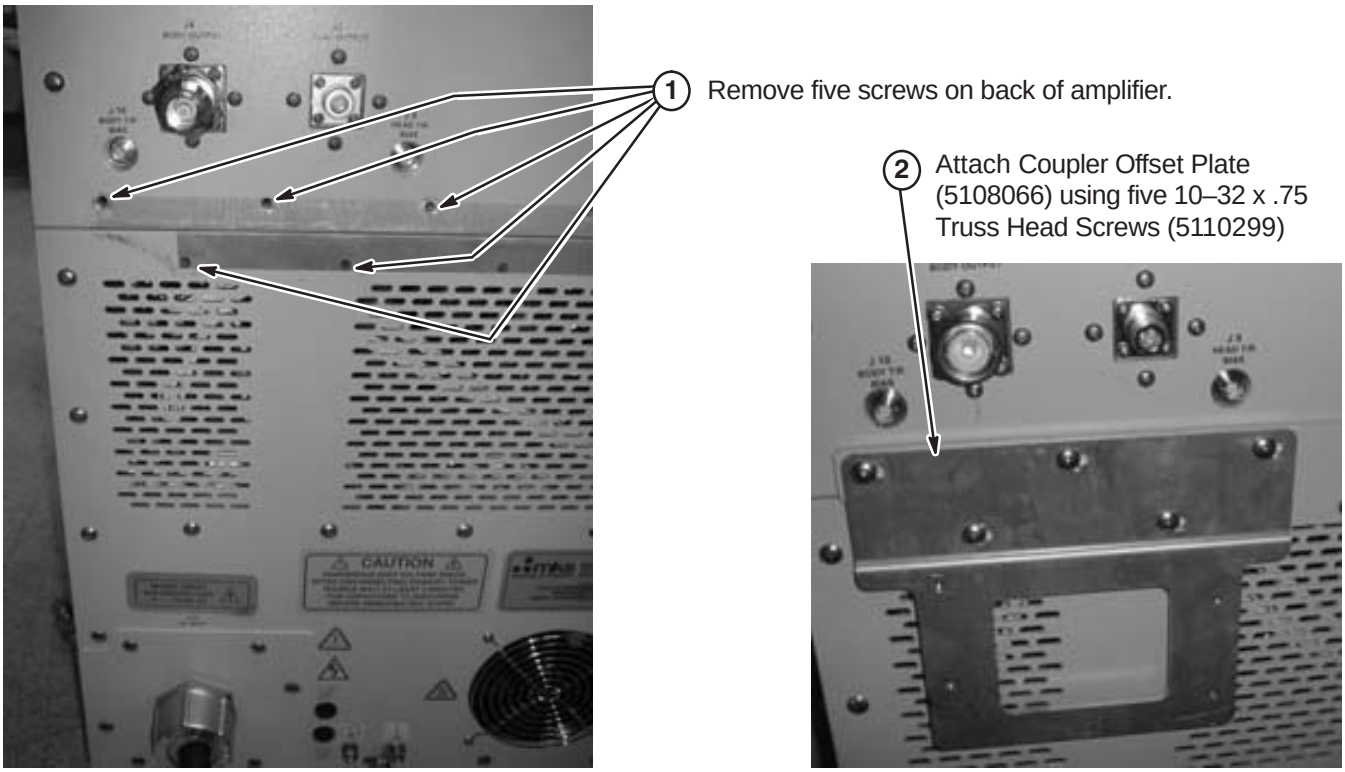
1. Check the amplifer serial number at back of cabinet.
2. If the serial number is any of the following:
   GE0136003
   GE0151025
   GE0205028
   GE0208031
   GE0209032
   GE0213038
   - **DO NOT** complete Section **12–2–2**.
   - **DO NOT** power up system.
   - Contact OLC immediately for required hardware updates.

12–2–2 INSTALL NEW EPROM



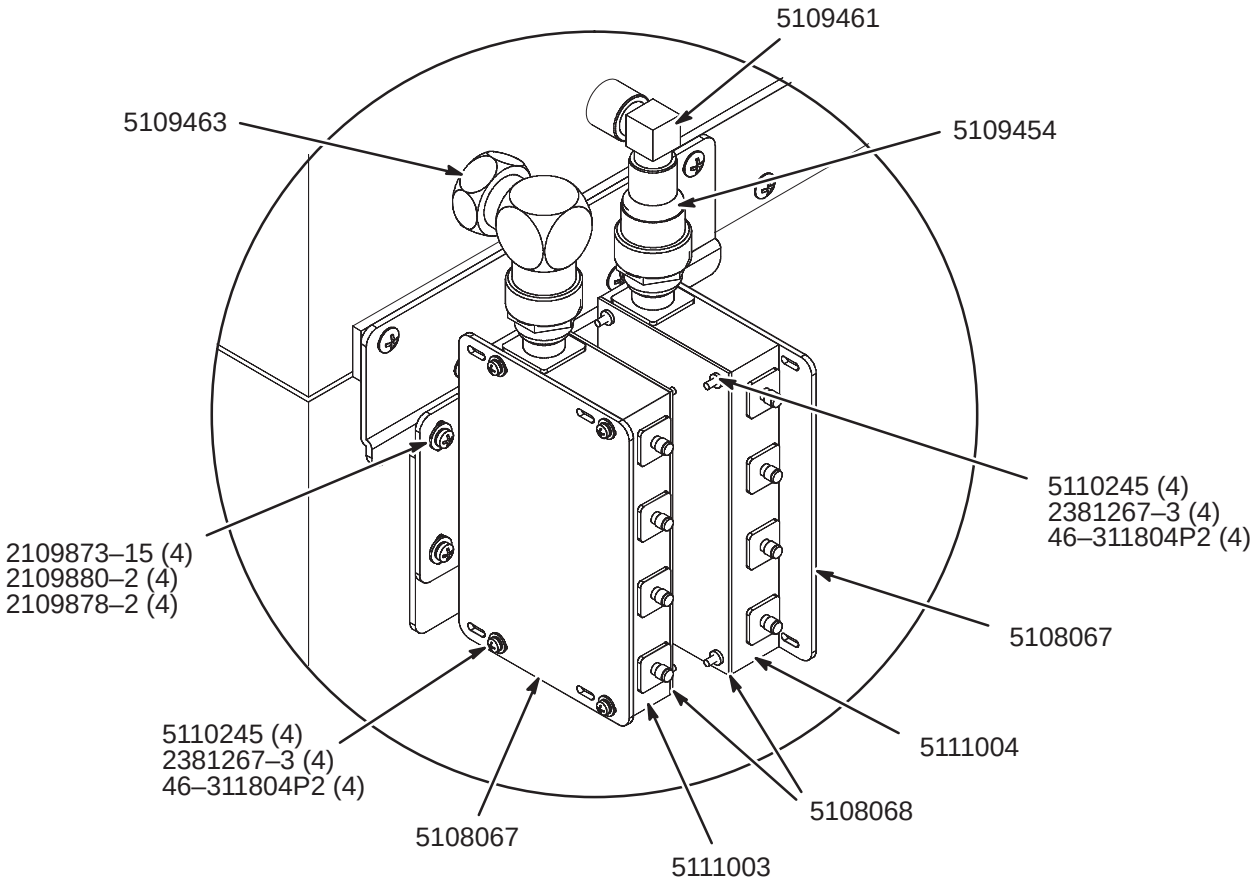
12–2–3 INSTALL DUAL DIRECTIONAL MOUNTING PLATE

The parts for installing the Dual Directional Couplers are located in the E2 Coupler and Bracket Kit (5110677).

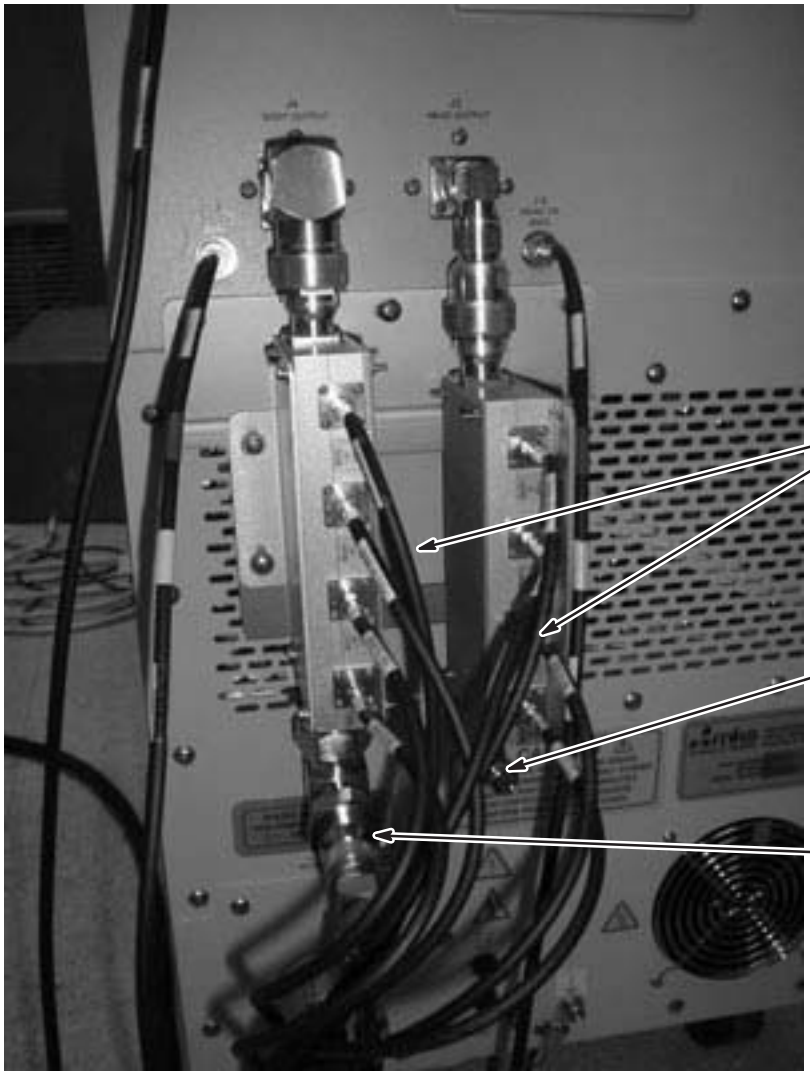


12–2–4 INSTALL HEAD AND BODY DUAL DIRECTIONAL COUPLERS

1. Assemble the 45DB–40DB Dual Directional Head RF Coupler (5111004), E2 Coupler Mounting Bracket (5108067) and the E2 Coupler Backing Plate (5108068) as shown below using four of each of the following: M3 x 40 Screw (5110245), Lock Washer (2381267–3), and Flat Washer (46–311804P2). Leave screws loose.
2. Assemble the 55DB–50DB Dual Directional Body RF Coupler (5111003), E2 Coupler Mounting Bracket (5108067) and the E2 Coupler Backing Plate (5108068) as shown below using four of each of the following: M3 x 40 Screw (5110245), Lock Washer (2381267–3), and Flat Washer (46–311804P2). Leave screws loose.
3. Attach both module assemblies to Coupler Offset Plate using four each M4 x 12 Screws (2109873–15), Lock Washer (2109880–2), and Flat Washer (2109878–2). Leave screws loose.
4. Install 7–16 female to N male Adapter (5109454)to J1 INPUT at top of Head Coupler. Install N female to N male 90 degree Adapter (5109461) between J3 on cabinet and 5109454 Adapter.
5. Install N female to N male 90 degree Adapter (5109461) at J2 INPUT at bottom of Head Coupler.
6. Install 7–16 female to 7–16 male 90 degree Adapter (5109463) between J1 OUTPUT at top of Body Coupler and J4 on cabinet.
7. Install 7–16 female to 7–16 male 90 degree Adapter (5109463) at J2 INPUT at bottom of Body Coupler.
8. Align couplers and tighten all screws.



12–2–5 CONNECT CABLES TO DUAL DIRECTIONAL COUPLERS

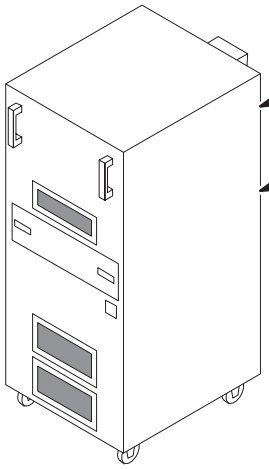


1. Connect the 8 cables that make up Run 1264 to the Head and Body Couplers as marked. Refer to cable map.
2. Connect Run 1249 to adapter at J2 INPUT at bottom of Head coupler.
3. Connect Run 1156 to adapter at J2 INPUT at bottom of Body coupler.

12–3 25/35Kw RF AMPLIFER INSTALLATION

If part of system, complete the following procedures to install a new 25/35kW RF Amplifer for upgrading to HDx.

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.



1. Position NB RF cabinet as indicated by site layout drawings.
2. Refer to cable map in Section 4 for information on connecting cables to cabinet.

Notes:

- Run 1232 is provided with the amplifer to replace Run 1177 shipped in the upgrade. Route and connect from J20 (RF Amp Cabinet) to J127 (RFS Cabinet).
- Run 1113 is provided with the RF Amp Cabinet for the MAIN INPUT (power) in place of Run 967 on cable map.
- All other runs per Section 4 cable map.

SECTION 13 – COMPLETION

13–1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non–GE personnel. Check off boxes at the front of each item below when complete.

- ❑ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ❑ Resolve shipment shortages
- ❑ Resolve omissions by mechanical contractors
- ❑ Make sure all rating plates are installed
- ❑ Recheck all wiring connections using cable map (Section 4)
- ❑ Correct for wiring errors if necessary
- ❑ Complete Sections 13–2, 13–3, and 13–4.

13–2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible.** Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 13–3. Recycling Operation will route them to appropriate location.

13–3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1–7. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

13–4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1–8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD–ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 14.x has been made before returning system to the users. Turn site over to applications who will instruct users.

13–5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD–ROM, Upgrade Calibration Flow, and perform the following:
   - Power On > PDU Voltage
   - Power On > Startup Checks
   - Power On > Transformer Taps
   - PDU E–Stop Check
4. Press the FULL ON button on the PDU front control panel.
5. Except for the HFD Gradient power modules, switch all cabinet circuit breakers to ON. The HFD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.